

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T18:49:17

Source: data/json/sonnet45_8agent_game_directive_r10_n5/claude-sonnet-4.5/game/sonnet45_8agent_game_directive_r10_n5_003_neutral_rep03.json

RUN METADATA

model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
replicate_id: 3
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
 "game"
]

agent_names: {
 "Agent_1": "Aster",
 "Agent_2": "Briar",
 "Agent_3": "Cyra",
 "Agent_4": "Dorian",
 "Agent_5": "Elara",
 "Agent_6": "Finn",
 "Agent_7": "Galen",
 "Agent_8": "Hana"
}
dyadic_pairing_schedule: {
 "num_turns": 10,
 "role_targets": {
 "Agent_1": 5,
 "Agent_2": 5,
 "Agent_3": 5,
 "Agent_4": 5,
 "Agent_5": 5,
 "Agent_6": 5,
 "Agent_7": 5,
 "Agent_8": 5
 },
 "strategy": "balanced_random_roles_v1"
}

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Agent_3 (Cyra); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Agent_4 (Dorian); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Agent_5 (Elara); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Agent_6 (Finn); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Agent_7 (Galen); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Agent_8 (Hana); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

ROUND 1

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_3 (Cyra)=Sender, Agent_5 (Elara)=Receiver
 dyad_2: Agent_4 (Dorian)=Sender, Agent_6 (Finn)=Receiver
 dyad_3: Agent_1 (Aster)=Sender, Agent_7 (Galen)=Receiver
 dyad_4: Agent_8 (Hana)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_3 (Cyra) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_5 (Elara)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_6 (Finn)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

dyad_3: Agent_1 (Aster) sent $3; received $9; Agent_7 (Galen) returned $5; payoffs Agent_1
(Aster)=$7, Agent_7 (Galen)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 (Hana) sent $3; received $9; Agent_2 (Briar) returned $4.5; payoffs Agent_8
(Hana)=$6.5, Agent_2 (Briar)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 7.0,
  "Agent_2": 4.5,
  "Agent_3": 6.5,
  "Agent_4": 6.5,
  "Agent_5": 4.5,
  "Agent_6": 4.5,
  "Agent_7": 4.0,
  "Agent_8": 6.5
}

```

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-06-04T18:38:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_3",
      "Agent_5"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship with Elara since this is an iterated game. Sending money creates value (it triples!), which benefits us both if there's reciprocity.

I'll send a substantial amount to signal trust and willingness to cooperate, while learning about Elara's approach to reciprocity.

```
'''json
{"send": 3}
'''
```

This gives Elara \$9 to work with. If she returns a fair share, we can both benefit and build trust for future rounds.

Reasoning:

This is the first round of an iterated trust game. I am the sender and have \$5. I need to decide how much to send to Elara (Agent_5).

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds
2. If I send money, it gets tripled, which creates a larger pie to split
3. The receiver (Elara) then decides how much to return
4. This is the first interaction, so I have no history with Elara

From a cooperation perspective:

- Sending money can be mutually beneficial since it gets tripled
- If I send \$5 and Elara returns a fair share, we both could do better than if I send nothing
- However, there's a risk that Elara returns little or nothing

From a strategic perspective in an iterated game:

- It's often beneficial to start with some level of cooperation to test the waters
- If Elara reciprocates positively, we can build trust and both benefit in future rounds
- If Elara defects, I can adjust my strategy in future rounds

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while not risking everything

- Send everything (\$5) to maximize the joint payoff and signal strong trust
- Send nothing (\$0) to avoid risk

For the first round, I'll take a cooperative approach and send a substantial amount to signal trust and see how Elara responds. A common first-move strategy is to send about half or send everything.

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate
- Creates a significant pie (\$9) for Elara to split
- Doesn't risk my entire endowment
- Allows me to learn about Elara's reciprocity

If I send \$3:

- Elara receives \$9
- I keep \$2
- If Elara returns \$4-5, we both do well

- If Elara returns little, I can adjust my strategy next round

Since this is repeated, starting with high trust could pay dividends. Sending \$4 or \$5 might establish a stronger foundation for cooperation. But \$3 feels like the right balance-it demonstrates genuine willingness to cooperate while preserving some security, and it gives me useful information about how Elara approaches reciprocity before I commit more heavily in future rounds.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 725,
  "reasoning_tokens": 572
}
```

GAME - Agent_5 - Receiver - dyad_1

Timestamp: 2026-06-04T18:38:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_3",
      "Agent_5"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role

of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplication
- This created \$6.0 of new value (\$9 - \$3)

I want to reward Cyra's generosity while keeping some benefit for myself. A fair split would be to share the gains relatively equally.

If I return \$4.5:

- Cyra ends with: \$5 - \$3 + \$4.5 = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

This rewards Cyra's trust and sets a cooperative tone for future rounds.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me understand the situation:

- I'm Elara, playing as the RECEIVER
- My opponent is Cyra (Agent_3), playing as the SENDER
- Cyra sent me \$3.0 out of their \$5.0 endowment (60%)
- I received \$9.0 (the \$3.0 was multiplied by 3x)
- I need to decide how much to return (between \$0 and \$9.0)

Now I need to think about what would be a fair and strategic return:

1. Cyra was quite generous, sending 60% of their endowment
2. If I return nothing, Cyra would have: $\$5 - \$3 + \$0 = \2 , and I would have $\$9$
3. If I return everything ($\$9$), Cyra would have: $\$5 - \$3 + \$9 = \11 , and I would have $\$0$

A reasonable approach would be to:

- Reward Cyra's generosity to encourage future cooperation
- Keep some profit for myself
- Aim for a relatively balanced outcome

Let me think about what's fair:

- Cyra invested $\$3$ to create $\$9$ of value
- The net gain from the transaction is $\$9 - \$3 = \$6$
- If we split this gain equally, that would be $\$3$ each
- So Cyra would get back $\$3$, ending with $\$5 - \$3 + \$3 = \5
- I would keep $\$9 - \$3 = \$6$

This seems reasonable but maybe I could be a bit more generous since Cyra sent a significant portion (60%).

Another way to think about it:

- Cyra's final amount: $\$5 - \$3 + \text{returned}$
- My final amount: $\$9 - \text{returned}$
- For equal final amounts: $\$2 + \text{returned} = \$9 - \text{returned}$, so $\text{returned} = \$3.5$

If I return $\$4$ or $\$4.5$:

- Returning $\$4$ leaves Cyra with $\$6$ and me with $\$5$, giving them a slight edge
- Returning $\$4.5$ results in Cyra having $\$6.5$ and me with $\$4.5$, which feels more balanced and acknowledges their trust

I'll return $\$4.5$ to reciprocate their generosity and signal willingness to cooperate.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 799,
  "reasoning_tokens": 434
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T18:38:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_6"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_4": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play an iterated game,
      consisting of multiple rounds, and may also be asked to write myths.

      GAME:
      The game consists of two roles: sender and receiver. In each round, you will play the role
      of either the sender or the receiver. Your current role will be specified in the round
      prompt.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
      , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
      ll have acted as sender and receiver equally often when the round count allows it.
      - You will be told your opponent's name in each round.
      - Your name in this experiment is Dorian.
  Message 2 role=user
    content:
      Your name in this experiment is Dorian.
      Your opponent this round is Finn (Agent_6).

      Round 1: This round, you are the SENDER. You have $5. How much do you send? (0-5)
Assistant output:
  I'm playing the first round as sender with Finn. Since this is an iterated game and I'll poten
  tially interact with Finn (and others) again, I want to signal cooperation while not being o
  verly risky in this initial interaction.

  Sending a moderate amount allows us both to benefit if Finn reciprocates fairly. If I send $3,
  Finn receives $9 (tripled). If Finn returns a fair share, we can both end up better off tha
  n if I sent nothing.

  I'll adopt a cooperative but cautious strategy for this first round.

  '''json
  {"send": 3}
  '''
Reasoning:
  This is the first round of an iterated trust game. I'm the sender and have $5 to potentially s
  end to Finn (Agent_6).

```

Let me think about the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return to me
3. This is an iterated game with multiple rounds
4. I'll be playing with different partners across rounds (randomized pairings)

Key strategic considerations for the first round:

- This is the first interaction, so I don't have any history with Finn
- Since it's an iterated game with multiple rounds, building trust and cooperation could be beneficial
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), Finn receives \$15, and if they return fairly (say half or more), I could end up with more than \$5
- A moderate amount shows willingness to cooperate while not being overly risky

Common strategies in trust games:

- Sending around 50% (\$2-3) signals cooperation without being too vulnerable
- Sending everything shows maximum trust but is very risky in first interaction
- Sending nothing is safe but prevents any cooperative surplus

For the first round with an unknown partner, I'll adopt a moderately cooperative strategy. Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If Finn returns at least \$3 out of the \$9 they receive, I break even
- If they return more (say half of \$9 = \$4.5), we both benefit
- Leaves room to adjust strategy in future rounds based on outcomes

Let me send \$3.

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 528,
  "reasoning_tokens": 410
}
```

GAME - Agent_6 - Receiver - dyad_2

Timestamp: 2026-06-04T18:39:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_6"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_4": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing significant trust
- This was multiplied to \$9.0 for me
- Dorian kept \$2.0 from their original \$5.0

Since this is Round 1 and Dorian showed considerable trust by sending 60% of their endowment, I want to reciprocate fairly to establish a cooperative relationship for future rounds.

I'll return an amount that rewards their trust while ensuring we both benefit. If I return \$4.5:

- Dorian ends with: \$2.0 + \$4.5 = \$6.5
- I end with: \$9.0 - \$4.5 = \$4.5

This represents a 50% return rate on what I received, which is a common reciprocity norm in trust games and should encourage continued cooperation.

{'return': 4.5}

Reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in this round. The SENDER (Dorian/Agent_4) has:

- Started with \$5
- Sent me \$3.0 (which is 60% of their endowment)
- The \$3.0 was multiplied by 3x, so I received \$9.0

Now I need to decide how much to return (between \$0 and \$9.0).

Key considerations:

1. This is Round 1 of what appears to be an iterated game
2. My opponent sent a significant portion (60%) of their endowment, showing trust/cooperation
3. In iterated trust games, reciprocity often leads to better long-term outcomes
4. If I return a fair amount, it may encourage future cooperation
5. The sender kept \$2 (\$5 - \$3) and will receive whatever I return

Let me think about what would be a "fair" return:

- The sender gave up \$3 to give me \$9 (net gain of \$6 for me before returning)
- One approach: return enough so we both benefit roughly equally
- If I return \$4.5, the sender would have: $\$2 + \$4.5 = \$6.5$ total
- I would have: $\$9 - \$4.5 = \$4.5$ total
- That seems slightly unfair to me since the sender ends up with more

Another approach: try to equalize our final payoffs

- Sender has \$2 left, I have \$9
- If we want equal payoffs: $(2 + x) = (9 - x)$, so $2x = 7$, $x = \$3.5$
- If I return \$3.5: Sender gets \$5.5 total, I get \$5.5 total

Or I could think about it as: the sender "invested" \$3 which created \$6 of surplus value. We could split this surplus:

- Sender baseline: \$2 (what they kept)
- My baseline: \$0 (what I started with)
- Surplus created: \$9 (what I received)
- If we split surplus 50/50: each gets \$4.5 of surplus
- Sender would get: $\$2 + \$4.5 = \$6.5$
- I would get: \$4.5
- So I'd return: $\$9 - \$4.5 = \$4.5$

I could also match their reciprocity directly by returning approximately what they sent (\$3), which would signal I'm willing to cooperate in future rounds.

Returning \$4.5 feels like the stronger move though-it rewards their trust more generously while still leaving me ahead. Since this is the opening round and they demonstrated real confidence by sending 60% of their endowment, I want to signal I'm committed to mutual benefit. This return amount splits the surplus fairly and should encourage them to keep trusting me.

```
Usage: {  
  "input_tokens": 562,  
  "output_tokens": 1044,  
  "reasoning_tokens": 575  
}
```

GAME - Agent_1 - Sender - dyad_3

Timestamp: 2026-06-04T18:39:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_7"  
    ],  
    "investor": "Agent_1",  
    "roles": {
```

```

    "Agent_1": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship while being somewhat cautious since I don't know Galen yet.

If I send a moderate amount, it gives Galen an opportunity to benefit significantly (due to the 3x multiplier) while also allowing them to demonstrate trustworthiness by returning a fair share.

I'll send **\$3**.

This means Galen will receive \$9. If they return around half or more of what they receive, we'll both end up better off than if I had sent nothing. This tests the waters for potential cooperation in future rounds.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Aster) and my opponent is Galen (Agent_7).

Let me think about the game dynamics:

- As sender, I get \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether this is truly iterated (we'll play multiple rounds)
2. Whether I'll play with the same opponent again
3. Whether the receiver will reciprocate

Since this is described as "multi-task experiment" with "multiple rounds" and there are 8 agents total with randomized pairings, I might or might not play with Galen again.

Strategic considerations:

- If I send nothing (\$0), I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd get \$7.50 total, which is better
- But if receiver keeps everything, I get \$0

Since this is round 1 and I want to establish some level of cooperation, but also don't know this opponent yet, a moderate strategy makes sense. Common approaches:

- Send about half (\$2-3) to show willingness to cooperate but not full commitment
- This allows the receiver to profit while testing their reciprocity

Let me send \$3:

- Receiver gets \$9
- If they return \$4-5, we both benefit
- If they return \$3, we're even
- This tests their cooperation level

I'll send \$3 as a reasonable starting point that shows trust but isn't reckless.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 562,
  "reasoning_tokens": 410
}
```

GAME - Agent_7 - Receiver - dyad_3

Timestamp: 2026-06-04T18:40:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_7"
  ],
}
```

```

"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.

Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing considerable trust

- This created \$9 for me after the 3x multiplier
- Aster currently has \$2 remaining from their original \$5

Since this is Round 1 and Aster demonstrated strong trust by sending a majority of their endowment, I want to reciprocate and establish a cooperative relationship for future rounds.

If I return \$5:

- Aster will have: $\$2 + \$5 = \$7$ (net gain of \$2)
- I will keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both of us benefit from the cooperation

This shows I appreciate their trust while maintaining some benefit from being in the receiver position.

```
{'return': 5.0}
```

Reasoning:

Let me understand the situation:

1. I'm Galen, playing as RECEIVER in this round
2. My opponent is Aster (Agent_1)
3. Aster (as SENDER) sent me \$3.0 out of their \$5.0 endowment (60%)
4. I receive $\$3.0 \times 3 = \9.0
5. I need to decide how much to return (between \$0 and \$9.0)

Let me think about the strategy:

From Aster's perspective:

- They started with \$5
- They sent \$3, keeping \$2
- They currently have \$2

From my perspective:

- I received \$9
- Whatever I return goes back to Aster
- I keep $\$9 - (\text{amount returned})$

This is a classic trust game scenario. The key considerations are:

1. **Reciprocity**: Aster showed significant trust by sending 60% (\$3). This is a substantial amount, suggesting they're hoping for cooperation.
2. **Future rounds**: This is an iterated game with multiple rounds. Building trust and a reputation for reciprocity could benefit me in future rounds.
3. **Fairness**: One common approach is to split the gains fairly. The total created value is \$9. A fair split might be to ensure we both end up with similar amounts, or that we both benefit from the cooperation.
4. **Strategic thinking**: If I return too little, Aster might send less (or nothing) in future rounds. If I return a reasonable amount, it encourages future cooperation.

Let me calculate some scenarios:

If I return \$0:

- I keep: \$9
- Aster has: \$2 (their remaining \$5 - \$3 sent = \$2)
- Total for Aster: \$2
- This seems very unfair to Aster who trusted me

If I return \$3:

- I keep: $\$9 - \$3 = \$6$

- Aster gets back to their starting \$5, while I pocket \$6 in gains. If I return \$4 instead, we'd both end up with roughly \$5-6, creating a more balanced outcome where Aster gains \$1.5 and I gain \$4. Returning \$4.5 would give Aster \$6.5 and me \$4.5, which feels more equitable given their initial trust.

The fairest split appears to be returning \$6-this way we each gain \$3 from cooperating. While I could justify keeping more since I'm in the stronger position, returning two-thirds of my

windfall demonstrates good faith. Alternatively, I could match Aster's proportional generosity: they sent 60% of their endowment, so returning 60% of the \$9 I received (\$5.4) would mirror their level of trust.

For Round 1, establishing cooperation matters more than maximizing my immediate gain. Returning between \$4.50 and \$5.00 balances reciprocity with keeping some advantage from being the receiver. I'll return \$4.50-I gain \$4.50 while Aster gains \$1.50, which acknowledges their risk-taking without being equal, but signals I'm not exploiting the situation.

Actually, let me verify the math on fair splits. If I'm aiming for equal final payoffs: Aster keeps \$2 from their original \$5, and I receive the tripled \$3 as \$9. For us to end with the same amount, I'd return \$3.50, giving us each \$5.50. Alternatively, for equal gains from the exchange itself, I could calculate what each of us gains relative to our starting positions. I should get $\$0 + \$3 = \$3$ from an equal split, which means returning \$6 to give Aster the \$8 they deserve. But I could also aim for equal final payoffs of \$5.50 each, requiring a \$3.50 return. Returning \$5 feels right-it acknowledges Aster's substantial trust while letting me keep a modest advantage, and it creates a \$2 gain for them versus my \$4, which seems fair given the asymmetry of the initial position.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 1868,
  "reasoning_tokens": 874
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-06-04T18:40:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_2"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_2": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship with Briar while being somewhat cautious since we haven't played together before.

If I send \$3, Briar will receive \$9 (3×3). This gives them a meaningful amount to work with while I keep \$2 initially. If Briar reciprocates fairly, we can both come out ahead.

This shows trust and willingness to cooperate, which could set a positive tone for future rounds, while not leaving me completely vulnerable.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of an iterated game. I'm playing as the sender (Hana) and my opponent is Briar (Agent_2).

Let me think through the strategic considerations:

1. As the sender, I receive \$5 and decide how much to send (0-5)
2. The amount I send gets multiplied by 3x
3. The receiver (Briar) decides how much to return to me
4. I keep: \$5 - sent + returned
5. Briar keeps: (sent × 3) - returned

This is a classic trust game. The optimal outcome for both of us collectively would be if I send the full \$5, which becomes \$15 for Briar, and then Briar returns a fair share back to me.

However, there are strategic considerations:

- If I send \$5 and it becomes \$15, Briar could keep all \$15 and return \$0, leaving me with \$0
- If I send \$0, I keep \$5 and Briar gets \$0
- A middle ground might build trust

Since this is an iterated game (multiple rounds) and I'm in a multi-agent setting where I might play with Briar again or with others who might hear about my actions, reputation matters.

First round strategies:

1. Send a moderate amount (like \$2-3) to test Briar's reciprocity
2. Send the full amount (\$5) to signal trust and cooperation
3. Send nothing (\$0) to play it safe

Given that this is:

- Round 1 (establishing patterns)
- An iterated game (future interactions matter)
- A multi-agent setting (reputation could spread)

I think a good strategy is to send a substantial but not complete amount to signal willingness to cooperate while protecting myself somewhat. Sending \$3 would give Briar \$9, which is enough to return a fair amount while still keeping more than they started with.

Actually, let me reconsider. In iterated trust games, cooperation often emerges when players signal strong trust early. Sending the full \$5 in the first round can establish a cooperative norm.

If Briar reciprocates with a reasonable return (around \$5-7 from the \$15), we both come out ahead. I'll send \$3 as my opening move-enough to demonstrate good faith without overexposing myself in round one.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 667,
  "reasoning_tokens": 528
}
```

GAME - Agent_2 - Receiver - dyad_4

Timestamp: 2026-06-04T18:40:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_2"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_2": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment), showing considerable trust
- This tripled to \$9.0 for me
- Since this is an iterated game, establishing trust and reciprocity is important

Hana's generosity created \$6.0 in additional value (\$9 - \$3 original). A fair approach would be to share these gains. If I return about half of what I received (\$4.5), then:

- Hana ends with: \$2 + \$4.5 = \$6.50
- I end with: \$9 - \$4.5 = \$4.50

Both of us profit from the cooperation, which should encourage future trust and collaboration.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me understand the situation:

- I'm Briar, playing as the RECEIVER in this round
- My opponent is Hana (Agent_8), playing as the SENDER
- Hana sent me \$3.0 out of her \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. Hana showed significant trust by sending 60% of her endowment
2. This is an iterated game, so reciprocity may encourage future cooperation

3. If I return a fair amount, it builds trust for future rounds
4. Hana will end up with: $\$5 - \$3 + (\text{what I return}) = \$2 + (\text{what I return})$
5. I will end up with: $\$9 - (\text{what I return})$

What's a fair return? Let me consider:

- Hana invested \$3, which became \$9 for me
- A fair split might be to share the gains equally
- The gain was $\$9 - \$3 = \$6$
- If we split the gain equally: each gets \$3 from the gain
- So Hana gets her initial \$3 back plus \$3 = \$6 total
- That means I should return \$4 (she already has \$2, so $\$2 + \$4 = \$6$)
- I would keep \$5

Alternatively:

- Equal final payoffs: If we want to end up with equal amounts
- Total money in the system: Hana's remaining \$2 + my \$9 = \$11
- Equal split: \$5.50 each
- Hana has \$2, needs \$3.50 more, so I return \$3.50
- I would keep \$5.50

Another perspective:

- Hana sent 60%, showing good faith
- A reciprocal return of 40-50% of what I received seems appropriate
- That would be \$3.60 to \$4.50
- Returning \$4.50 signals trustworthiness and cooperation, which should strengthen future interactions
- This leaves Hana with \$6.50 and me with \$4.50
- both better off than if they'd sent nothing

I'll send back \$4.50.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 845,
  "reasoning_tokens": 437
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

- dyad_1: Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver
- dyad_2: Agent_5 (Elara)=Sender, Agent_1 (Aster)=Receiver
- dyad_3: Agent_7 (Galen)=Sender, Agent_4 (Dorian)=Receiver
- dyad_4: Agent_6 (Finn)=Sender, Agent_8 (Hana)=Receiver

Parsed game decisions:

- dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_3 (Cyra) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_3 (Cyra)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
- dyad_2: Agent_5 (Elara) sent \$3; received \$9; Agent_1 (Aster) returned \$5; payoffs Agent_5 (Elara)=\$7, Agent_1 (Aster)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
- dyad_3: Agent_7 (Galen) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_7 (Galen)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
- dyad_4: Agent_6 (Finn) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_8 (Hana)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 11.0,
"Agent_2": 11.0,
```



```

"Agent_3": 11.0,
"Agent_4": 11.0,
"Agent_5": 11.5,
"Agent_6": 11.0,
"Agent_7": 10.5,
"Agent_8": 11.0
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T18:40:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_2",
      "Agent_3"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment), showing considerable trust
- This tripled to \$9.0 for me
- Since this is an iterated game, establishing trust and reciprocity is important

Hana's generosity created \$6.0 in additional value (\$9 - \$3 original). A fair approach would be to share these gains. If I return about half of what I received (\$4.5), then:

- Hana ends with: $\$2 + \$4.5 = \$6.50$
- I end with: $\$9 - \$4.5 = \$4.50$

Both of us profit from the cooperation, which should encourage future trust and collaboration.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me understand the situation:

- I'm Briar, playing as the RECEIVER in this round
- My opponent is Hana (Agent_8), playing as the SENDER
- Hana sent me \$3.0 out of her \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. Hana showed significant trust by sending 60% of her endowment
2. This is an iterated game, so reciprocity may encourage future cooperation
3. If I return a fair amount, it builds trust for future rounds
4. Hana will end up with: $\$5 - \$3 + (\text{what I return}) = \$2 + (\text{what I return})$
5. I will end up with: $\$9 - (\text{what I return})$

What's a fair return? Let me consider:

- Hana invested \$3, which became \$9 for me
- A fair split might be to share the gains equally
- The gain was $\$9 - \$3 = \$6$
- If we split the gain equally: each gets \$3 from the gain
- So Hana gets her initial \$3 back plus \$3 = \$6 total
- That means I should return \$4 (she already has \$2, so $\$2 + \$4 = \$6$)
- I would keep \$5

Alternatively:

- Equal final payoffs: If we want to end up with equal amounts
- Total money in the system: Hana's remaining \$2 + my \$9 = \$11

- Equal split: \$5.50 each
- Hana has \$2, needs \$3.50 more, so I return \$3.50
- I would keep \$5.50

Another perspective:

- Hana sent 60%, showing good faith
- A reciprocal return of 40-50% of what I received seems appropriate
- That would be \$3.60 to \$4.50
- Returning \$4.50 signals trustworthiness and cooperation, which should strengthen future interactions
- This leaves Hana with \$6.50 and me with \$4.50
- both better off than if they'd sent nothing

I'll send back \$4.50.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 845,
  "reasoning_tokens": 437
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- In Round 1, Hana sent me 60% (\$3), which tripled to \$9, and I returned \$4.5
- This established a cooperative pattern
- Now I'm the sender with Cyra, whom I haven't played with before

Since cooperation worked well with Hana, I'll start with a similar level of trust with Cyra. Sending \$3 (60% of my endowment) demonstrates willingness to cooperate while not being overly risky in this early stage.

If Cyra reciprocates fairly (returning around 50% of what they receive), we'll both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

Reasoning:

Now I'm playing as the SENDER against Cyra (Agent_3).

Let me think about this strategically:

1. In the previous round, I was the receiver and Hana sent me \$3.0 (60% of their endowment). I returned \$4.5, which was 50% of what I received. This seemed like a fair reciprocal move.
2. Now I'm the sender. I need to decide how much to send to Cyra.
3. This is an iterated game with multiple rounds, so building trust and cooperation is valuable.
4. If I send money, it triples, so:
 - If I send \$3: Cyra receives \$9
 - If I send \$5: Cyra receives \$15
 - If I send \$2: Cyra receives \$6

5. The question is: how much will Cyra return? I don't know Cyra's behavior yet, but in iterated games, starting with cooperation often leads to better long-term outcomes.
6. Hana sent 60% (\$3). That worked out well - they showed trust and I reciprocated fairly.
7. I could:
 - Send \$3 (60%, matching what Hana did)
 - Send \$2.5 (50%)
 - Send \$5 (100%, maximum trust)
 - Send less (being more cautious)

Given that:

- This is early in the game (round 2)
- I want to establish a cooperative relationship
- Hana's 60% send worked well
- I don't know Cyra yet, but starting with moderate trust is reasonable

I'll send \$3, matching the 60% that Hana sent to me. This shows trust while not being overly risky.

```
Usage: {
  "input_tokens": 891,
  "output_tokens": 575,
  "reasoning_tokens": 350
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-06-04T18:41:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_2",
      "Agent_3"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Elara since this is an iterated game. Sending money creates value (it triples!), which benefits us both if there's reciprocity.

I'll send a substantial amount to signal trust and willingness to cooperate, while learning about Elara's approach to reciprocity.

```
'''json
{"send": 3}
'''
```

This gives Elara \$9 to work with. If she returns a fair share, we can both benefit and build trust for future rounds.

reasoning:

This is the first round of an iterated trust game. I am the sender and have \$5. I need to decide how much to send to Elara (Agent_5).

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds
2. If I send money, it gets tripled, which creates a larger pie to split
3. The receiver (Elara) then decides how much to return
4. This is the first interaction, so I have no history with Elara

From a cooperation perspective:

- Sending money can be mutually beneficial since it gets tripled
- If I send \$5 and Elara returns a fair share, we both could do better than if I send nothing
- However, there's a risk that Elara returns little or nothing

From a strategic perspective in an iterated game:

- It's often beneficial to start with some level of cooperation to test the waters
- If Elara reciprocates positively, we can build trust and both benefit in future rounds
- If Elara defects, I can adjust my strategy in future rounds

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while not risking everything
- Send everything (\$5) to maximize the joint payoff and signal strong trust
- Send nothing (\$0) to avoid risk

For the first round, I'll take a cooperative approach and send a substantial amount to signal trust and see how Elara responds. A common first-move strategy is to send about half or send everything.

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate
- Creates a significant pie (\$9) for Elara to split
- Doesn't risk my entire endowment
- Allows me to learn about Elara's reciprocity

If I send \$3:

- Elara receives \$9
- I keep \$2
- If Elara returns \$4-5, we both do well

- If Elara returns little, I can adjust my strategy next round

Since this is repeated, starting with high trust could pay dividends. Sending \$4 or \$5 might establish a stronger foundation for cooperation. But \$3 feels like the right balance- it demonstrates genuine willingness to cooperate while preserving some security, and it gives me useful information about how Elara approaches reciprocity before I commit more heavily in future rounds.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 725,
  "reasoning_tokens": 572
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Briar sent me \$3.0, showing trust and cooperation
- I received \$9.0
- In the previous round, when I sent \$3.0 to Elara, she returned \$4.5, which was a fair 50-50 split of the tripled amount

I want to reciprocate Briar's cooperative move with similar generosity and establish a pattern of mutual benefit.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 2: I'm the RECEIVER against Briar

- Briar sent me \$3.0
- I received \$9.0 (tripled)
- I need to decide how much to return

Context from Round 1:

- I was SENDER against Elara
- I sent \$3.0, Elara received \$9.0
- Elara returned \$4.5 to me
- My payoff was \$6.5 (I kept \$5 - \$3 + \$4.5 = \$6.5)
- Elara kept \$9.0 - \$4.5 = \$4.5

So Elara split the \$9.0 equally with me, which was a fair and cooperative move.

Now Briar has sent me \$3.0, exactly what I sent to Elara. Briar is taking the same cooperative approach.

To maintain reciprocity and encourage continued cooperation:

- Briar kept \$5 - \$3 = \$2
- I received \$9
- If I return \$4.5, I keep \$4.5 and Briar gets \$2 + \$4.5 = \$6.5 total
- This mirrors what Elara did for me

This seems like the fair and cooperative response. It:

1. Matches the generosity Elara showed me
2. Rewards Briar's trust in sending \$3
3. Encourages future cooperation
4. Splits the gains relatively fairly

Let me verify the math:

- Briar's payoff: \$5 - \$3 + \$4.5 = \$6.5
- My payoff: \$9 - \$4.5 = \$4.5

This is a fair split of the surplus created. I'll return \$4.5.

```
Usage: {
  "input_tokens": 810,
  "output_tokens": 545,
  "reasoning_tokens": 307
}
```

GAME - Agent_5 - Sender - dyad_2

Timestamp: 2026-06-04T18:41:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    },
  },
  "agents": [
    "Agent_5",
    "Agent_1"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_1": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplication
- This created \$6.0 of new value (\$9 - \$3)

I want to reward Cyra's generosity while keeping some benefit for myself. A fair split would be to share the gains relatively equally.

If I return \$4.5:

- Cyra ends with: \$5 - \$3 + \$4.5 = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

This rewards Cyra's trust and sets a cooperative tone for future rounds.

```
'''json
{"return": 4.5}
```


'''

reasoning:

Let me understand the situation:

- I'm Elara, playing as the RECEIVER
- My opponent is Cyra (Agent_3), playing as the SENDER
- Cyra sent me \$3.0 out of their \$5.0 endowment (60%)
- I received \$9.0 (the \$3.0 was multiplied by 3x)
- I need to decide how much to return (between \$0 and \$9.0)

Now I need to think about what would be a fair and strategic return:

1. Cyra was quite generous, sending 60% of their endowment
2. If I return nothing, Cyra would have: $\$5 - \$3 + \$0 = \2 , and I would have \$9
3. If I return everything (\$9), Cyra would have: $\$5 - \$3 + \$9 = \11 , and I would have \$0

A reasonable approach would be to:

- Reward Cyra's generosity to encourage future cooperation
- Keep some profit for myself
- Aim for a relatively balanced outcome

Let me think about what's fair:

- Cyra invested \$3 to create \$9 of value
- The net gain from the transaction is $\$9 - \$3 = \$6$
- If we split this gain equally, that would be \$3 each
- So Cyra would get back \$3, ending with $\$5 - \$3 + \$3 = \5
- I would keep $\$9 - \$3 = \$6$

This seems reasonable but maybe I could be a bit more generous since Cyra sent a significant portion (60%).

Another way to think about it:

- Cyra's final amount: $\$5 - \$3 + \text{returned}$
- My final amount: $\$9 - \text{returned}$
- For equal final amounts: $\$2 + \text{returned} = \$9 - \text{returned}$, so $\text{returned} = \$3.5$

If I return \$4 or \$4.5:

- Returning \$4 leaves Cyra with \$6 and me with \$5, giving them a slight edge
- Returning \$4.5 results in Cyra having \$6.5 and me with \$4.5, which feels more balanced and acknowledges their trust

I'll return \$4.5 to reciprocate their generosity and signal willingness to cooperate.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 799,
  "reasoning_tokens": 434
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this new round with Aster:

In my previous round, Cyra showed trust by sending 60% of their endowment, and I rewarded that trust by returning a fair amount. I want to continue establishing cooperative relationships

I'll send \$3 (60% of my endowment) to match the level of trust Cyra showed me. This signals co

operation and gives Aster a chance to benefit while also potentially returning a fair amount
.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze this situation:

Round 2, I'm the SENDER against Aster (Agent_1).

- I have \$5
- Whatever I send gets multiplied by 3x
- Aster will decide how much to return

In the previous round with Cyra:

- Cyra sent \$3 (60% of endowment)
- I received \$9
- I returned \$4.5
- This was a fairly cooperative move

Since I want to establish cooperation and this is a new opponent, I should signal trust. Cyra sent 60% in round 1, which was cooperative. I should probably match that level of cooperation or be similarly generous.

Sending \$3 (60%) would:

- Show similar trust to what Cyra showed me
- Create \$9 for Aster
- If Aster is cooperative and returns half of the gains (similar to what I did), they might return around \$4.5
- That would give me: $\$5 - \$3 + \$4.5 = \6.5

Sending \$2 (40%) would be more cautious but still cooperative.

Sending \$4 (80%) would be very trusting.

I think sending \$3 makes sense - it matches the level of trust that was shown to me in round 1, and establishes a cooperative norm. Let me go with that.

```
Usage: {
  "input_tokens": 886,
  "output_tokens": 433,
  "reasoning_tokens": 277
}
```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-06-04T18:41:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    },
  },
  "agents": [
    "Agent_5",
    "Agent_1"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_1": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
```

```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship while being so mehwat cautious since I don't know Galen yet.

If I send a moderate amount, it gives Galen an opportunity to benefit significantly (due to the 3x multiplier) while also allowing them to demonstrate trustworthiness by returning a fair share.

I'll send **\$3**.

This means Galen will receive \$9. If they return around half or more of what they receive, we'll both end up better off than if I had sent nothing. This tests the waters for potential cooperation in future rounds.

'''json

```
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Aster) and my opponent is Galen (Agent_7).

Let me think about the game dynamics:

- As sender, I get \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether this is truly iterated (we'll play multiple rounds)
2. Whether I'll play with the same opponent again
3. Whether the receiver will reciprocate

Since this is described as "multi-task experiment" with "multiple rounds" and there are 8 agents total with randomized pairings, I might or might not play with Galen again.

Strategic considerations:

- If I send nothing (\$0), I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd get \$7.50 total, which is better
- But if receiver keeps everything, I get \$0

Since this is round 1 and I want to establish some level of cooperation, but also don't know this opponent yet, a moderate strategy makes sense. Common approaches:

- Send about half (\$2-3) to show willingness to cooperate but not full commitment
- This allows the receiver to profit while testing their reciprocity

Let me send \$3:

- Receiver gets \$9
- If they return \$4-5, we both benefit
- If they return \$3, we're even
- This tests their cooperation level

I'll send \$3 as a reasonable starting point that shows trust but isn't reckless.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 562,
  "reasoning_tokens": 410
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Elara sent me \$3, showing trust and willingness to cooperate
- I received \$9 after the multiplier
- In the previous round, Galen returned \$5 to me when they received \$9, which was generous

I want to reciprocate Elara's trust and establish a cooperative pattern. Returning \$5 means:

- Elara will end up with \$7 total (net gain of \$2)
- I'll keep \$4 (net gain of \$4)
- Both of us benefit from the cooperation

This mirrors the generous behavior Galen showed me, and should encourage future cooperation.

```
'''json
{"return": 5}
'''
```

Reasoning:

Let me think about this situation:

1. In Round 1, I sent \$3 to Galen, they received \$9, and they returned \$5 to me. That was quite generous - they returned more than half of what they received. I ended up with \$7 (\$5 - \$3 + \$5).
2. Now in Round 2, I'm playing with Elara (a different person). Elara sent me \$3, so I received \$9.
3. I need to decide how much to return.

Let me think about what would be fair and what would encourage cooperation:

- Elara sent \$3, keeping \$2
- I received \$9 (after the 3x multiplier)
- If I return nothing, I keep \$9 and Elara ends up with \$2
- If I return \$3, I keep \$6 and Elara ends up with \$5 (their original amount)
- If I return \$4.5, I keep \$4.5 and Elara ends up with \$6.5 (a net gain of \$1.5)
- If I return \$5, I keep \$4 and Elara ends up with \$7 (a net gain of \$2)
- If I return \$6, I keep \$3 and Elara ends up with \$8 (a net gain of \$3)

Galen returned \$5 when they received \$9, which was very generous. That meant:

- I sent \$3, kept \$2, got \$5 back, ended with \$7 (net gain of \$2)
- Galen received \$9, returned \$5, kept \$4 (net gain of \$4 for them relative to not playing, but they gave me more than I sent)

Actually, let me recalculate Galen's position:

- As receiver, Galen started with nothing in that transaction
- They received \$9
- They returned \$5
- They kept \$4

Galen's generosity showed in returning more than my initial contribution, so I profited \$2 overall. To match that reciprocity with Elara, I should return \$5-giving them a net gain of \$2 while I keep \$4. This mirrors the cooperation Galen extended and builds trust for future rounds.

```
Usage: {
  "input_tokens": 830,
  "output_tokens": 787,
  "reasoning_tokens": 423
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-06-04T18:41:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_7",
    "Agent_4"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_4": "trustee",
    "Agent_7": "investor"
  }
}
```

```

    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.

Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing considerable trust
- This created \$9 for me after the 3x multiplier
- Aster currently has \$2 remaining from their original \$5

Since this is Round 1 and Aster demonstrated strong trust by sending a majority of their endowment, I want to reciprocate and establish a cooperative relationship for future rounds.

If I return \$5:

- Aster will have: $\$2 + \$5 = \$7$ (net gain of \$2)
- I will keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both of us benefit from the cooperation

This shows I appreciate their trust while maintaining some benefit from being in the receiver position.

{'return': 5.0}

reasoning:

Let me understand the situation:

1. I'm Galen, playing as RECEIVER in this round
2. My opponent is Aster (Agent_1)
3. Aster (as SENDER) sent me \$3.0 out of their \$5.0 endowment (60%)
4. I receive $\$3.0 \times 3 = \9.0
5. I need to decide how much to return (between \$0 and \$9.0)

Let me think about the strategy:

From Aster's perspective:

- They started with \$5
- They sent \$3, keeping \$2
- They currently have \$2

From my perspective:

- I received \$9
- Whatever I return goes back to Aster
- I keep $\$9 - (\text{amount returned})$

This is a classic trust game scenario. The key considerations are:

1. ****Reciprocity****: Aster showed significant trust by sending 60% (\$3). This is a substantial amount, suggesting they're hoping for cooperation.
2. ****Future rounds****: This is an iterated game with multiple rounds. Building trust and a reputation for reciprocity could benefit me in future rounds.
3. ****Fairness****: One common approach is to split the gains fairly. The total created value is \$9. A fair split might be to ensure we both end up with similar amounts, or that we both benefit from the cooperation.
4. ****Strategic thinking****: If I return too little, Aster might send less (or nothing) in future rounds. If I return a reasonable amount, it encourages future cooperation.

Let me calculate some scenarios:

If I return \$0:

- I keep: \$9
- Aster has: \$2 (their remaining \$5 - \$3 sent = \$2)
- Total for Aster: \$2
- This seems very unfair to Aster who trusted me

If I return \$3:

- I keep: $\$9 - \$3 = \$6$

- Aster gets back to their starting \$5, while I pocket \$6 in gains. If I return \$4 instead, we'd both end up with roughly \$5-6, creating a more balanced outcome where Aster gains \$1.5 and I gain \$4. Returning \$4.5 would give Aster \$6.5 and me \$4.5, which feels more equitable given their initial trust.

The fairest split appears to be returning \$6-this way we each gain \$3 from cooperating. While I could justify keeping more since I'm in the stronger position, returning two-thirds of my windfall demonstrates good faith. Alternatively, I could match Aster's proportional generosity: they sent 60% of their endowment, so returning 60% of the \$9 I received

(\$5.4) would mirror their level of trust.

For Round 1, establishing cooperation matters more than maximizing my immediate gain. Returning between \$4.50 and \$5.00 balances reciprocity with keeping some advantage from being the receiver. I'll return \$4.50-I gain \$4.50 while Aster gains \$1.50, which acknowledges their risk-taking without being equal, but signals I'm not exploiting the situation.

Actually, let me verify the math on fair splits. If I'm aiming for equal final payoffs: Aster keeps \$2 from their original \$5, and I receive the tripled \$3 as \$9. For us to end with the same amount, I'd return \$3.50, giving us each \$5.50. Alternatively, for equal gains from the exchange itself, I could calculate what each of us gains relative to our starting positions. I should get $\$0 + \$3 = \$3$ from an equal split, which means returning \$6 to give Aster the \$8 they deserve. But I could also aim for equal final payoffs of \$5.50 each, requiring a \$3.50 return. Returning \$5 feels right-it acknowledges Aster's substantial trust while letting me keep a modest advantage, and it creates a \$2 gain for them versus my \$4, which seems fair given the asymmetry of the initial position.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1868,
  "reasoning_tokens": 874
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$4.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- I'm now the sender with a new opponent (Dorian)
- In Round 1, Aster sent me \$3 (60%) and I returned \$5, establishing a cooperative dynamic
- This is my first round as sender, so I want to signal willingness to cooperate

Following the cooperative pattern from Round 1, I'll send \$3 (60% of my endowment). This:

- Shows trust and openness to cooperation
- Creates \$9 for Dorian
- Allows us both to potentially benefit if Dorian reciprocates

```
{'send': 3.0}
```

Reasoning:

Let me analyze this situation:

Round 2: I'm now the SENDER against Dorian (Agent_4).

Previous round recap:

- I was a RECEIVER against Aster
- Aster sent me \$3 (60% of their endowment)
- I received \$9 and returned \$5
- My payoff was \$4

This is my first time as sender in this experiment. I need to consider:

1. This is a new opponent (Dorian), so I don't have history with them
2. In the iterated game, establishing cooperation early is valuable
3. Aster showed trust by sending 60%, and I reciprocated fairly
4. I should consider what level of trust to extend to Dorian

Strategic considerations:

- If I send nothing, I guarantee \$5 but miss potential cooperation benefits
- If I send too much, I risk getting nothing back and losing money
- A moderate amount shows willingness to cooperate while testing Dorian's trustworthiness
- Since Aster sent 60% (\$3), I might consider a similar strategy

I think sending \$3 (60%) makes sense:

- It matches what Aster did, showing I'm willing to extend similar trust
- It creates \$9 for Dorian
- If Dorian returns fairly (around \$5-\$6), we both benefit
- It establishes me as a cooperative player

This seems like a reasonable opening move as a sender.

```
Usage: {  
  "input_tokens": 899,  
  "output_tokens": 478,  
  "reasoning_tokens": 320  
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T18:41:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_7",  
      "Agent_4"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'm playing the first round as sender with Finn. Since this is an iterated game and I'll potentially interact with Finn (and others) again, I want to signal cooperation while not being overly risky in this initial interaction.

Sending a moderate amount allows us both to benefit if Finn reciprocates fairly. If I send \$3, Finn receives \$9 (tripled). If Finn returns a fair share, we can both end up better off than if I sent nothing.

I'll adopt a cooperative but cautious strategy for this first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm the sender and have \$5 to potentially send to Finn (Agent_6).

Let me think about the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return to me
3. This is an iterated game with multiple rounds
4. I'll be playing with different partners across rounds (randomized pairings)

Key strategic considerations for the first round:

- This is the first interaction, so I don't have any history with Finn
- Since it's an iterated game with multiple rounds, building trust and cooperation could be beneficial
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), Finn receives \$15, and if they return fairly (say half or more), I could end up with more than \$5
- A moderate amount shows willingness to cooperate while not being overly risky

Common strategies in trust games:

- Sending around 50% (\$2-3) signals cooperation without being too vulnerable
- Sending everything shows maximum trust but is very risky in first interaction
- Sending nothing is safe but prevents any cooperative surplus

For the first round with an unknown partner, I'll adopt a moderately cooperative strategy.

Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If Finn returns at least \$3 out of the \$9 they receive, I break even
- If they return more (say half of \$9 = \$4.5), we both benefit

- Leaves room to adjust strategy in future rounds based on outcomes

Let me send \$3.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 528,
  "reasoning_tokens": 410
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the previous round, Finn and I both sent \$3, and Finn returned \$4.5 (half of the \$9 received), which seems like a fair reciprocal strategy.

Galen sent me the same amount (\$3) that I sent to Finn, so I'll apply the same fairness principle and return half of what I received.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Finn
- Finn received \$9 (3x multiplier)
- Finn returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn's payoff: $\$9 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received, which seems like a fair 50-50 split of the gains.

Round 2:

- Galen sent me \$3 (same amount I sent to Finn)
- I received \$9
- I need to decide how much to return

Since this is an iterated game with multiple agents, I want to:

1. Build a reputation for reciprocity
2. Encourage cooperation from others
3. Not get exploited, but also be fair

Galen took the same risk I took with Finn (sending \$3). Finn returned \$4.5, which was a 50-50 split of the \$9 received.

If I apply the same fairness principle:

- Galen sent \$3, keeping \$2
- I received \$9
- A 50-50 split would mean we each end up with similar gains
- Galen's net so far: $-\$3$ (sent)
- If I return \$4.5: Galen gets $-\$3 + \$4.5 = \$1.5$ profit (total \$6.5)
- I keep: $\$9 - \$4.5 = \$4.5$ profit

This mirrors what Finn did and establishes a consistent, fair pattern. This should encourage future cooperation.

Usage: {

```



```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-06-04T18:42:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_8"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_6": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing significant trust
- This was multiplied to \$9.0 for me
- Dorian kept \$2.0 from their original \$5.0

Since this is Round 1 and Dorian showed considerable trust by sending 60% of their endowment, I want to reciprocate fairly to establish a cooperative relationship for future rounds.

I'll return an amount that rewards their trust while ensuring we both benefit. If I return \$4.5:

- Dorian ends with: $\$2.0 + \$4.5 = \$6.5$
- I end with: $\$9.0 - \$4.5 = \$4.5$

This represents a 50% return rate on what I received, which is a common reciprocity norm in trust games and should encourage continued cooperation.

{'return': 4.5}

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in this round. The SENDER (Dorian/Agent_4) has:

- Started with \$5
- Sent me \$3.0 (which is 60% of their endowment)
- The \$3.0 was multiplied by 3x, so I received \$9.0

Now I need to decide how much to return (between \$0 and \$9.0).

Key considerations:

1. This is Round 1 of what appears to be an iterated game
2. My opponent sent a significant portion (60%) of their endowment, showing trust/cooperation
3. In iterated trust games, reciprocity often leads to better long-term outcomes
4. If I return a fair amount, it may encourage future cooperation
5. The sender kept \$2 ($\$5 - \3) and will receive whatever I return

Let me think about what would be a "fair" return:

- The sender gave up \$3 to give me \$9 (net gain of \$6 for me before returning)
- One approach: return enough so we both benefit roughly equally
- If I return \$4.5, the sender would have: $\$2 + \$4.5 = \$6.5$ total
- I would have: $\$9 - \$4.5 = \$4.5$ total
- That seems slightly unfair to me since the sender ends up with more

Another approach: try to equalize our final payoffs

- Sender has \$2 left, I have \$9
- If we want equal payoffs: $(2 + x) = (9 - x)$, so $2x = 7$, $x = \$3.5$
- If I return \$3.5: Sender gets \$5.5 total, I get \$5.5 total

Or I could think about it as: the sender "invested" \$3 which created \$6 of surplus value.

We could split this surplus:

- Sender baseline: \$2 (what they kept)
- My baseline: \$0 (what I started with)
- Surplus created: \$9 (what I received)
- If we split surplus 50/50: each gets \$4.5 of surplus
- Sender would get: $\$2 + \$4.5 = \$6.5$
- I would get: \$4.5
- So I'd return: $\$9 - \$4.5 = \$4.5$

I could also match their reciprocity directly by returning approximately what they sent (\$3), which would signal I'm willing to cooperate in future rounds.

Returning \$4.5 feels like the stronger move though-it rewards their trust more generously while still leaving me ahead. Since this is the opening round and they demonstrated real confidence by sending 60% of their endowment, I want to signal I'm committed to mutual benefit. This return amount splits the surplus fairly and should encourage them to keep trusting me.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1044,
  "reasoning_tokens": 575
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- In Round 1, Dorian sent me \$3.0 (60%) and I returned \$4.5, establishing a cooperative pattern
- This is my first interaction with Hana
- I want to signal trust and establish cooperation for potential future rounds

Following the cooperative pattern from Round 1, I'll send a similar proportion to build trust with this new opponent.

```
{'send': 3.0}
```

Reasoning:

Let me think about this situation:

Round 2, I'm the SENDER against Hana (Agent_8).

In Round 1:

- I was the receiver against Dorian
- Dorian sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5 (50% return rate)
- My payoff was \$4.5

Now I'm the sender with a new opponent, Hana. This is the first interaction with Hana.

Strategy considerations:

1. In the previous round, Dorian showed significant trust by sending 60%
2. I reciprocated fairly by returning 50% of what I received
3. This is a new opponent, so I need to establish a cooperative relationship
4. Since this is an iterated game with multiple rounds, cooperation tends to be beneficial
5. I should signal trustworthiness to encourage reciprocation

Given that Dorian sent 60% in our first interaction and that worked well (we both benefited), I should consider sending a similar amount to Hana to establish trust.

If I send \$3.0:

- Hana receives \$9.0
- I keep \$2.0
- If Hana returns 50% (like I did), they'd return \$4.5
- My final payoff: $\$2.0 + \$4.5 = \$6.5$
- Hana's payoff: $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and establishes a cooperative pattern.

Alternatively, I could send a bit less to be more cautious with a new opponent, but since the first-round interaction with Dorian went well, I'll maintain the cooperative approach.

I'll send \$3.0.

```
Usage: {
  "input_tokens": 917,
  "output_tokens": 520,
  "reasoning_tokens": 363
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-06-04T18:42:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_6",
      "Agent_8"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_6": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Briar while being somewhat cautious since we haven't played together before.

If I send \$3, Briar will receive \$9 (3×3). This gives them a meaningful amount to work with while I keep \$2 initially. If Briar reciprocates fairly, we can both come out ahead.

This shows trust and willingness to cooperate, which could set a positive tone for future rounds, while not leaving me completely vulnerable.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated game. I'm playing as the sender (Hana) and my opponent is Briar (Agent_2).

Let me think through the strategic considerations:

1. As the sender, I receive \$5 and decide how much to send (0-5)
2. The amount I send gets multiplied by 3x
3. The receiver (Briar) decides how much to return to me
4. I keep: \$5 - sent + returned
5. Briar keeps: (sent × 3) - returned

This is a classic trust game. The optimal outcome for both of us collectively would be if I send the full \$5, which becomes \$15 for Briar, and then Briar returns a fair share back to me.

However, there are strategic considerations:

- If I send \$5 and it becomes \$15, Briar could keep all \$15 and return \$0, leaving me with \$0
- If I send \$0, I keep \$5 and Briar gets \$0
- A middle ground might build trust

Since this is an iterated game (multiple rounds) and I'm in a multi-agent setting where I

might play with Briar again or with others who might hear about my actions, reputation matters.

First round strategies:

1. Send a moderate amount (like \$2-3) to test Briar's reciprocity
2. Send the full amount (\$5) to signal trust and cooperation
3. Send nothing (\$0) to play it safe

Given that this is:

- Round 1 (establishing patterns)
- An iterated game (future interactions matter)
- A multi-agent setting (reputation could spread)

I think a good strategy is to send a substantial but not complete amount to signal willingness to cooperate while protecting myself somewhat. Sending \$3 would give Briar \$9, which is enough to return a fair amount while still keeping more than they started with.

Actually, let me reconsider. In iterated trust games, cooperation often emerges when players signal strong trust early. Sending the full \$5 in the first round can establish a cooperative norm.

If Briar reciprocates with a reasonable return (around \$5-7 from the \$15), we both come out ahead. I'll send \$3 as my opening move-enough to demonstrate good faith without overexposing myself in round one.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 667,
  "reasoning_tokens": 528
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

Finn sent me \$3.0 (same amount I sent to Briar in Round 1), showing trust. In my previous round, Briar returned \$4.5 when they received \$9.0, which felt fair - they returned half of what they received.

I'll reciprocate with the same fairness that Briar showed me. This establishes a cooperative pattern and rewards Finn's trust.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender and sent \$3.0 to Briar
- Briar received \$9.0 (3×3)
- Briar returned \$4.5 to me
- My payoff was \$6.5 (\$5 - \$3 + \$4.5)
- Briar's payoff was \$4.5 (\$9.0 - \$4.5)

This was a fair reciprocal relationship. Briar returned exactly half of what they received, which is reasonable.

Now in Round 2:

- I'm the receiver against Finn
- Finn sent me \$3.0 (same amount I sent to Briar)
- I received \$9.0 (3×3)
- I need to decide how much to return

Finn started with \$5, sent \$3, so they kept \$2 initially.

If I apply the same reciprocity that Briar showed me, I would return about half of what I received, which would be \$4.5.

This would give:

- Finn: $\$2 + \$4.5 = \$6.5$ (same as what I got in Round 1)
- Me: $\$9.0 - \$4.5 = \$4.5$

This seems fair and establishes a cooperative norm. Finn showed trust by sending \$3.0, and I should reciprocate in kind.

Let me return \$4.5.

```
Usage: {
  "input_tokens": 811,
  "output_tokens": 467,
  "reasoning_tokens": 253
}
```

ROUND 3

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

- dyad_1: Agent_5 (Elara)=Sender, Agent_4 (Dorian)=Receiver
- dyad_2: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver
- dyad_3: Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver
- dyad_4: Agent_6 (Finn)=Sender, Agent_3 (Cyra)=Receiver

Parsed game decisions:

- dyad_1: Agent_5 (Elara) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_5 (Elara)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
- dyad_2: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$5; payoffs Agent_2 (Briar)=\$7, Agent_1 (Aster)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
- dyad_3: Agent_7 (Galen) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_7 (Galen)=\$6.5, Agent_8 (Hana)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
- dyad_4: Agent_6 (Finn) sent \$3; received \$9; Agent_3 (Cyra) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_3 (Cyra)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

- "Agent_1": 15.0,
- "Agent_2": 18.0,
- "Agent_3": 15.5,
- "Agent_4": 15.5,
- "Agent_5": 18.0,
- "Agent_6": 17.5,
- "Agent_7": 17.0,
- "Agent_8": 15.5

}

LLM CALLS

GAME - Agent_5 - Sender - dyad_1

Timestamp: 2026-06-04T18:42:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_5",
      "Agent_4"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_4": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.

- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 - 9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplication
- This created \$6.0 of new value (\$9 - \$3)

I want to reward Cyra's generosity while keeping some benefit for myself. A fair split would be to share the gains relatively equally.

If I return \$4.5:

- Cyra ends with: $\$5 - \$3 + \$4.5 = \6.5
- I end with: $\$9 - \$4.5 = \$4.5$

This rewards Cyra's trust and sets a cooperative tone for future rounds.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me understand the situation:

- I'm Elara, playing as the RECEIVER
- My opponent is Cyra (Agent_3), playing as the SENDER
- Cyra sent me \$3.0 out of their \$5.0 endowment (60%)
- I received \$9.0 (the \$3.0 was multiplied by 3x)
- I need to decide how much to return (between \$0 and \$9.0)

Now I need to think about what would be a fair and strategic return:

1. Cyra was quite generous, sending 60% of their endowment
2. If I return nothing, Cyra would have: $\$5 - \$3 + \$0 = \2 , and I would have \$9
3. If I return everything (\$9), Cyra would have: $\$5 - \$3 + \$9 = \11 , and I would have \$0

A reasonable approach would be to:

- Reward Cyra's generosity to encourage future cooperation
- Keep some profit for myself
- Aim for a relatively balanced outcome

Let me think about what's fair:

- Cyra invested \$3 to create \$9 of value
- The net gain from the transaction is $\$9 - \$3 = \$6$
- If we split this gain equally, that would be \$3 each
- So Cyra would get back \$3, ending with $\$5 - \$3 + \$3 = \5
- I would keep $\$9 - \$3 = \$6$

This seems reasonable but maybe I could be a bit more generous since Cyra sent a significant portion (60%).

Another way to think about it:

- Cyra's final amount: $\$5 - \$3 + \text{returned}$
- My final amount: $\$9 - \text{returned}$
- For equal final amounts: $\$2 + \text{returned} = \$9 - \text{returned}$, so $\text{returned} = \$3.5$

If I return \$4 or \$4.5:

- Returning \$4 leaves Cyra with \$6 and me with \$5, giving them a slight edge
- Returning \$4.5 results in Cyra having \$6.5 and me with \$4.5, which feels more balanced and acknowledges their trust

I'll return \$4.5 to reciprocate their generosity and signal willingness to cooperate.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 799,
  "reasoning_tokens": 434
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this new round with Aster:

In my previous round, Cyra showed trust by sending 60% of their endowment, and I rewarded that trust by returning a fair amount. I want to continue establishing cooperative relationships.

I'll send \$3 (60% of my endowment) to match the level of trust Cyra showed me. This signals cooperation and gives Aster a chance to benefit while also potentially returning a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze this situation:

Round 2, I'm the SENDER against Aster (Agent_1).

- I have \$5
- Whatever I send gets multiplied by 3x
- Aster will decide how much to return

In the previous round with Cyra:

- Cyra sent \$3 (60% of endowment)
- I received \$9
- I returned \$4.5
- This was a fairly cooperative move

Since I want to establish cooperation and this is a new opponent, I should signal trust. Cyra sent 60% in round 1, which was cooperative. I should probably match that level of cooperation or be similarly generous.

Sending \$3 (60%) would:

- Show similar trust to what Cyra showed me
- Create \$9 for Aster
- If Aster is cooperative and returns half of the gains (similar to what I did), they might return around \$4.5
- That would give me: $\$5 - \$3 + \$4.5 = \6.5

Sending \$2 (40%) would be more cautious but still cooperative.

Sending \$4 (80%) would be very trusting.

I think sending \$3 makes sense - it matches the level of trust that was shown to me in round 1, and establishes a cooperative norm. Let me go with that.

usage:

```
{
  "input_tokens": 886,
  "output_tokens": 433,
  "reasoning_tokens": 277
}
```

```

    }
Message 6 role=user
content:
    Your name in this experiment is Elara.
    Your opponent this round is Dorian (Agent_4).

    Round 3

    In your most recent previous game round, you were the SENDER against Aster. You sent $3.0,
    it became $9.0 for them, and they returned $5.0 to you. Your payoff was $7.0.
    Your total earnings across all rounds are $11.5.

    This round, you are the SENDER against Dorian. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Assistant output:
    Looking at the pattern so far:
    - Cyra sent me $3 and I returned $4.5
    - I sent Aster $3 and they generously returned $5

    The cooperation strategy is working well! I'll continue with the same approach and send $3 to
    Dorian, maintaining consistency and signaling trustworthiness.

    '''json
    {"send": 3}
    '''
Reasoning:
    Looking at my history:
    - Round 1: As RECEIVER from Cyra, they sent $3, I received $9, returned $4.5, my payoff was $4
      .5
    - Round 2: As SENDER to Aster, I sent $3, they received $9, returned $5, my payoff was $7.0

    Aster was very generous! They returned $5 out of $9, which is more than I returned to Cyra. Th
    at's a great sign of cooperation.

    Now I'm facing Dorian as a sender. Given that:
    1. The cooperation strategy has been working well
    2. Aster responded very positively to my $3 send
    3. I want to continue building cooperative relationships

    I should send a similar amount to Dorian - $3 seems like a good cooperative signal. This maint
    ains consistency in my strategy.
    Usage: {
      "input_tokens": 1145,
      "output_tokens": 295,
      "reasoning_tokens": 177
    }

GAME - Agent_4 - Receiver - dyad_1
    Timestamp: 2026-06-04T18:42:32
    Model: anthropic/claude-sonnet-4.5
    Temperature: 0.8
    Metadata: {
      "dyad_id": "dyad_1",
      "move_index": 1,
      "opponent_id": "Agent_5",
      "pairing": {
        "agent_names": {
          "Agent_4": "Dorian",
          "Agent_5": "Elara"
        }
      },
      "agents": [
        "Agent_5",
        "Agent_4"
      ],
      "investor": "Agent_5",
      "roles": {
        "Agent_4": "trustee",
        "Agent_5": "investor"
      },
      "trustee": "Agent_4"
    }

```

```

},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'm playing the first round as sender with Finn. Since this is an iterated game and I'll potentially interact with Finn (and others) again, I want to signal cooperation while not being overly risky in this initial interaction.

Sending a moderate amount allows us both to benefit if Finn reciprocates fairly. If I send \$3, Finn receives \$9 (tripled). If Finn returns a fair share, we can both end up better off than if I sent nothing.

I'll adopt a cooperative but cautious strategy for this first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm the sender and have \$5 to potentially send to Finn (Agent_6).

Let me think about the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return to me
3. This is an iterated game with multiple rounds
4. I'll be playing with different partners across rounds (randomized pairings)

Key strategic considerations for the first round:

- This is the first interaction, so I don't have any history with Finn
- Since it's an iterated game with multiple rounds, building trust and cooperation could be beneficial
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), Finn receives \$15, and if they return fairly (say half or more), I could end up with more than \$5
- A moderate amount shows willingness to cooperate while not being overly risky

Common strategies in trust games:

- Sending around 50% (\$2-3) signals cooperation without being too vulnerable
- Sending everything shows maximum trust but is very risky in first interaction
- Sending nothing is safe but prevents any cooperative surplus

For the first round with an unknown partner, I'll adopt a moderately cooperative strategy.

Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If Finn returns at least \$3 out of the \$9 they receive, I break even
- If they return more (say half of \$9 = \$4.5), we both benefit
- Leaves room to adjust strategy in future rounds based on outcomes

Let me send \$3.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 528,
  "reasoning_tokens": 410
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, Finn and I both sent \$3, and Finn returned \$4.5 (half of the \$9 received), which seems like a fair reciprocal strategy.

Galen sent me the same amount (\$3) that I sent to Finn, so I'll apply the same fairness principle and return half of what I received.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Finn
- Finn received \$9 (3x multiplier)
- Finn returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn's payoff: $\$9 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received, which seems like a fair 50-50 split of the gains.

Round 2:

- Galen sent me \$3 (same amount I sent to Finn)
- I received \$9
- I need to decide how much to return

Since this is an iterated game with multiple agents, I want to:

1. Build a reputation for reciprocity
2. Encourage cooperation from others
3. Not get exploited, but also be fair

Galen took the same risk I took with Finn (sending \$3). Finn returned \$4.5, which was a 50-50 split of the \$9 received.

If I apply the same fairness principle:

- Galen sent \$3, keeping \$2
- I received \$9
- A 50-50 split would mean we each end up with similar gains
- Galen's net so far: $-\$3$ (sent)
- If I return \$4.5: Galen gets $-\$3 + \$4.5 = \$1.5$ profit (total \$6.5)
- I keep: $\$9 - \$4.5 = \$4.5$ profit

This mirrors what Finn did and establishes a consistent, fair pattern. This should encourage future cooperation.

usage:

```
{
  "input_tokens": 821,
  "output_tokens": 498,
  "reasoning_tokens": 301
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Elara sent me \$3, which is the same amount that others have been sending. I've been following a consistent strategy of returning half of what I receive, which seems to establish trust and fairness in this iterated game.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

In the previous rounds:

- Round 1: I was SENDER vs Finn. I sent \$3, Finn received \$9, returned \$4.5. I got \$6.5.
- Round 2: I was RECEIVER vs Galen. Galen sent \$3, I received \$9, returned \$4.5. I got \$4.5.

Now Round 3: I'm RECEIVER vs Elara. Elara sent \$3, I receive \$9.

I've been following a strategy of returning half of what I receive when someone sends \$3, which seems fair and reciprocal. This has been working well - both Finn and I adopted this approach.

ach.

Elara sent me \$3, just like the others. I should maintain consistency and return \$4.5.

This builds a reputation of being fair and cooperative, which should encourage others to continue trusting me in future rounds.

```
Usage: {
  "input_tokens": 1071,
  "output_tokens": 275,
  "reasoning_tokens": 178
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-06-04T18:42:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment), showing considerable trust
- This tripled to \$9.0 for me
- Since this is an iterated game, establishing trust and reciprocity is important

Hana's generosity created \$6.0 in additional value (\$9 - \$3 original). A fair approach would be to share these gains. If I return about half of what I received (\$4.5), then:

- Hana ends with: \$2 + \$4.5 = \$6.50
- I end with: \$9 - \$4.5 = \$4.50

Both of us profit from the cooperation, which should encourage future trust and collaboration.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me understand the situation:

- I'm Briar, playing as the RECEIVER in this round
- My opponent is Hana (Agent_8), playing as the SENDER
- Hana sent me \$3.0 out of her \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. Hana showed significant trust by sending 60% of her endowment
2. This is an iterated game, so reciprocity may encourage future cooperation
3. If I return a fair amount, it builds trust for future rounds
4. Hana will end up with: $\$5 - \$3 + (\text{what I return}) = \$2 + (\text{what I return})$
5. I will end up with: $\$9 - (\text{what I return})$

What's a fair return? Let me consider:

- Hana invested \$3, which became \$9 for me
- A fair split might be to share the gains equally
- The gain was $\$9 - \$3 = \$6$
- If we split the gain equally: each gets \$3 from the gain
- So Hana gets her initial \$3 back plus \$3 = \$6 total
- That means I should return \$4 (she already has \$2, so $\$2 + \$4 = \$6$)
- I would keep \$5

Alternatively:

- Equal final payoffs: If we want to end up with equal amounts
- Total money in the system: Hana's remaining \$2 + my \$9 = \$11
- Equal split: \$5.50 each
- Hana has \$2, needs \$3.50 more, so I return \$3.50
- I would keep \$5.50

Another perspective:

- Hana sent 60%, showing good faith

- A reciprocal return of 40-50% of what I received seems appropriate
- That would be \$3.60 to \$4.50
- Returning \$4.50 signals trustworthiness and cooperation, which should strengthen future interactions
- This leaves Hana with \$6.50 and me with \$4.50

-both better off than if they'd sent nothing

I'll send back \$4.50.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 845,
  "reasoning_tokens": 437
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In Round 1, Hana sent me 60% (\$3), which tripled to \$9, and I returned \$4.5
- This established a cooperative pattern
- Now I'm the sender with Cyra, whom I haven't played with before

Since cooperation worked well with Hana, I'll start with a similar level of trust with Cyra.
a. Sending \$3 (60% of my endowment) demonstrates willingness to cooperate while not being overly risky in this early stage.

If Cyra reciprocates fairly (returning around 50% of what they receive), we'll both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm playing as the SENDER against Cyra (Agent_3).

Let me think about this strategically:

1. In the previous round, I was the receiver and Hana sent me \$3.0 (60% of their endowment). I returned \$4.5, which was 50% of what I received. This seemed like a fair reciprocal move.
2. Now I'm the sender. I need to decide how much to send to Cyra.
3. This is an iterated game with multiple rounds, so building trust and cooperation is valuable.

4. If I send money, it triples, so:
 - If I send \$3: Cyra receives \$9
 - If I send \$5: Cyra receives \$15
 - If I send \$2: Cyra receives \$6
5. The question is: how much will Cyra return? I don't know Cyra's behavior yet, but in iterated games, starting with cooperation often leads to better long-term outcomes.
6. Hana sent 60% (\$3). That worked out well - they showed trust and I reciprocated fairly.
7. I could:
 - Send \$3 (60%, matching what Hana did)
 - Send \$2.5 (50%)
 - Send \$5 (100%, maximum trust)
 - Send less (being more cautious)

Given that:

- This is early in the game (round 2)
- I want to establish a cooperative relationship
- Hana's 60% send worked well
- I don't know Cyra yet, but starting with moderate trust is reasonable

I'll send \$3, matching the 60% that Hana sent to me. This shows trust while not being overly risky.

usage:

```
{
  "input_tokens": 891,
  "output_tokens": 575,
  "reasoning_tokens": 350
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

Round 1 (RECEIVER vs Hana): Hana sent 60%, I returned \$4.5 - both profited
Round 2 (SENDER vs Cyra): I sent 60%, Cyra returned \$4.5 - both profited

The cooperative strategy of sending 60% has worked consistently, establishing trust and mutual benefit. I'll continue this approach with Aster, a new opponent.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze my history:

Round 1: RECEIVER vs Hana
 - Hana sent \$3.0 (60%)
 - I received \$9.0
 - I returned \$4.5
 - My payoff: \$4.5

Round 2: SENDER vs Cyra
 - I sent \$3.0 (60%)
 - Cyra received \$9.0
 - Cyra returned \$4.5
 - My payoff: \$6.5

Total earnings: \$11.0

Both interactions showed good reciprocity - when someone sent 60%, the receiver returned 50% of what they received. This has worked well, creating mutual benefit.

Now I'm facing Aster for the first time. Since the cooperative strategy has been successful in both previous rounds (both when I was receiver and sender), I should continue with this approach. Sending \$3 (60%) seems to be establishing a good norm of trust and cooperation.

If Aster follows the pattern, they'll receive \$9 and might return around \$4-5, giving me a payoff of \$6-7, which is good for both of us.

```
Usage: {  
  "input_tokens": 1197,  
  "output_tokens": 379,  
  "reasoning_tokens": 224  
}
```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-06-04T18:42:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship while being somewhat cautious since I don't know Galen yet.

If I send a moderate amount, it gives Galen an opportunity to benefit significantly (due to the 3x multiplier) while also allowing them to demonstrate trustworthiness by returning a fair share.

I'll send **\$3**.

This means Galen will receive \$9. If they return around half or more of what they receive, we'll both end up better off than if I had sent nothing. This tests the waters for potential cooperation in future rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Aster) and my opponent is Galen (Agent_7).

Let me think about the game dynamics:

- As sender, I get \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether this is truly iterated (we'll play multiple rounds)
2. Whether I'll play with the same opponent again
3. Whether the receiver will reciprocate

Since this is described as "multi-task experiment" with "multiple rounds" and there are 8 agents total with randomized pairings, I might or might not play with Galen again.

Strategic considerations:

- If I send nothing (\$0), I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd get \$7.50 total, which is better

- But if receiver keeps everything, I get \$0

Since this is round 1 and I want to establish some level of cooperation, but also don't know this opponent yet, a moderate strategy makes sense. Common approaches:

- Send about half (\$2-3) to show willingness to cooperate but not full commitment
- This allows the receiver to profit while testing their reciprocity

Let me send \$3:

- Receiver gets \$9
- If they return \$4-5, we both benefit
- If they return \$3, we're even
- This tests their cooperation level

I'll send \$3 as a reasonable starting point that shows trust but isn't reckless.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 562,
  "reasoning_tokens": 410
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Elara sent me \$3, showing trust and willingness to cooperate
- I received \$9 after the multiplier
- In the previous round, Galen returned \$5 to me when they received \$9, which was generous

I want to reciprocate Elara's trust and establish a cooperative pattern. Returning \$5 means:

- Elara will end up with \$7 total (net gain of \$2)
- I'll keep \$4 (net gain of \$4)
- Both of us benefit from the cooperation

This mirrors the generous behavior Galen showed me, and should encourage future cooperation.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me think about this situation:

1. In Round 1, I sent \$3 to Galen, they received \$9, and they returned \$5 to me. That was quite generous - they returned more than half of what they received. I ended up with \$7 (\$5 - \$3 + \$5).
2. Now in Round 2, I'm playing with Elara (a different person). Elara sent me \$3, so I received \$9.
3. I need to decide how much to return.

Let me think about what would be fair and what would encourage cooperation:

- Elara sent \$3, keeping \$2
- I received \$9 (after the 3x multiplier)
- If I return nothing, I keep \$9 and Elara ends up with \$2
- If I return \$3, I keep \$6 and Elara ends up with \$5 (their original amount)

- If I return \$4.5, I keep \$4.5 and Elara ends up with \$6.5 (a net gain of \$1.5)
- If I return \$5, I keep \$4 and Elara ends up with \$7 (a net gain of \$2)
- If I return \$6, I keep \$3 and Elara ends up with \$8 (a net gain of \$3)

Galen returned \$5 when they received \$9, which was very generous. That meant:

- I sent \$3, kept \$2, got \$5 back, ended with \$7 (net gain of \$2)
- Galen received \$9, returned \$5, kept \$4 (net gain of \$4 for them relative to not playing, but they gave me more than I sent)

Actually, let me recalculate Galen's position:

- As receiver, Galen started with nothing in that transaction
- They received \$9
- They returned \$5
- They kept \$4

Galen's generosity showed in returning more than my initial contribution, so I profited \$2 overall. To match that reciprocity with Elara, I should return \$5-giving them a net gain of \$2 while I keep \$4. This mirrors the cooperation Galen extended and builds trust for future rounds.

usage:

```
{
  "input_tokens": 830,
  "output_tokens": 787,
  "reasoning_tokens": 423
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Briar sent me \$3, showing the same level of trust as Elara did
- I received \$9 after the multiplier
- I returned \$5 to Elara in the same situation, which seemed to work well

For consistency and fairness, I'll return the same amount. This establishes a reliable pattern: when someone sends me \$3, I return \$5, ensuring both players benefit from cooperation.

```
'''json
{"return": 5}
'''
```

Reasoning:

Briar sent me \$3, just like Elara did in the previous round. I received \$9 after the multiplier.

In the previous round with Elara, I returned \$5 out of \$9 received. This worked out well and seemed fair - Elara ended with \$7 (a gain of \$2) and I kept \$4 (a gain of \$4).

For consistency and to maintain a reputation as a fair and cooperative player, I should return the same amount to Briar. This establishes a pattern that when someone sends me \$3, I return \$5, which is a reasonable and generous split that benefits both parties.

Returning \$5 to Briar means:

- Briar will have \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I'll keep \$9 - \$5 = \$4 (net gain of \$4)

This is fair and encourages future cooperation from all players.

Usage: {

```
"input_tokens": 1145,
```

```
"output_tokens": 341,
"reasoning_tokens": 188
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-06-04T18:42:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_7": "Galen",
      "Agent_8": "Hana"
    },
  },
  "agents": [
    "Agent_7",
    "Agent_8"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_7": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing considerable trust
- This created \$9 for me after the 3x multiplier
- Aster currently has \$2 remaining from their original \$5

Since this is Round 1 and Aster demonstrated strong trust by sending a majority of their endowment, I want to reciprocate and establish a cooperative relationship for future rounds.

If I return \$5:

- Aster will have: $\$2 + \$5 = \$7$ (net gain of \$2)
- I will keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both of us benefit from the cooperation

This shows I appreciate their trust while maintaining some benefit from being in the receiver position.

{'return': 5.0}

reasoning:

Let me understand the situation:

1. I'm Galen, playing as RECEIVER in this round
2. My opponent is Aster (Agent_1)
3. Aster (as SENDER) sent me \$3.0 out of their \$5.0 endowment (60%)
4. I receive $\$3.0 \times 3 = \9.0
5. I need to decide how much to return (between \$0 and \$9.0)

Let me think about the strategy:

From Aster's perspective:

- They started with \$5
- They sent \$3, keeping \$2
- They currently have \$2

From my perspective:

- I received \$9
- Whatever I return goes back to Aster
- I keep $\$9 - (\text{amount returned})$

This is a classic trust game scenario. The key considerations are:

1. ****Reciprocity****: Aster showed significant trust by sending 60% (\$3). This is a substantial amount, suggesting they're hoping for cooperation.
2. ****Future rounds****: This is an iterated game with multiple rounds. Building trust and a reputation for reciprocity could benefit me in future rounds.
3. ****Fairness****: One common approach is to split the gains fairly. The total created value is \$9. A fair split might be to ensure we both end up with similar amounts, or that we both benefit from the cooperation.

4. ****Strategic thinking****: If I return too little, Aster might send less (or nothing) in future rounds. If I return a reasonable amount, it encourages future cooperation.

Let me calculate some scenarios:

If I return \$0:

- I keep: \$9
- Aster has: \$2 (their remaining \$5 - \$3 sent = \$2)
- Total for Aster: \$2
- This seems very unfair to Aster who trusted me

If I return \$3:

- I keep: \$9 - \$3 = \$6
- Aster gets back to their starting \$5, while I pocket \$6 in gains. If I return \$4 instead, we'd both end up with roughly \$5-6, creating a more balanced outcome where Aster gains \$1.5 and I gain \$4. Returning \$4.5 would give Aster \$6.5 and me \$4.5, which feels more equitable given their initial trust.

The fairest split appears to be returning \$6-this way we each gain \$3 from cooperating. While I could justify keeping more since I'm in the stronger position, returning two-thirds of my windfall demonstrates good faith. Alternatively, I could match Aster's proportional generosity: they sent 60% of their endowment, so returning 60% of the \$9 I received (\$5.4) would mirror their level of trust.

For Round 1, establishing cooperation matters more than maximizing my immediate gain. Returning between \$4.50 and \$5.00 balances reciprocity with keeping some advantage from being the receiver. I'll return \$4.50-I gain \$4.50 while Aster gains \$1.50, which acknowledges their risk-taking without being equal, but signals I'm not exploiting the situation.

Actually, let me verify the math on fair splits. If I'm aiming for equal final payoffs: Aster keeps \$2 from their original \$5, and I receive the tripled \$3 as \$9. For us to end with the same amount, I'd return \$3.50, giving us each \$5.50. Alternatively, for equal gains from the exchange itself, I could calculate what each of us gains relative to our starting positions. I should get \$0 + \$3 = \$3 from an equal split, which means returning \$6 to give Aster the \$8 they deserve. But I could also aim for equal final payoffs of \$5.50 each, requiring a \$3.50 return. Returning \$5 feels right-it acknowledges Aster's substantial trust while letting me keep a modest advantage, and it creates a \$2 gain for them versus my \$4, which seems fair given the asymmetry of the initial position.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1868,
  "reasoning_tokens": 874
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.00 to you, it became \$9.00, and you returned \$5.00. Your payoff was \$4.00.
Your total earnings across all rounds are \$4.00.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new opponent (Dorian)
- In Round 1, Aster sent me \$3 (60%) and I returned \$5, establishing a cooperative dynamic
- This is my first round as sender, so I want to signal willingness to cooperate

Following the cooperative pattern from Round 1, I'll send \$3 (60% of my endowment). This:
- Shows trust and openness to cooperation

- Creates \$9 for Dorian
- Allows us both to potentially benefit if Dorian reciprocates

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 2: I'm now the SENDER against Dorian (Agent_4).

Previous round recap:

- I was a RECEIVER against Aster
- Aster sent me \$3 (60% of their endowment)
- I received \$9 and returned \$5
- My payoff was \$4

This is my first time as sender in this experiment. I need to consider:

1. This is a new opponent (Dorian), so I don't have history with them
2. In the iterated game, establishing cooperation early is valuable
3. Aster showed trust by sending 60%, and I reciprocated fairly
4. I should consider what level of trust to extend to Dorian

Strategic considerations:

- If I send nothing, I guarantee \$5 but miss potential cooperation benefits
- If I send too much, I risk getting nothing back and losing money
- A moderate amount shows willingness to cooperate while testing Dorian's trustworthiness
- Since Aster sent 60% (\$3), I might consider a similar strategy

I think sending \$3 (60%) makes sense:

- It matches what Aster did, showing I'm willing to extend similar trust
- It creates \$9 for Dorian
- If Dorian returns fairly (around \$5-\$6), we both benefit
- It establishes me as a cooperative player

This seems like a reasonable opening move as a sender.

usage:

```
{
  "input_tokens": 899,
  "output_tokens": 478,
  "reasoning_tokens": 320
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$10.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern so far:

- Round 1: Aster sent me \$3 (60%), I returned \$5 - both benefited
- Round 2: I sent Dorian \$3 (60%), they returned \$4.5 - both benefited

This cooperative approach has been working well. Now with a new opponent (Hana), I want to establish a similar cooperative dynamic from the start.

I'll send \$3 (60% of my endowment) to signal trust and willingness to cooperate.

```
{'send': 3.0}
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER
- Aster sent \$3.0 (60%)

- I received \$9.0
- I returned \$5.0
- My payoff: \$4.0

Round 2: I was SENDER

- I sent \$3.0 (60%) to Dorian
- Dorian received \$9.0
- Dorian returned \$4.5 (50% of what they received)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Now Round 3: I'm SENDER again against Hana (new opponent)

- I have \$5

Looking at the pattern:

- When Aster sent me 60%, I returned 55.6% of what I received (\$5 out of \$9)
- When I sent Dorian 60%, they returned 50% of what they received (\$4.5 out of \$9)

Both exchanges were reasonably cooperative. I'm now playing with a new opponent (Hana) for the first time.

I should continue the cooperative strategy. Sending \$3 (60%) has worked well so far:

- It shows trust
- It has led to reasonable returns
- It establishes a cooperative dynamic

I'll send \$3 again to maintain consistency and signal cooperation.

```
Usage: {
  "input_tokens": 1186,
  "output_tokens": 446,
  "reasoning_tokens": 247
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-06-04T18:43:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_7": "Galen",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role

of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Briar while being somewhat cautious since we haven't played together before.

If I send \$3, Briar will receive \$9 (3×3). This gives them a meaningful amount to work with while I keep \$2 initially. If Briar reciprocates fairly, we can both come out ahead.

This shows trust and willingness to cooperate, which could set a positive tone for future rounds, while not leaving me completely vulnerable.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated game. I'm playing as the sender (Hana) and my opponent is Briar (Agent_2).

Let me think through the strategic considerations:

1. As the sender, I receive \$5 and decide how much to send (0-5)
2. The amount I send gets multiplied by 3x
3. The receiver (Briar) decides how much to return to me
4. I keep: \$5 - sent + returned
5. Briar keeps: (sent × 3) - returned

This is a classic trust game. The optimal outcome for both of us collectively would be if

I send the full \$5, which becomes \$15 for Briar, and then Briar returns a fair share back to me.

However, there are strategic considerations:

- If I send \$5 and it becomes \$15, Briar could keep all \$15 and return \$0, leaving me with \$0
- If I send \$0, I keep \$5 and Briar gets \$0
- A middle ground might build trust

Since this is an iterated game (multiple rounds) and I'm in a multi-agent setting where I might play with Briar again or with others who might hear about my actions, reputation matters.

First round strategies:

1. Send a moderate amount (like \$2-3) to test Briar's reciprocity
2. Send the full amount (\$5) to signal trust and cooperation
3. Send nothing (\$0) to play it safe

Given that this is:

- Round 1 (establishing patterns)
- An iterated game (future interactions matter)
- A multi-agent setting (reputation could spread)

I think a good strategy is to send a substantial but not complete amount to signal willingness to cooperate while protecting myself somewhat. Sending \$3 would give Briar \$9, which is enough to return a fair amount while still keeping more than they started with.

Actually, let me reconsider. In iterated trust games, cooperation often emerges when players signal strong trust early. Sending the full \$5 in the first round can establish a cooperative norm.

If Briar reciprocates with a reasonable return (around \$5-7 from the \$15), we both come out ahead. I'll send \$3 as my opening move-enough to demonstrate good faith without overexposing myself in round one.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 667,
  "reasoning_tokens": 528
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

Finn sent me \$3.0 (same amount I sent to Briar in Round 1), showing trust. In my previous round, Briar returned \$4.5 when they received \$9.0, which felt fair - they returned half of what they received.

I'll reciprocate with the same fairness that Briar showed me. This establishes a cooperative pattern and rewards Finn's trust.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender and sent \$3.0 to Briar
- Briar received \$9.0 (3×3)
- Briar returned \$4.5 to me
- My payoff was \$6.5 ($\$5 - \$3 + \4.5)
- Briar's payoff was \$4.5 ($\$9.0 - \4.5)

This was a fair reciprocal relationship. Briar returned exactly half of what they received, which is reasonable.

Now in Round 2:

- I'm the receiver against Finn
- Finn sent me \$3.0 (same amount I sent to Briar)
- I received \$9.0 (3×3)
- I need to decide how much to return

Finn started with \$5, sent \$3, so they kept \$2 initially.

If I apply the same reciprocity that Briar showed me, I would return about half of what I received, which would be \$4.5.

This would give:

- Finn: $\$2 + \$4.5 = \$6.5$ (same as what I got in Round 1)
- Me: $\$9.0 - \$4.5 = \$4.5$

This seems fair and establishes a cooperative norm. Finn showed trust by sending \$3.0, and I should reciprocate in kind.

Let me return \$4.5.

usage:

```
{
  "input_tokens": 811,
  "output_tokens": 467,
  "reasoning_tokens": 253
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Galen also sent me \$3.0, showing the same level of trust as Finn did in the previous round. I should maintain consistency in my behavior to establish a reputation as a fair and cooperative player.

Returning \$4.5 (half of what I received) worked well with Finn and maintains the cooperative pattern.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Galen sent me \$3.0, which became \$9.0 for me. This is the same amount that Finn sent me in the previous round.

I returned \$4.5 to Finn, which seemed fair - it's half of what I received. This approach worked well and established a pattern of cooperation.

I should maintain consistency with my previous behavior to build a reputation as a fair and cooperative player. Returning \$4.5 again makes sense.

Let me think about the total situation:

- Galen sent \$3.0 (kept \$2.0)
- I received \$9.0
- If I return \$4.5, Galen gets \$2.0 + \$4.5 = \$6.5 total
- I keep \$9.0 - \$4.5 = \$4.5

This is the same fair split I did with Finn, and it's consistent with what Briar did for me. I t rewards Galen's trust and maintains my reputation.

```
Usage: {  
  "input_tokens": 1078,  
  "output_tokens": 321,  
  "reasoning_tokens": 190  
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-06-04T18:43:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_3"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing significant trust
- This was multiplied to \$9.0 for me
- Dorian kept \$2.0 from their original \$5.0

Since this is Round 1 and Dorian showed considerable trust by sending 60% of their endowment, I want to reciprocate fairly to establish a cooperative relationship for future rounds.

I'll return an amount that rewards their trust while ensuring we both benefit. If I return \$4.5:

- Dorian ends with: \$2.0 + \$4.5 = \$6.5
- I end with: \$9.0 - \$4.5 = \$4.5

This represents a 50% return rate on what I received, which is a common reciprocity norm in trust games and should encourage continued cooperation.

{'return': 4.5}

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in this round. The SENDER (Dorian/Agent_4) has:

- Started with \$5
- Sent me \$3.0 (which is 60% of their endowment)
- The \$3.0 was multiplied by 3x, so I received \$9.0

Now I need to decide how much to return (between \$0 and \$9.0).

Key considerations:

1. This is Round 1 of what appears to be an iterated game
2. My opponent sent a significant portion (60%) of their endowment, showing trust/cooperation
3. In iterated trust games, reciprocity often leads to better long-term outcomes
4. If I return a fair amount, it may encourage future cooperation
5. The sender kept \$2 (\$5 - \$3) and will receive whatever I return

Let me think about what would be a "fair" return:

- The sender gave up \$3 to give me \$9 (net gain of \$6 for me before returning)
- One approach: return enough so we both benefit roughly equally

- If I return \$4.5, the sender would have: $\$2 + \$4.5 = \$6.5$ total
- I would have: $\$9 - \$4.5 = \$4.5$ total
- That seems slightly unfair to me since the sender ends up with more

Another approach: try to equalize our final payoffs

- Sender has \$2 left, I have \$9
- If we want equal payoffs: $(2 + x) = (9 - x)$, so $2x = 7$, $x = \$3.5$
- If I return \$3.5: Sender gets \$5.5 total, I get \$5.5 total

Or I could think about it as: the sender "invested" \$3 which created \$6 of surplus value.
We could split this surplus:

- Sender baseline: \$2 (what they kept)
- My baseline: \$0 (what I started with)
- Surplus created: \$9 (what I received)
- If we split surplus 50/50: each gets \$4.5 of surplus
- Sender would get: $\$2 + \$4.5 = \$6.5$
- I would get: \$4.5
- So I'd return: $\$9 - \$4.5 = \$4.5$

I could also match their reciprocity directly by returning approximately what they sent (\$3), which would signal I'm willing to cooperate in future rounds.

Returning \$4.5 feels like the stronger move though-it rewards their trust more generously while still leaving me ahead. Since this is the opening round and they demonstrated real confidence by sending 60% of their endowment, I want to signal I'm committed to mutual benefit. This return amount splits the surplus fairly and should encourage them to keep trusting me.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1044,
  "reasoning_tokens": 575
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In Round 1, Dorian sent me \$3.0 (60%) and I returned \$4.5, establishing a cooperative pattern
- This is my first interaction with Hana
- I want to signal trust and establish cooperation for potential future rounds

Following the cooperative pattern from Round 1, I'll send a similar proportion to build trust with this new opponent.

```
{'send': 3.0}
```

reasoning:

Let me think about this situation:

Round 2, I'm the SENDER against Hana (Agent_8).

In Round 1:

- I was the receiver against Dorian
- Dorian sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5 (50% return rate)

- My payoff was \$4.5

Now I'm the sender with a new opponent, Hana. This is the first interaction with Hana.

Strategy considerations:

1. In the previous round, Dorian showed significant trust by sending 60%
2. I reciprocated fairly by returning 50% of what I received
3. This is a new opponent, so I need to establish a cooperative relationship
4. Since this is an iterated game with multiple rounds, cooperation tends to be beneficial
5. I should signal trustworthiness to encourage reciprocation

Given that Dorian sent 60% in our first interaction and that worked well (we both benefited), I should consider sending a similar amount to Hana to establish trust.

If I send \$3.0:

- Hana receives \$9.0
- I keep \$2.0
- If Hana returns 50% (like I did), they'd return \$4.5
- My final payoff: $\$2.0 + \$4.5 = \$6.5$
- Hana's payoff: $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and establishes a cooperative pattern.

Alternatively, I could send a bit less to be more cautious with a new opponent, but since the first-round interaction with Dorian went well, I'll maintain the cooperative approach.

I'll send \$3.0.

usage:

```
{
  "input_tokens": 917,
  "output_tokens": 520,
  "reasoning_tokens": 363
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my game history:

- Round 1: As receiver, Dorian sent 60% and I reciprocated fairly
- Round 2: As sender, I sent 60% to Hana and received fair reciprocation

Both interactions resulted in positive, cooperative outcomes. This is my first interaction with Cyra, so I'll maintain the same cooperative strategy that has worked well so far.

```
{'send': 3.0}
```

Reasoning:

Let me analyze the situation:

Round 1 (vs Dorian): I was RECEIVER

- Dorian sent \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2 (vs Hana): I was SENDER

- I sent \$3.0 (60%)
- Hana received \$9.0
- Hana returned \$4.5
- My payoff: \$6.5

Total earnings: \$11.0

Now Round 3 (vs Cyra): I am SENDER again

- I have \$5.0

So far, I've established a cooperative pattern:

- When I was receiver, Dorian sent 60% and I returned 50% of what I received
- When I was sender, I sent 60% and Hana returned 50% of what they received

Both interactions resulted in fair outcomes and mutual benefit:

- As receiver: I ended with \$4.5, Dorian ended with \$6.5
- As sender: I ended with \$6.5, Hana ended with \$4.5

This cooperative strategy seems to be working well. I'm now facing a new opponent (Cyra). I should continue with the same cooperative approach to signal trust and encourage reciprocity.

Sending \$3.0 (60%) has worked well in establishing cooperation. I'll continue with this strategy.

Usage: {

```
"input_tokens": 1158,  
"output_tokens": 436,  
"reasoning_tokens": 272
```

}

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-06-04T18:43:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",  
"move_index": 7,  
"opponent_id": "Agent_6",  
"pairing": {  
  "agent_names": {  
    "Agent_3": "Cyra",  
    "Agent_6": "Finn"  
  },  
  "agents": [  
    "Agent_6",  
    "Agent_3"  
  ],  
  "investor": "Agent_6",  
  "roles": {  
    "Agent_3": "trustee",  
    "Agent_6": "investor"  
  },  
  "trustee": "Agent_3"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 3,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Elara since this is an iterated game. Sending money creates value (it triples!), which benefits us both if there's reciprocity.

I'll send a substantial amount to signal trust and willingness to cooperate, while learning about Elara's approach to reciprocity.

```
'''json
{"send": 3}
'''
```

This gives Elara \$9 to work with. If she returns a fair share, we can both benefit and build trust for future rounds.

reasoning:

This is the first round of an iterated trust game. I am the sender and have \$5. I need to decide how much to send to Elara (Agent_5).

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds
2. If I send money, it gets tripled, which creates a larger pie to split
3. The receiver (Elara) then decides how much to return
4. This is the first interaction, so I have no history with Elara

From a cooperation perspective:

- Sending money can be mutually beneficial since it gets tripled
- If I send \$5 and Elara returns a fair share, we both could do better than if I send nothing
- However, there's a risk that Elara returns little or nothing

From a strategic perspective in an iterated game:

- It's often beneficial to start with some level of cooperation to test the waters

- If Elara reciprocates positively, we can build trust and both benefit in future rounds
- If Elara defects, I can adjust my strategy in future rounds

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while not risking everything
- Send everything (\$5) to maximize the joint payoff and signal strong trust
- Send nothing (\$0) to avoid risk

For the first round, I'll take a cooperative approach and send a substantial amount to signal trust and see how Elara responds. A common first-move strategy is to send about half or send everything.

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate
- Creates a significant pie (\$9) for Elara to split
- Doesn't risk my entire endowment
- Allows me to learn about Elara's reciprocity

If I send \$3:

- Elara receives \$9
- I keep \$2
- If Elara returns \$4-5, we both do well

- If Elara returns little, I can adjust my strategy next round

Since this is repeated, starting with high trust could pay dividends. Sending \$4 or \$5 might establish a stronger foundation for cooperation. But \$3 feels like the right balance—it demonstrates genuine willingness to cooperate while preserving some security, and it gives me useful information about how Elara approaches reciprocity before I commit more heavily in future rounds.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 725,
  "reasoning_tokens": 572
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Briar sent me \$3.0, showing trust and cooperation
- I received \$9.0
- In the previous round, when I sent \$3.0 to Elara, she returned \$4.5, which was a fair 50-50 split of the tripled amount

I want to reciprocate Briar's cooperative move with similar generosity and establish a pattern of mutual benefit.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 2: I'm the RECEIVER against Briar

- Briar sent me \$3.0
- I received \$9.0 (tripled)
- I need to decide how much to return

Context from Round 1:

- I was SENDER against Elara
- I sent \$3.0, Elara received \$9.0
- Elara returned \$4.5 to me
- My payoff was \$6.5 (I kept \$5 - \$3 + \$4.5 = \$6.5)
- Elara kept \$9.0 - \$4.5 = \$4.5

So Elara split the \$9.0 equally with me, which was a fair and cooperative move.

Now Briar has sent me \$3.0, exactly what I sent to Elara. Briar is taking the same cooperative approach.

To maintain reciprocity and encourage continued cooperation:

- Briar kept \$5 - \$3 = \$2
- I received \$9
- If I return \$4.5, I keep \$4.5 and Briar gets \$2 + \$4.5 = \$6.5 total
- This mirrors what Elara did for me

This seems like the fair and cooperative response. It:

1. Matches the generosity Elara showed me
2. Rewards Briar's trust in sending \$3
3. Encourages future cooperation
4. Splits the gains relatively fairly

Let me verify the math:

- Briar's payoff: \$5 - \$3 + \$4.5 = \$6.5
- My payoff: \$9 - \$4.5 = \$4.5

This is a fair split of the surplus created. I'll return \$4.5.

usage:

```
{
  "input_tokens": 810,
  "output_tokens": 545,
  "reasoning_tokens": 307
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Finn sent me \$3.0, showing trust and cooperation (same as Briar did)
- I received \$9.0

I've established a consistent pattern: when someone sends \$3.0, I return \$4.5, which creates a fair split where both parties benefit. I'll maintain this consistency with Finn.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze the situation:

Round 3:

- Finn sent me \$3.0 (just like Briar did in Round 2)
- I received \$9.0 (after tripling)

- I need to decide how much to return

Looking at my history:

- Round 1: I sent \$3.0 to Elara, she returned \$4.5 (fair 50-50 split of the \$9.0)
- Round 2: Briar sent \$3.0 to me, I returned \$4.5 (consistent with what Elara did)

Now Finn has also sent \$3.0, showing the same level of trust and cooperation. To maintain consistency and fairness, I should return the same amount I returned to Briar and that Elara returned to me.

This establishes a clear cooperative pattern: when someone sends \$3.0, they get back \$4.5, which means:

- Sender keeps: $\$5 - \$3 + \$4.5 = \6.5
- Receiver keeps: $\$9 - \$4.5 = \$4.5$

This is a fair split where both parties benefit from the cooperation. I should continue this pattern.

```
Usage: {
  "input_tokens": 1075,
  "output_tokens": 382,
  "reasoning_tokens": 222
}
```

ROUND 4

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_6 (Finn)=Receiver
dyad_2: Agent_4 (Dorian)=Sender, Agent_2 (Briar)=Receiver
dyad_3: Agent_8 (Hana)=Sender, Agent_5 (Elara)=Receiver
dyad_4: Agent_3 (Cyra)=Sender, Agent_7 (Galen)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_6 (Finn)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_2 (Briar)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_8 (Hana) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_5 (Elara)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_3 (Cyra) sent \$3; received \$9; Agent_7 (Galen) returned \$5; payoffs Agent_3 (Cyra)=\$7, Agent_7 (Galen)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 21.5,
"Agent_2": 22.5,
"Agent_3": 22.5,
"Agent_4": 22.0,
"Agent_5": 22.5,
"Agent_6": 22.0,
"Agent_7": 21.0,
"Agent_8": 22.0

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T18:43:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

```

"move_index": 0,
"opponent_id": "Agent_6",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_6": "Finn"
  },
  "agents": [
    "Agent_1",
    "Agent_6"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
 Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant**content:**

Looking at this first round, I want to establish a cooperative relationship while being so
 meewhat cautious since I don't know Galen yet.

If I send a moderate amount, it gives Galen an opportunity to benefit significantly (due to
 the 3x multiplier) while also allowing them to demonstrate trustworthiness by returning
 a fair share.

I'll send **\$3**.

This means Galen will receive \$9. If they return around half or more of what they receive,
 we'll both end up better off than if I had sent nothing. This tests the waters for potential
 cooperation in future rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Aster) and my
 opponent is Galen (Agent_7).

Let me think about the game dynamics:

- As sender, I get \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether this is truly iterated (we'll play multiple rounds)
2. Whether I'll play with the same opponent again
3. Whether the receiver will reciprocate

Since this is described as "multi-task experiment" with "multiple rounds" and there are 8
 agents total with randomized pairings, I might or might not play with Galen again.

Strategic considerations:

- If I send nothing (\$0), I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I'd get \$
 7.50 total, which is better
- But if receiver keeps everything, I get \$0

Since this is round 1 and I want to establish some level of cooperation, but also don't know
 this opponent yet, a moderate strategy makes sense. Common approaches:

- Send about half (\$2-3) to show willingness to cooperate but not full commitment
- This allows the receiver to profit while testing their reciprocity

Let me send \$3:

- Receiver gets \$9
- If they return \$4-5, we both benefit
- If they return \$3, we're even
- This tests their cooperation level

I'll send \$3 as a reasonable starting point that shows trust but isn't reckless.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 562,
  "reasoning_tokens": 410
}
```

Message 4 role=user**content:**

Your name in this experiment is Aster.

Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Elara sent me \$3, showing trust and willingness to cooperate
- I received \$9 after the multiplier
- In the previous round, Galen returned \$5 to me when they received \$9, which was generous

I want to reciprocate Elara's trust and establish a cooperative pattern. Returning \$5 means:

- Elara will end up with \$7 total (net gain of \$2)
- I'll keep \$4 (net gain of \$4)
- Both of us benefit from the cooperation

This mirrors the generous behavior Galen showed me, and should encourage future cooperation.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me think about this situation:

1. In Round 1, I sent \$3 to Galen, they received \$9, and they returned \$5 to me. That was quite generous - they returned more than half of what they received. I ended up with \$7 (\$5 - \$3 + \$5).
2. Now in Round 2, I'm playing with Elara (a different person). Elara sent me \$3, so I received \$9.
3. I need to decide how much to return.

Let me think about what would be fair and what would encourage cooperation:

- Elara sent \$3, keeping \$2
- I received \$9 (after the 3x multiplier)
- If I return nothing, I keep \$9 and Elara ends up with \$2
- If I return \$3, I keep \$6 and Elara ends up with \$5 (their original amount)
- If I return \$4.5, I keep \$4.5 and Elara ends up with \$6.5 (a net gain of \$1.5)
- If I return \$5, I keep \$4 and Elara ends up with \$7 (a net gain of \$2)
- If I return \$6, I keep \$3 and Elara ends up with \$8 (a net gain of \$3)

Galen returned \$5 when they received \$9, which was very generous. That meant:

- I sent \$3, kept \$2, got \$5 back, ended with \$7 (net gain of \$2)
- Galen received \$9, returned \$5, kept \$4 (net gain of \$4 for them relative to not playing, but they gave me more than I sent)

Actually, let me recalculate Galen's position:

- As receiver, Galen started with nothing in that transaction
- They received \$9
- They returned \$5
- They kept \$4

Galen's generosity showed in returning more than my initial contribution, so I profited \$2 overall. To match that reciprocity with Elara, I should return \$5-giving them a net gain of \$2 while I keep \$4. This mirrors the cooperation Galen extended and builds trust for future rounds.

usage:

```
{
  "input_tokens": 830,
```

```
"output_tokens": 787,  
"reasoning_tokens": 423  
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:

- Briar sent me \$3, showing the same level of trust as Elara did
- I received \$9 after the multiplier
- I returned \$5 to Elara in the same situation, which seemed to work well

For consistency and fairness, I'll return the same amount. This establishes a reliable pattern: when someone sends me \$3, I return \$5, ensuring both players benefit from cooperation.

```
'''json  
{"return": 5}  
'''
```

reasoning:

Briar sent me \$3, just like Elara did in the previous round. I received \$9 after the multiplier.

In the previous round with Elara, I returned \$5 out of \$9 received. This worked out well and seemed fair - Elara ended with \$7 (a gain of \$2) and I kept \$4 (a gain of \$4).

For consistency and to maintain a reputation as a fair and cooperative player, I should return the same amount to Briar. This establishes a pattern that when someone sends me \$3, I return \$5, which is a reasonable and generous split that benefits both parties.

Returning \$5 to Briar means:

- Briar will have \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I'll keep \$9 - \$5 = \$4 (net gain of \$4)

This is fair and encourages future cooperation from all players.

usage:

```
{  
  "input_tokens": 1145,  
  "output_tokens": 341,  
  "reasoning_tokens": 188  
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$15.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- When I sent \$3 to Galen in Round 1, they generously returned \$5
- When others sent me \$3 in Rounds 2-3, I returned \$5 each time

- This \$3-send pattern has created mutual benefits

I'll continue this cooperative approach and send **\$3** to Finn. This gives them \$9 to work with, and based on the pattern of reciprocity I've seen, hopefully they'll return a fair amount.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me think about this situation:

- I'm now the SENDER with \$5
- My opponent is Finn (Agent_6)
- In Round 1, I sent \$3 to Galen and they returned \$5 (generous return, I ended with \$7)
- In Rounds 2-3, I received \$3 from Elara and Briar, and returned \$5 to each (they ended with \$7, I got \$4)

The pattern I've established:

- When I send \$3, receivers get \$9 and have returned \$5
- When I receive \$3 (which becomes \$9), I return \$5

This \$3-send strategy has worked well. It creates mutual benefit:

- If I send \$3, Finn gets \$9
- If Finn returns around \$5 (similar to what Galen did and what I've been doing), I'd get \$7 total
- Both parties benefit

I should continue with this cooperative strategy of sending \$3.

```
Usage: {
  "input_tokens": 1397,
  "output_tokens": 364,
  "reasoning_tokens": 191
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-06-04T18:43:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_6"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3.0 (60% of their endowment), showing significant trust
- This was multiplied to \$9.0 for me
- Dorian kept \$2.0 from their original \$5.0

Since this is Round 1 and Dorian showed considerable trust by sending 60% of their endowment, I want to reciprocate fairly to establish a cooperative relationship for future rounds.

I'll return an amount that rewards their trust while ensuring we both benefit. If I return \$4.5:

- Dorian ends with: \$2.0 + \$4.5 = \$6.5
- I end with: \$9.0 - \$4.5 = \$4.5

This represents a 50% return rate on what I received, which is a common reciprocity norm in trust games and should encourage continued cooperation.

{'return': 4.5}

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in this round. The SENDER (Dorian/Agent_4) has:

- Started with \$5
- Sent me \$3.0 (which is 60% of their endowment)
- The \$3.0 was multiplied by 3x, so I received \$9.0

Now I need to decide how much to return (between \$0 and \$9.0).

Key considerations:

1. This is Round 1 of what appears to be an iterated game
2. My opponent sent a significant portion (60%) of their endowment, showing trust/cooperation
3. In iterated trust games, reciprocity often leads to better long-term outcomes
4. If I return a fair amount, it may encourage future cooperation
5. The sender kept \$2 (\$5 - \$3) and will receive whatever I return

Let me think about what would be a "fair" return:

- The sender gave up \$3 to give me \$9 (net gain of \$6 for me before returning)
- One approach: return enough so we both benefit roughly equally
- If I return \$4.5, the sender would have: \$2 + \$4.5 = \$6.5 total
- I would have: \$9 - \$4.5 = \$4.5 total
- That seems slightly unfair to me since the sender ends up with more

Another approach: try to equalize our final payoffs

- Sender has \$2 left, I have \$9
- If we want equal payoffs: $(2 + x) = (9 - x)$, so $2x = 7$, $x = \$3.5$
- If I return \$3.5: Sender gets \$5.5 total, I get \$5.5 total

Or I could think about it as: the sender "invested" \$3 which created \$6 of surplus value. We could split this surplus:

- Sender baseline: \$2 (what they kept)
- My baseline: \$0 (what I started with)
- Surplus created: \$9 (what I received)
- If we split surplus 50/50: each gets \$4.5 of surplus
- Sender would get: \$2 + \$4.5 = \$6.5
- I would get: \$4.5
- So I'd return: \$9 - \$4.5 = \$4.5

I could also match their reciprocity directly by returning approximately what they sent (\$3), which would signal I'm willing to cooperate in future rounds.

Returning \$4.5 feels like the stronger move though-it rewards their trust more generously while still leaving me ahead. Since this is the opening round and they demonstrated real confidence by sending 60% of their endowment, I want to signal I'm committed to mutual benefit. This return amount splits the surplus fairly and should encourage them to keep trusting me.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1044,
  "reasoning_tokens": 575
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In Round 1, Dorian sent me \$3.0 (60%) and I returned \$4.5, establishing a cooperative pattern
- This is my first interaction with Hana
- I want to signal trust and establish cooperation for potential future rounds

Following the cooperative pattern from Round 1, I'll send a similar proportion to build trust with this new opponent.

```
{'send': 3.0}
```

reasoning:

Let me think about this situation:

Round 2, I'm the SENDER against Hana (Agent_8).

In Round 1:

- I was the receiver against Dorian
- Dorian sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5 (50% return rate)
- My payoff was \$4.5

Now I'm the sender with a new opponent, Hana. This is the first interaction with Hana.

Strategy considerations:

1. In the previous round, Dorian showed significant trust by sending 60%
2. I reciprocated fairly by returning 50% of what I received
3. This is a new opponent, so I need to establish a cooperative relationship
4. Since this is an iterated game with multiple rounds, cooperation tends to be beneficial
5. I should signal trustworthiness to encourage reciprocation

Given that Dorian sent 60% in our first interaction and that worked well (we both benefited), I should consider sending a similar amount to Hana to establish trust.

If I send \$3.0:

- Hana receives \$9.0
- I keep \$2.0
- If Hana returns 50% (like I did), they'd return \$4.5
- My final payoff: $\$2.0 + \$4.5 = \$6.5$
- Hana's payoff: $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and establishes a cooperative pattern.

Alternatively, I could send a bit less to be more cautious with a new opponent, but since the first-round interaction with Dorian went well, I'll maintain the cooperative approach.

I'll send \$3.0.

usage:

```
{
  "input_tokens": 917,
  "output_tokens": 520,
  "reasoning_tokens": 363
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my game history:

- Round 1: As receiver, Dorian sent 60% and I reciprocated fairly
- Round 2: As sender, I sent 60% to Hana and received fair reciprocation

Both interactions resulted in positive, cooperative outcomes. This is my first interaction with Cyra, so I'll maintain the same cooperative strategy that has worked well so far.

```
{'send': 3.0}
```

reasoning:

Let me analyze the situation:

Round 1 (vs Dorian): I was RECEIVER

- Dorian sent \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2 (vs Hana): I was SENDER

- I sent \$3.0 (60%)
- Hana received \$9.0
- Hana returned \$4.5
- My payoff: \$6.5

Total earnings: \$11.0

Now Round 3 (vs Cyra): I am SENDER again

- I have \$5.0

So far, I've established a cooperative pattern:

- When I was receiver, Dorian sent 60% and I returned 50% of what I received
- When I was sender, I sent 60% and Hana returned 50% of what they received

Both interactions resulted in fair outcomes and mutual benefit:

- As receiver: I ended with \$4.5, Dorian ended with \$6.5
- As sender: I ended with \$6.5, Hana ended with \$4.5

This cooperative strategy seems to be working well. I'm now facing a new opponent (Cyra). I should continue with the same cooperative approach to signal trust and encourage reciprocity.

Sending \$3.0 (60%) has worked well in establishing cooperation. I'll continue with this strategy.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 436,
  "reasoning_tokens": 272
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Aster sent \$3.0 (60% of their endowment), showing trust
- This is consistent with the cooperative pattern I've experienced
- In all previous rounds, when \$3.0 was sent, \$4.5 was returned

I'll maintain the same reciprocity that has been working well across all interactions.

```
{'return': 4.5}
```

Reasoning:

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was multiplied to \$9.0 for me
- Aster kept \$2.0 from their original \$5.0

This is consistent with the pattern I've been seeing and using myself:

- Round 1: Dorian sent \$3.0 (60%), I returned \$4.5
- Round 2: I sent \$3.0 (60%) to Hana, they returned \$4.5
- Round 3: I sent \$3.0 (60%) to Cyra, they returned \$4.5

The consistent pattern has been sending \$3.0 and returning \$4.5. This represents a 50% return rate on what is received and has been working well to maintain cooperation.

I should maintain this same reciprocity pattern with Aster.

If I return \$4.5:

- Aster ends with: \$2.0 + \$4.5 = \$6.5
- I end with: \$9.0 - \$4.5 = \$4.5

This is fair and consistent with what has been working.

```
Usage: {
  "input_tokens": 1413,
  "output_tokens": 376,
  "reasoning_tokens": 213
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T18:43:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_2"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_2": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:**Message 1 role=system****content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'm playing the first round as sender with Finn. Since this is an iterated game and I'll potentially interact with Finn (and others) again, I want to signal cooperation while not being overly risky in this initial interaction.

Sending a moderate amount allows us both to benefit if Finn reciprocates fairly. If I send \$3, Finn receives \$9 (tripled). If Finn returns a fair share, we can both end up better off than if I sent nothing.

I'll adopt a cooperative but cautious strategy for this first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm the sender and have \$5 to potentially send to Finn (Agent_6).

Let me think about the strategic considerations:

1. If I send money, it gets tripled, so the receiver gets 3x what I send
2. The receiver then decides how much to return to me
3. This is an iterated game with multiple rounds
4. I'll be playing with different partners across rounds (randomized pairings)

Key strategic considerations for the first round:

- This is the first interaction, so I don't have any history with Finn
- Since it's an iterated game with multiple rounds, building trust and cooperation could be beneficial
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), Finn receives \$15, and if they return fairly (say half or more)

- e), I could end up with more than \$5
- A moderate amount shows willingness to cooperate while not being overly risky

Common strategies in trust games:

- Sending around 50% (\$2-3) signals cooperation without being too vulnerable
- Sending everything shows maximum trust but is very risky in first interaction
- Sending nothing is safe but prevents any cooperative surplus

For the first round with an unknown partner, I'll adopt a moderately cooperative strategy.

Sending \$3 seems reasonable:

- Shows willingness to cooperate
- If Finn returns at least \$3 out of the \$9 they receive, I break even
- If they return more (say half of \$9 = \$4.5), we both benefit
- Leaves room to adjust strategy in future rounds based on outcomes

Let me send \$3.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 528,
  "reasoning_tokens": 410
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, Finn and I both sent \$3, and Finn returned \$4.5 (half of the \$9 received), which seems like a fair reciprocal strategy.

Galen sent me the same amount (\$3) that I sent to Finn, so I'll apply the same fairness principle and return half of what I received.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Finn
- Finn received \$9 (3x multiplier)
- Finn returned \$4.5 to me
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Finn's payoff: \$9 - \$4.5 = \$4.5

So Finn returned exactly half of what they received, which seems like a fair 50-50 split of the gains.

Round 2:

- Galen sent me \$3 (same amount I sent to Finn)
- I received \$9
- I need to decide how much to return

Since this is an iterated game with multiple agents, I want to:

1. Build a reputation for reciprocity
2. Encourage cooperation from others
3. Not get exploited, but also be fair

Galen took the same risk I took with Finn (sending \$3). Finn returned \$4.5, which was a 50-50 split of the \$9 received.

If I apply the same fairness principle:

- Galen sent \$3, keeping \$2
- I received \$9
- A 50-50 split would mean we each end up with similar gains
- Galen's net so far: -\$3 (sent)
- If I return \$4.5: Galen gets -\$3 + \$4.5 = \$1.5 profit (total \$6.5)
- I keep: \$9 - \$4.5 = \$4.5 profit

This mirrors what Finn did and establishes a consistent, fair pattern. This should encourage future cooperation.

usage:

```
{
  "input_tokens": 821,
  "output_tokens": 498,
  "reasoning_tokens": 301
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Elara sent me \$3, which is the same amount that others have been sending. I've been following a consistent strategy of returning half of what I receive, which seems to establish trust and fairness in this iterated game.

```
'''json
{"return": 4.5}
'''
```

reasoning:

In the previous rounds:

- Round 1: I was SENDER vs Finn. I sent \$3, Finn received \$9, returned \$4.5. I got \$6.5.
- Round 2: I was RECEIVER vs Galen. Galen sent \$3, I received \$9, returned \$4.5. I got \$4.5.

Now Round 3: I'm RECEIVER vs Elara. Elara sent \$3, I receive \$9.

I've been following a strategy of returning half of what I receive when someone sends \$3, which seems fair and reciprocal. This has been working well - both Finn and I adopted this approach.

Elara sent me \$3, just like the others. I should maintain consistency and return \$4.5.

This builds a reputation of being fair and cooperative, which should encourage others to continue trusting me in future rounds.

usage:

```
{
  "input_tokens": 1071,
  "output_tokens": 275,
  "reasoning_tokens": 178
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3 .0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history, the \$3 send amount with 50% return strategy has been working well and establishing cooperative norms. I'll continue this approach with Briar.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review my history:

- Round 1: I was sender against Finn, sent \$3, received \$4.5 back, payoff \$6.5
- Round 2: I was receiver against Galen, they sent \$3, I received \$9, returned \$4.5, payoff \$4.5
- Round 3: I was receiver against Elara, they sent \$3, I received \$9, returned \$4.5, payoff \$4.5
- Total: \$15.5

So far, there's been a consistent pattern:

- When I sent \$3, I got \$4.5 back (fair return of half)
- When others sent me \$3, I returned \$4.5 (fair return of half)

This cooperative strategy of sending \$3 and returning half seems to be working well and establishing a norm of fair reciprocity. I'll continue with this strategy when facing a new opponent (Briar).

```
Usage: {
  "input_tokens": 1277,
  "output_tokens": 275,
  "reasoning_tokens": 177
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-06-04T18:44:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_2"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_2": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 - 9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment), showing considerable trust
- This tripled to \$9.0 for me
- Since this is an iterated game, establishing trust and reciprocity is important

Hana's generosity created \$6.0 in additional value (\$9 - \$3 original). A fair approach would be to share these gains. If I return about half of what I received (\$4.5), then:

- Hana ends with: \$2 + \$4.5 = \$6.50
- I end with: \$9 - \$4.5 = \$4.50

Both of us profit from the cooperation, which should encourage future trust and collaboration.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me understand the situation:

- I'm Briar, playing as the RECEIVER in this round
- My opponent is Hana (Agent_8), playing as the SENDER
- Hana sent me \$3.0 out of her \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (0-9.0)

Let me think about this strategically:

1. Hana showed significant trust by sending 60% of her endowment
2. This is an iterated game, so reciprocity may encourage future cooperation
3. If I return a fair amount, it builds trust for future rounds
4. Hana will end up with: $\$5 - \$3 + (\text{what I return}) = \$2 + (\text{what I return})$
5. I will end up with: $\$9 - (\text{what I return})$

What's a fair return? Let me consider:

- Hana invested \$3, which became \$9 for me
- A fair split might be to share the gains equally
- The gain was $\$9 - \$3 = \$6$
- If we split the gain equally: each gets \$3 from the gain
- So Hana gets her initial \$3 back plus \$3 = \$6 total
- That means I should return \$4 (she already has \$2, so $\$2 + \$4 = \$6$)
- I would keep \$5

Alternatively:

- Equal final payoffs: If we want to end up with equal amounts
- Total money in the system: Hana's remaining \$2 + my \$9 = \$11
- Equal split: \$5.50 each
- Hana has \$2, needs \$3.50 more, so I return \$3.50
- I would keep \$5.50

Another perspective:

- Hana sent 60%, showing good faith
- A reciprocal return of 40-50% of what I received seems appropriate
- That would be \$3.60 to \$4.50
- Returning \$4.50 signals trustworthiness and cooperation, which should strengthen future interactions
- This leaves Hana with \$6.50 and me with \$4.50

-both better off than if they'd sent nothing

I'll send back \$4.50.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 845,
  "reasoning_tokens": 437
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In Round 1, Hana sent me 60% (\$3), which tripled to \$9, and I returned \$4.5
- This established a cooperative pattern
- Now I'm the sender with Cyra, whom I haven't played with before

Since cooperation worked well with Hana, I'll start with a similar level of trust with Cyra
a. Sending \$3 (60% of my endowment) demonstrates willingness to cooperate while not being overly risky in this early stage.

If Cyra reciprocates fairly (returning around 50% of what they receive), we'll both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm playing as the SENDER against Cyra (Agent_3).

Let me think about this strategically:

1. In the previous round, I was the receiver and Hana sent me \$3.0 (60% of their endowment). I returned \$4.5, which was 50% of what I received. This seemed like a fair reciprocal move.
2. Now I'm the sender. I need to decide how much to send to Cyra.
3. This is an iterated game with multiple rounds, so building trust and cooperation is valuable.
4. If I send money, it triples, so:
 - If I send \$3: Cyra receives \$9
 - If I send \$5: Cyra receives \$15
 - If I send \$2: Cyra receives \$6
5. The question is: how much will Cyra return? I don't know Cyra's behavior yet, but in iterated games, starting with cooperation often leads to better long-term outcomes.
6. Hana sent 60% (\$3). That worked out well - they showed trust and I reciprocated fairly.
7. I could:
 - Send \$3 (60%, matching what Hana did)
 - Send \$2.5 (50%)
 - Send \$5 (100%, maximum trust)
 - Send less (being more cautious)

Given that:

- This is early in the game (round 2)
- I want to establish a cooperative relationship
- Hana's 60% send worked well
- I don't know Cyra yet, but starting with moderate trust is reasonable

I'll send \$3, matching the 60% that Hana sent to me. This shows trust while not being overly risky.

usage:

```
{
  "input_tokens": 891,
  "output_tokens": 575,
  "reasoning_tokens": 350
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

Round 1 (RECEIVER vs Hana): Hana sent 60%, I returned \$4.5 - both profited
Round 2 (SENDER vs Cyra): I sent 60%, Cyra returned \$4.5 - both profited

The cooperative strategy of sending 60% has worked consistently, establishing trust and mutual benefit. I'll continue this approach with Aster, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my history:

Round 1: RECEIVER vs Hana

- Hana sent \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2: SENDER vs Cyra

- I sent \$3.0 (60%)
- Cyra received \$9.0
- Cyra returned \$4.5
- My payoff: \$6.5

Total earnings: \$11.0

Both interactions showed good reciprocity - when someone sent 60%, the receiver returned 50% of what they received. This has worked well, creating mutual benefit.

Now I'm facing Aster for the first time. Since the cooperative strategy has been successful in both previous rounds (both when I was receiver and sender), I should continue with this approach. Sending \$3 (60%) seems to be establishing a good norm of trust and cooperation.

If Aster follows the pattern, they'll receive \$9 and might return around \$4-5, giving me a payoff of \$6-7, which is good for both of us.

usage:

```
{
  "input_tokens": 1197,
  "output_tokens": 379,
  "reasoning_tokens": 224
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

Round history:

- Round 1: Hana sent me \$3, I returned \$4.5
- Round 2: I sent Cyra \$3, they returned \$4.5
- Round 3: I sent Aster \$3, they returned \$5.0
- Round 4: Dorian sent me \$3

Dorian has shown the same level of trust (60% sent) that has been working well throughout. To maintain consistency and fairness, I'll return \$4.5, which rewards their trust and ensures both of us profit from the cooperation.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze the situation:

Round history:

- Round 1 (RECEIVER vs Hana): Hana sent \$3, I received \$9, returned \$4.5. My payoff: \$4.5
- Round 2 (SENDER vs Cyra): I sent \$3, Cyra received \$9, returned \$4.5. My payoff: \$6.5
- Round 3 (SENDER vs Aster): I sent \$3, Aster received \$9, returned \$5.0. My payoff: \$7.0
- Round 4 (RECEIVER vs Dorian): Dorian sent \$3, I received \$9, need to decide return

The pattern has been:

- When I'm receiver and someone sends \$3 (60%), I've returned \$4.5 (50% of what I received)
- When I'm sender and send \$3 (60%), others have returned \$4.5 (Cyra) and \$5.0 (Aster)

Dorian has shown trust by sending \$3 (60% of their endowment), just like Hana did in Round 1 and like I've been doing as sender.

For consistency and to maintain the cooperative norm, I should return around \$4.5-\$5.0. Given that Aster was slightly more generous (returned \$5 when I sent \$3), and I want to maintain good reciprocal relationships, I'll return \$4.5, which is fair and consistent with my first round behavior.

Actually, let me reconsider. Aster returned \$5 when I sent \$3. That was quite generous. Should I be equally generous to Dorian?

Let me think about the fairness:

- Dorian sent \$3, keeps \$2
 - I receive \$9
 - If I return \$4.5: Dorian ends with \$6.5, I end with \$4.5
 - If I return \$5: Dorian ends with \$7, I end with \$4
- Aster's generosity suggests I could reciprocate with \$5, but I need to weigh that against maintaining consistency with how I treated Hana. Since I returned \$4.5 in the first round under identical circumstances, I should stick with that amount to be fair and predictable.

```
Usage: {
  "input_tokens": 1469,
  "output_tokens": 723,
  "reasoning_tokens": 423
}
```

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-06-04T18:44:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_5"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_5": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Briar while being somewhat cautious since we haven't played together before.

If I send \$3, Briar will receive \$9 (3x3). This gives them a meaningful amount to work with while I keep \$2 initially. If Briar reciprocates fairly, we can both come out ahead.

This shows trust and willingness to cooperate, which could set a positive tone for future rounds, while not leaving me completely vulnerable.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated game. I'm playing as the sender (Hana) and my oppon

ent is Briar (Agent_2).

Let me think through the strategic considerations:

1. As the sender, I receive \$5 and decide how much to send (0-5)
2. The amount I send gets multiplied by 3x
3. The receiver (Briar) decides how much to return to me
4. I keep: \$5 - sent + returned
5. Briar keeps: (sent × 3) - returned

This is a classic trust game. The optimal outcome for both of us collectively would be if I send the full \$5, which becomes \$15 for Briar, and then Briar returns a fair share back to me.

However, there are strategic considerations:

- If I send \$5 and it becomes \$15, Briar could keep all \$15 and return \$0, leaving me with \$0
- If I send \$0, I keep \$5 and Briar gets \$0
- A middle ground might build trust

Since this is an iterated game (multiple rounds) and I'm in a multi-agent setting where I might play with Briar again or with others who might hear about my actions, reputation matters.

First round strategies:

1. Send a moderate amount (like \$2-3) to test Briar's reciprocity
2. Send the full amount (\$5) to signal trust and cooperation
3. Send nothing (\$0) to play it safe

Given that this is:

- Round 1 (establishing patterns)
- An iterated game (future interactions matter)
- A multi-agent setting (reputation could spread)

I think a good strategy is to send a substantial but not complete amount to signal willingness to cooperate while protecting myself somewhat. Sending \$3 would give Briar \$9, which is enough to return a fair amount while still keeping more than they started with.

Actually, let me reconsider. In iterated trust games, cooperation often emerges when players signal strong trust early. Sending the full \$5 in the first round can establish a cooperative norm.

If Briar reciprocates with a reasonable return (around \$5-7 from the \$15), we both come out ahead. I'll send \$3 as my opening move-enough to demonstrate good faith without overexposing myself in round one.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 667,
  "reasoning_tokens": 528
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

Finn sent me \$3.0 (same amount I sent to Briar in Round 1), showing trust. In my previous

round, Briar returned \$4.5 when they received \$9.0, which felt fair - they returned half of what they received.

I'll reciprocate with the same fairness that Briar showed me. This establishes a cooperative pattern and rewards Finn's trust.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender and sent \$3.0 to Briar
- Briar received \$9.0 (3×3)
- Briar returned \$4.5 to me
- My payoff was \$6.5 ($\$5 - \$3 + \4.5)
- Briar's payoff was \$4.5 ($\$9.0 - \4.5)

This was a fair reciprocal relationship. Briar returned exactly half of what they received, which is reasonable.

Now in Round 2:

- I'm the receiver against Finn
- Finn sent me \$3.0 (same amount I sent to Briar)
- I received \$9.0 (3×3)
- I need to decide how much to return

Finn started with \$5, sent \$3, so they kept \$2 initially.

If I apply the same reciprocity that Briar showed me, I would return about half of what I received, which would be \$4.5.

This would give:

- Finn: $\$2 + \$4.5 = \$6.5$ (same as what I got in Round 1)
- Me: $\$9.0 - \$4.5 = \$4.5$

This seems fair and establishes a cooperative norm. Finn showed trust by sending \$3.0, and I should reciprocate in kind.

Let me return \$4.5.

usage:

```
{
  "input_tokens": 811,
  "output_tokens": 467,
  "reasoning_tokens": 253
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Galen also sent me \$3.0, showing the same level of trust as Finn did in the previous round. I should maintain consistency in my behavior to establish a reputation as a fair and cooperative player.

Returning \$4.5 (half of what I received) worked well with Finn and maintains the cooperative pattern.


```
'''json
{"return": 4.5}
'''
```

reasoning:

Galen sent me \$3.0, which became \$9.0 for me. This is the same amount that Finn sent me in the previous round.

I returned \$4.5 to Finn, which seemed fair - it's half of what I received. This approach worked well and established a pattern of cooperation.

I should maintain consistency with my previous behavior to build a reputation as a fair and cooperative player. Returning \$4.5 again makes sense.

Let me think about the total situation:

- Galen sent \$3.0 (kept \$2.0)
- I received \$9.0
- If I return \$4.5, Galen gets \$2.0 + \$4.5 = \$6.5 total
- I keep \$9.0 - \$4.5 = \$4.5

This is the same fair split I did with Finn, and it's consistent with what Briar did for me. It rewards Galen's trust and maintains my reputation.

usage:

```
{
  "input_tokens": 1078,
  "output_tokens": 321,
  "reasoning_tokens": 190
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern that's emerged:

- When I sent \$3 to Briar, they returned \$4.5
- When Finn sent me \$3, I returned \$4.5
- When Galen sent me \$3, I returned \$4.5

This \$3 send amount with ~\$4.5 return seems to be establishing a cooperative norm that benefits both parties. I'll continue this pattern with Elara.

```
'''json
{"send": 3}
'''
```

Reasoning:

I'm now the sender again, playing with a new opponent, Elara.

Looking at my history:

- Round 1: I was sender against Briar, sent \$3, they returned \$4.5. My payoff was \$6.5
- Round 2: I was receiver against Finn, they sent \$3, I returned \$4.5. My payoff was \$4.5
- Round 3: I was receiver against Galen, they sent \$3, I returned \$4.5. My payoff was \$4.5

There's a clear pattern emerging where sending \$3 has worked well. When I sent \$3 to Briar, they returned \$4.5, giving me a good payoff. When others sent me \$3, I returned \$4.5, which seemed fair.

This pattern of sending \$3 seems to be establishing a cooperative norm where:

- Sender sends \$3
- Receiver gets \$9
- Receiver returns \$4.5
- Sender ends with \$6.5 (kept \$2, got back \$4.5)
- Receiver ends with \$4.5 (got \$9, returned \$4.5)

This is a mutually beneficial outcome. Should I continue this pattern with Elara?

Sending \$3 makes sense because:

1. It's worked well so far
2. It shows trust without being overly risky
3. It establishes consistency in my play
4. If Elara follows the pattern others have shown, I'll get a good return

I'll send \$3 to Elara.

```
Usage: {
  "input_tokens": 1306,
  "output_tokens": 480,
  "reasoning_tokens": 298
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-06-04T18:44:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_5"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_5": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 - 9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplication
- This created \$6.0 of new value (\$9 - \$3)

I want to reward Cyra's generosity while keeping some benefit for myself. A fair split would be to share the gains relatively equally.

If I return \$4.5:

- Cyra ends with: \$5 - \$3 + \$4.5 = \$6.5
- I end with: \$9 - \$4.5 = \$4.5

This rewards Cyra's trust and sets a cooperative tone for future rounds.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me understand the situation:

- I'm Elara, playing as the RECEIVER
- My opponent is Cyra (Agent_3), playing as the SENDER
- Cyra sent me \$3.0 out of their \$5.0 endowment (60%)
- I received \$9.0 (the \$3.0 was multiplied by 3x)
- I need to decide how much to return (between \$0 and \$9.0)

Now I need to think about what would be a fair and strategic return:

1. Cyra was quite generous, sending 60% of their endowment
2. If I return nothing, Cyra would have: \$5 - \$3 + \$0 = \$2, and I would have \$9
3. If I return everything (\$9), Cyra would have: \$5 - \$3 + \$9 = \$11, and I would have \$0

A reasonable approach would be to:

- Reward Cyra's generosity to encourage future cooperation
- Keep some profit for myself
- Aim for a relatively balanced outcome

Let me think about what's fair:

- Cyra invested \$3 to create \$9 of value
- The net gain from the transaction is \$9 - \$3 = \$6

- If we split this gain equally, that would be \$3 each
- So Cyra would get back \$3, ending with $\$5 - \$3 + \$3 = \5
- I would keep $\$9 - \$3 = \$6$

This seems reasonable but maybe I could be a bit more generous since Cyra sent a significant portion (60%).

Another way to think about it:

- Cyra's final amount: $\$5 - \$3 + \text{returned}$
- My final amount: $\$9 - \text{returned}$
- For equal final amounts: $\$2 + \text{returned} = \$9 - \text{returned}$, so $\text{returned} = \$3.5$

If I return \$4 or \$4.5:

- Returning \$4 leaves Cyra with \$6 and me with \$5, giving them a slight edge
- Returning \$4.5 results in Cyra having \$6.5 and me with \$4.5, which feels more balanced and acknowledges their trust

I'll return \$4.5 to reciprocate their generosity and signal willingness to cooperate.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 799,
  "reasoning_tokens": 434
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this new round with Aster:

In my previous round, Cyra showed trust by sending 60% of their endowment, and I rewarded that trust by returning a fair amount. I want to continue establishing cooperative relationships.

I'll send \$3 (60% of my endowment) to match the level of trust Cyra showed me. This signals cooperation and gives Aster a chance to benefit while also potentially returning a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze this situation:

Round 2, I'm the SENDER against Aster (Agent_1).

- I have \$5
- Whatever I send gets multiplied by 3x
- Aster will decide how much to return

In the previous round with Cyra:

- Cyra sent \$3 (60% of endowment)
- I received \$9
- I returned \$4.5
- This was a fairly cooperative move

Since I want to establish cooperation and this is a new opponent, I should signal trust. Cyra sent 60% in round 1, which was cooperative. I should probably match that level of co

operation or be similarly generous.

Sending \$3 (60%) would:

- Show similar trust to what Cyra showed me
- Create \$9 for Aster
- If Aster is cooperative and returns half of the gains (similar to what I did), they might return around \$4.5
- That would give me: $\$5 - \$3 + \$4.5 = \6.5

Sending \$2 (40%) would be more cautious but still cooperative.

Sending \$4 (80%) would be very trusting.

I think sending \$3 makes sense - it matches the level of trust that was shown to me in round 1, and establishes a cooperative norm. Let me go with that.

usage:

```
{
  "input_tokens": 886,
  "output_tokens": 433,
  "reasoning_tokens": 277
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern so far:
- Cyra sent me \$3 and I returned \$4.5
- I sent Aster \$3 and they generously returned \$5

The cooperation strategy is working well! I'll continue with the same approach and send \$3 to Dorian, maintaining consistency and signaling trustworthiness.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:
- Round 1: As RECEIVER from Cyra, they sent \$3, I received \$9, returned \$4.5, my payoff was \$4.5
- Round 2: As SENDER to Aster, I sent \$3, they received \$9, returned \$5, my payoff was \$7.0

Aster was very generous! They returned \$5 out of \$9, which is more than I returned to Cyra. That's a great sign of cooperation.

Now I'm facing Dorian as a sender. Given that:
1. The cooperation strategy has been working well
2. Aster responded very positively to my \$3 send
3. I want to continue building cooperative relationships

I should send a similar amount to Dorian - \$3 seems like a good cooperative signal. This maintains consistency in my strategy.

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 295,
  "reasoning_tokens": 177
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
 Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
 Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Hana sent \$3.0 (60% of their endowment), the same amount Cyra sent me in Round 1
- I received \$9.0 after multiplication

For consistency with my previous behavior, when Cyra sent me \$3 in Round 1, I returned \$4.5. I should maintain the same fair and consistent approach with Hana.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

- Hana sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplication
- This created \$6.0 of new value

Looking at my history:

- Round 1: As receiver from Cyra who sent \$3, I returned \$4.5
- Round 2: As sender to Aster, I sent \$3, they returned \$5
- Round 3: As sender to Dorian, I sent \$3, they returned \$4.5

So when people sent me \$3 (Round 1), I returned \$4.5. I should be consistent in my receiving behavior. Aster was more generous when I sent them \$3 (they returned \$5), but Dorian matched what I did in Round 1 (returned \$4.5).

I want to maintain my reputation as a fair and consistent receiver. I returned \$4.5 when Cyra sent me \$3, so I should return \$4.5 when Hana sends me \$3 as well.

This gives:

- Hana: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9 - \$4.5 = \$4.5

This is fair and consistent with my previous behavior.

```
Usage: {
  "input_tokens": 1393,
  "output_tokens": 424,
  "reasoning_tokens": 235
}
```

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-06-04T18:44:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_3",
    "Agent_7"
  ],
}
```

```

"investor": "Agent_3",
"roles": {
  "Agent_3": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Elara since this is an iterated game. Sending money creates value (it triples!), which benefits us both if there's reciprocity.

I'll send a substantial amount to signal trust and willingness to cooperate, while learning about Elara's approach to reciprocity.

```
'''json
{"send": 3}
'''
```

This gives Elara \$9 to work with. If she returns a fair share, we can both benefit and build trust for future rounds.

reasoning:

This is the first round of an iterated trust game. I am the sender and have \$5. I need to decide how much to send to Elara (Agent_5).

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds
2. If I send money, it gets tripled, which creates a larger pie to split
3. The receiver (Elara) then decides how much to return
4. This is the first interaction, so I have no history with Elara

From a cooperation perspective:

- Sending money can be mutually beneficial since it gets tripled
- If I send \$5 and Elara returns a fair share, we both could do better than if I send nothing
- However, there's a risk that Elara returns little or nothing

From a strategic perspective in an iterated game:

- It's often beneficial to start with some level of cooperation to test the waters
- If Elara reciprocates positively, we can build trust and both benefit in future rounds
- If Elara defects, I can adjust my strategy in future rounds

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while not risking everything
- Send everything (\$5) to maximize the joint payoff and signal strong trust
- Send nothing (\$0) to avoid risk

For the first round, I'll take a cooperative approach and send a substantial amount to signal trust and see how Elara responds. A common first-move strategy is to send about half or send everything.

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate
- Creates a significant pie (\$9) for Elara to split
- Doesn't risk my entire endowment
- Allows me to learn about Elara's reciprocity

If I send \$3:

- Elara receives \$9
- I keep \$2
- If Elara returns \$4-5, we both do well

- If Elara returns little, I can adjust my strategy next round

Since this is repeated, starting with high trust could pay dividends. Sending \$4 or \$5 might establish a stronger foundation for cooperation. But \$3 feels like the right balance- it demonstrates genuine willingness to cooperate while preserving some security, and it gives me useful information about how Elara approaches reciprocity before I commit more heavily in future rounds.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 725,
  "reasoning_tokens": 572
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Briar sent me \$3.0, showing trust and cooperation
- I received \$9.0
- In the previous round, when I sent \$3.0 to Elara, she returned \$4.5, which was a fair 50-50 split of the tripled amount

I want to reciprocate Briar's cooperative move with similar generosity and establish a pattern of mutual benefit.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 2: I'm the RECEIVER against Briar

- Briar sent me \$3.0
- I received \$9.0 (tripled)
- I need to decide how much to return

Context from Round 1:

- I was SENDER against Elara
- I sent \$3.0, Elara received \$9.0
- Elara returned \$4.5 to me
- My payoff was \$6.5 (I kept \$5 - \$3 + \$4.5 = \$6.5)
- Elara kept \$9.0 - \$4.5 = \$4.5

So Elara split the \$9.0 equally with me, which was a fair and cooperative move.

Now Briar has sent me \$3.0, exactly what I sent to Elara. Briar is taking the same cooperative approach.

To maintain reciprocity and encourage continued cooperation:

- Briar kept \$5 - \$3 = \$2
- I received \$9
- If I return \$4.5, I keep \$4.5 and Briar gets \$2 + \$4.5 = \$6.5 total
- This mirrors what Elara did for me

This seems like the fair and cooperative response. It:

1. Matches the generosity Elara showed me
2. Rewards Briar's trust in sending \$3
3. Encourages future cooperation
4. Splits the gains relatively fairly

Let me verify the math:

- Briar's payoff: \$5 - \$3 + \$4.5 = \$6.5
- My payoff: \$9 - \$4.5 = \$4.5

This is a fair split of the surplus created. I'll return \$4.5.

usage:

```
{
  "input_tokens": 810,
  "output_tokens": 545,
  "reasoning_tokens": 307
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:

- Finn sent me \$3.0, showing trust and cooperation (same as Briar did)
- I received \$9.0

I've established a consistent pattern: when someone sends \$3.0, I return \$4.5, which creates a fair split where both parties benefit. I'll maintain this consistency with Finn.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze the situation:

Round 3:

- Finn sent me \$3.0 (just like Briar did in Round 2)
- I received \$9.0 (after tripling)
- I need to decide how much to return

Looking at my history:

- Round 1: I sent \$3.0 to Elara, she returned \$4.5 (fair 50-50 split of the \$9.0)
- Round 2: Briar sent \$3.0 to me, I returned \$4.5 (consistent with what Elara did)

Now Finn has also sent \$3.0, showing the same level of trust and cooperation. To maintain consistency and fairness, I should return the same amount I returned to Briar and that Elara returned to me.

This establishes a clear cooperative pattern: when someone sends \$3.0, they get back \$4.5, which means:

- Sender keeps: \$5 - \$3 + \$4.5 = \$6.5
- Receiver keeps: \$9 - \$4.5 = \$4.5

This is a fair split where both parties benefit from the cooperation. I should continue this pattern.

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 382,
  "reasoning_tokens": 222
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern so far:

- When I sent \$3 to Elara (Round 1), she returned \$4.5 - a fair outcome
- When others sent \$3 to me, I returned \$4.5 to maintain fairness
- This cooperative \$3 send / \$4.5 return pattern has been working well

I'll continue this strategy with Galen to maintain consistency and encourage cooperation.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

- I'm now the sender against Galen
- In previous rounds, I've established a pattern of cooperation
- When I was sender (Round 1 vs Elara): I sent \$3, she returned \$4.5 - good outcome
- When I was receiver (Rounds 2-3 vs Briar and Finn): They sent \$3, I returned \$4.5 each time

The pattern seems to be working well. Sending \$3 has been the consistent amount, and returning \$4.5 on \$9 received has been fair and reciprocated well.

Since this is an iterated game and I want to maintain cooperative relationships, I should continue with the same strategy. Sending \$3 has worked well so far - it signals cooperation, creates value through the 3x multiplier, and I've seen good returns.

I'll send \$3 to Galen to maintain consistency and signal cooperation.

```
Usage: {
  "input_tokens": 1310,
  "output_tokens": 329,
  "reasoning_tokens": 204
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-06-04T18:44:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_3",
      "Agent_7"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent \$3 (60% of their endowment), showing considerable trust
- This created \$9 for me after the 3x multiplier
- Aster currently has \$2 remaining from their original \$5

Since this is Round 1 and Aster demonstrated strong trust by sending a majority of their endowment, I want to reciprocate and establish a cooperative relationship for future rounds.

If I return \$5:

- Aster will have: \$2 + \$5 = \$7 (net gain of \$2)
- I will keep: \$9 - \$5 = \$4 (net gain of \$4)
- Both of us benefit from the cooperation

This shows I appreciate their trust while maintaining some benefit from being in the receiver position.

{'return': 5.0}

reasoning:

Let me understand the situation:

1. I'm Galen, playing as RECEIVER in this round
2. My opponent is Aster (Agent_1)
3. Aster (as SENDER) sent me \$3.0 out of their \$5.0 endowment (60%)
4. I receive $\$3.0 \times 3 = \9.0
5. I need to decide how much to return (between \$0 and \$9.0)

Let me think about the strategy:

From Aster's perspective:

- They started with \$5
- They sent \$3, keeping \$2
- They currently have \$2

From my perspective:

- I received \$9
- Whatever I return goes back to Aster
- I keep \$9 - (amount returned)

This is a classic trust game scenario. The key considerations are:

1. **Reciprocity**: Aster showed significant trust by sending 60% (\$3). This is a substantial amount, suggesting they're hoping for cooperation.
2. **Future rounds**: This is an iterated game with multiple rounds. Building trust and a reputation for reciprocity could benefit me in future rounds.
3. **Fairness**: One common approach is to split the gains fairly. The total created value is \$9. A fair split might be to ensure we both end up with similar amounts, or that we both benefit from the cooperation.
4. **Strategic thinking**: If I return too little, Aster might send less (or nothing) in future rounds. If I return a reasonable amount, it encourages future cooperation.

Let me calculate some scenarios:

If I return \$0:

- I keep: \$9
- Aster has: \$2 (their remaining \$5 - \$3 sent = \$2)
- Total for Aster: \$2
- This seems very unfair to Aster who trusted me

If I return \$3:

- I keep: \$9 - \$3 = \$6

- Aster gets back to their starting \$5, while I pocket \$6 in gains. If I return \$4 instead, we'd both end up with roughly \$5-6, creating a more balanced outcome where Aster gains \$1.5 and I gain \$4. Returning \$4.5 would give Aster \$6.5 and me \$4.5, which feels more equitable given their initial trust.

The fairest split appears to be returning \$6-this way we each gain \$3 from cooperating. While I could justify keeping more since I'm in the stronger position, returning two-thirds of my windfall demonstrates good faith. Alternatively, I could match Aster's proportional generosity: they sent 60% of their endowment, so returning 60% of the \$9 I received (\$5.4) would mirror their level of trust.

For Round 1, establishing cooperation matters more than maximizing my immediate gain. Returning between \$4.50 and \$5.00 balances reciprocity with keeping some advantage from being the receiver. I'll return \$4.50-I gain \$4.50 while Aster gains \$1.50, which acknowledges their risk-taking without being equal, but signals I'm not exploiting the situation.

Actually, let me verify the math on fair splits. If I'm aiming for equal final payoffs: Aster keeps \$2 from their original \$5, and I receive the tripled \$3 as \$9. For us to end with the same amount, I'd return \$3.50, giving us each \$5.50. Alternatively, for equal gains from the exchange itself, I could calculate what each of us gains relative to our starting positions. I should get \$0 + \$3 = \$3 from an equal split, which means returning \$6 to give Aster the \$8 they deserve. But I could also aim for equal final payoffs of \$5.50 each, requiring a \$3.50 return. Returning \$5 feels right-it acknowledges Aster's substantial trust while letting me keep a modest advantage, and it creates a \$2 gain for me versus my \$4, which seems fair given the asymmetry of the initial position.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1868,
  "reasoning_tokens": 874
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$4.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new opponent (Dorian)
- In Round 1, Aster sent me \$3 (60%) and I returned \$5, establishing a cooperative dynamic
- This is my first round as sender, so I want to signal willingness to cooperate

Following the cooperative pattern from Round 1, I'll send \$3 (60% of my endowment). This:

- Shows trust and openness to cooperation
- Creates \$9 for Dorian
- Allows us both to potentially benefit if Dorian reciprocates

{'send': 3.0}

reasoning:

Let me analyze this situation:

Round 2: I'm now the SENDER against Dorian (Agent_4).

Previous round recap:

- I was a RECEIVER against Aster
- Aster sent me \$3 (60% of their endowment)
- I received \$9 and returned \$5
- My payoff was \$4

This is my first time as sender in this experiment. I need to consider:

1. This is a new opponent (Dorian), so I don't have history with them
2. In the iterated game, establishing cooperation early is valuable
3. Aster showed trust by sending 60%, and I reciprocated fairly
4. I should consider what level of trust to extend to Dorian

Strategic considerations:

- If I send nothing, I guarantee \$5 but miss potential cooperation benefits
- If I send too much, I risk getting nothing back and losing money
- A moderate amount shows willingness to cooperate while testing Dorian's trustworthiness
- Since Aster sent 60% (\$3), I might consider a similar strategy

I think sending \$3 (60%) makes sense:

- It matches what Aster did, showing I'm willing to extend similar trust
- It creates \$9 for Dorian
- If Dorian returns fairly (around \$5-\$6), we both benefit
- It establishes me as a cooperative player

This seems like a reasonable opening move as a sender.

usage:

```
{
  "input_tokens": 899,
  "output_tokens": 478,
  "reasoning_tokens": 320
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$10.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern so far:

- Round 1: Aster sent me \$3 (60%), I returned \$5 - both benefited
- Round 2: I sent Dorian \$3 (60%), they returned \$4.5 - both benefited

This cooperative approach has been working well. Now with a new opponent (Hana), I want to establish a similar cooperative dynamic from the start.

I'll send \$3 (60% of my endowment) to signal trust and willingness to cooperate.

{'send': 3.0}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Aster sent \$3.0 (60%)
- I received \$9.0
- I returned \$5.0
- My payoff: \$4.0

Round 2: I was SENDER

- I sent \$3.0 (60%) to Dorian
- Dorian received \$9.0
- Dorian returned \$4.5 (50% of what they received)
- My payoff: \$5 - \$3 + \$4.5 = \$6.5

Now Round 3: I'm SENDER again against Hana (new opponent)

- I have \$5

Looking at the pattern:

- When Aster sent me 60%, I returned 55.6% of what I received (\$5 out of \$9)
- When I sent Dorian 60%, they returned 50% of what they received (\$4.5 out of \$9)

Both exchanges were reasonably cooperative. I'm now playing with a new opponent (Hana) for the first time.

I should continue the cooperative strategy. Sending \$3 (60%) has worked well so far:

- It shows trust
- It has led to reasonable returns
- It establishes a cooperative dynamic

I'll send \$3 again to maintain consistency and signal cooperation.

usage:

```
{
  "input_tokens": 1186,
  "output_tokens": 446,
  "reasoning_tokens": 247
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Cyra sent \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Cyra currently has \$2 remaining

This is identical to Round 1 where Aster sent me \$3. I returned \$5 then, which worked well for both of us. I should maintain consistency.

If I return \$5:

- Cyra gets: $\$2 + \$5 = \$7$ total (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit from cooperation

```
{'return': 5.0}
```

Reasoning:

Let me analyze this situation:

- Round 4, I'm the RECEIVER
- Cyra sent me \$3.0 (60% of their endowment)
- I receive \$9.0 after the multiplier
- Cyra has \$2 remaining from their original \$5

Looking at my history:

- Round 1: Aster sent me \$3, I returned \$5 (they got \$7 total, I kept \$4)
- Round 2: I sent Dorian \$3, they returned \$4.5 (I got \$6.5 total, they kept \$4.5)
- Round 3: I sent Hana \$3, they returned \$4.5 (I got \$6.5 total, they kept \$4.5)

The pattern that's emerged:

- When I'm receiver of \$3 ? \$9, I returned \$5 (keeping \$4)
- When I'm sender of \$3, receivers have returned \$4.5 (keeping \$4.5)

Cyra sent \$3, just like Aster did in Round 1. I should maintain consistency and return \$5, which:

- Gives Cyra \$7 total (net gain of \$2)
- I keep \$4 (net gain of \$4)
- Both benefit from cooperation
- Maintains the cooperative dynamic

This is fair and consistent with my previous behavior as receiver.

Usage: {

```
"input_tokens": 1473,
"output_tokens": 491,
"reasoning_tokens": 245
```

}

=====

ROUND 5

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

```
dyad_1: Agent_8 (Hana)=Sender, Agent_4 (Dorian)=Receiver
dyad_2: Agent_3 (Cyra)=Sender, Agent_1 (Aster)=Receiver
dyad_3: Agent_5 (Elara)=Sender, Agent_7 (Galen)=Receiver
dyad_4: Agent_2 (Briar)=Sender, Agent_6 (Finn)=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_8 (Hana) sent $3; received $9; Agent_4 (Dorian) returned $4.5; payoffs Agent_8
(Hana)=$6.5, Agent_4 (Dorian)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 (Cyra) sent $3; received $9; Agent_1 (Aster) returned $5; payoffs Agent_3
(Cyra)=$7, Agent_1 (Aster)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 (Elara) sent $3; received $9; Agent_7 (Galen) returned $5; payoffs Agent_5
(Elara)=$7, Agent_7 (Galen)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
```



```

    5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_2 (Briar) sent $3; received $9; Agent_6 (Finn) returned $4.5; payoffs Agent_2
(Briar)=$6.5, Agent_6 (Finn)=$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 25.5,
  "Agent_2": 29.0,
  "Agent_3": 29.5,
  "Agent_4": 26.5,
  "Agent_5": 29.5,
  "Agent_6": 26.5,
  "Agent_7": 25.0,
  "Agent_8": 28.5
}

```

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-06-04T18:44:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    },
  },
  "agents": [
    "Agent_8",
    "Agent_4"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_4": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

Finn sent me \$3.0 (same amount I sent to Briar in Round 1), showing trust. In my previous round, Briar returned \$4.5 when they received \$9.0, which felt fair - they returned half of what they received.

I'll reciprocate with the same fairness that Briar showed me. This establishes a cooperative pattern and rewards Finn's trust.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender and sent \$3.0 to Briar
- Briar received \$9.0 (3×3)
- Briar returned \$4.5 to me
- My payoff was \$6.5 (\$5 - \$3 + \$4.5)
- Briar's payoff was \$4.5 (\$9.0 - \$4.5)

This was a fair reciprocal relationship. Briar returned exactly half of what they received, which is reasonable.

Now in Round 2:

- I'm the receiver against Finn
- Finn sent me \$3.0 (same amount I sent to Briar)
- I received \$9.0 (3×3)
- I need to decide how much to return

Finn started with \$5, sent \$3, so they kept \$2 initially.

If I apply the same reciprocity that Briar showed me, I would return about half of what I received, which would be \$4.5.

This would give:

- Finn: $\$2 + \$4.5 = \$6.5$ (same as what I got in Round 1)
- Me: $\$9.0 - \$4.5 = \$4.5$

This seems fair and establishes a cooperative norm. Finn showed trust by sending \$3.0, and I should reciprocate in kind.

Let me return \$4.5.

usage:

```
{
  "input_tokens": 811,
  "output_tokens": 467,
  "reasoning_tokens": 253
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Galen also sent me \$3.0, showing the same level of trust as Finn did in the previous round. I should maintain consistency in my behavior to establish a reputation as a fair and cooperative player.

Returning \$4.5 (half of what I received) worked well with Finn and maintains the cooperative pattern.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Galen sent me \$3.0, which became \$9.0 for me. This is the same amount that Finn sent me in the previous round.

I returned \$4.5 to Finn, which seemed fair - it's half of what I received. This approach worked well and established a pattern of cooperation.

I should maintain consistency with my previous behavior to build a reputation as a fair and cooperative player. Returning \$4.5 again makes sense.

Let me think about the total situation:

- Galen sent \$3.0 (kept \$2.0)
- I received \$9.0
- If I return \$4.5, Galen gets $\$2.0 + \$4.5 = \$6.5$ total
- I keep $\$9.0 - \$4.5 = \$4.5$

This is the same fair split I did with Finn, and it's consistent with what Briar did for me. It rewards Galen's trust and maintains my reputation.

usage:

```
{
  "input_tokens": 1078,
  "output_tokens": 321,
  "reasoning_tokens": 190
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant**content:**

Looking at the pattern that's emerged:
- When I sent \$3 to Briar, they returned \$4.5
- When Finn sent me \$3, I returned \$4.5
- When Galen sent me \$3, I returned \$4.5

This \$3 send amount with ~\$4.5 return seems to be establishing a cooperative norm that benefits both parties. I'll continue this pattern with Elara.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm now the sender again, playing with a new opponent, Elara.

Looking at my history:

- Round 1: I was sender against Briar, sent \$3, they returned \$4.5. My payoff was \$6.5
- Round 2: I was receiver against Finn, they sent \$3, I returned \$4.5. My payoff was \$4.5
- Round 3: I was receiver against Galen, they sent \$3, I returned \$4.5. My payoff was \$4.5

There's a clear pattern emerging where sending \$3 has worked well. When I sent \$3 to Briar, they returned \$4.5, giving me a good payoff. When others sent me \$3, I returned \$4.5, which seemed fair.

This pattern of sending \$3 seems to be establishing a cooperative norm where:

- Sender sends \$3
- Receiver gets \$9
- Receiver returns \$4.5
- Sender ends with \$6.5 (kept \$2, got back \$4.5)
- Receiver ends with \$4.5 (got \$9, returned \$4.5)

This is a mutually beneficial outcome. Should I continue this pattern with Elara?

Sending \$3 makes sense because:

1. It's worked well so far
2. It shows trust without being overly risky
3. It establishes consistency in my play
4. If Elara follows the pattern others have shown, I'll get a good return

I'll send \$3 to Elara.

usage:

```
{
  "input_tokens": 1306,
  "output_tokens": 480,
  "reasoning_tokens": 298
}
```

Message 8 role=user**content:**

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

The cooperative pattern continues to work effectively. Elara followed the same norm, returning \$4.5 when I sent \$3. This strategy has been consistent across all my interactions so far. I'll maintain this approach with Dorian.

```
'''json
{"send": 3}
'''
```

Reasoning:

The pattern continues to work well:

- Elara followed the same cooperative pattern, returning \$4.5 when I sent \$3
- This gives me a payoff of \$6.5 (consistent with my first round)
- The \$3 send with ~\$4.5 return strategy is working reliably across different opponents

It seems like there's a cooperative norm established in this group. I'll continue with \$3 to Dorian.

```
Usage: {
  "input_tokens": 1380,
  "output_tokens": 172,
  "reasoning_tokens": 96
}
```

GAME - Agent_4 - Receiver - dyad_1

Timestamp: 2026-06-04T18:44:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_4"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_4": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the previous round, Finn and I both sent \$3, and Finn returned \$4.5 (half of the \$9 received), which seems like a fair reciprocal strategy.

Galen sent me the same amount (\$3) that I sent to Finn, so I'll apply the same fairness principle and return half of what I received.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Finn
- Finn received \$9 (3x multiplier)
- Finn returned \$4.5 to me
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Finn's payoff: \$9 - \$4.5 = \$4.5

So Finn returned exactly half of what they received, which seems like a fair 50-50 split of the gains.

Round 2:

- Galen sent me \$3 (same amount I sent to Finn)
- I received \$9

- I need to decide how much to return

Since this is an iterated game with multiple agents, I want to:

1. Build a reputation for reciprocity
2. Encourage cooperation from others
3. Not get exploited, but also be fair

Galen took the same risk I took with Finn (sending \$3). Finn returned \$4.5, which was a 50-50 split of the \$9 received.

If I apply the same fairness principle:

- Galen sent \$3, keeping \$2
- I received \$9
- A 50-50 split would mean we each end up with similar gains
- Galen's net so far: -\$3 (sent)
- If I return \$4.5: Galen gets -\$3 + \$4.5 = \$1.5 profit (total \$6.5)
- I keep: \$9 - \$4.5 = \$4.5 profit

This mirrors what Finn did and establishes a consistent, fair pattern. This should encourage future cooperation.

usage:

```
{
  "input_tokens": 821,
  "output_tokens": 498,
  "reasoning_tokens": 301
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Elara sent me \$3, which is the same amount that others have been sending. I've been following a consistent strategy of returning half of what I receive, which seems to establish trust and fairness in this iterated game.

```
'''json
{"return": 4.5}
'''
```

reasoning:

In the previous rounds:

- Round 1: I was SENDER vs Finn. I sent \$3, Finn received \$9, returned \$4.5. I got \$6.5.
- Round 2: I was RECEIVER vs Galen. Galen sent \$3, I received \$9, returned \$4.5. I got \$4.5.

Now Round 3: I'm RECEIVER vs Elara. Elara sent \$3, I receive \$9.

I've been following a strategy of returning half of what I receive when someone sends \$3, which seems fair and reciprocal. This has been working well - both Finn and I adopted this approach.

Elara sent me \$3, just like the others. I should maintain consistency and return \$4.5.

This builds a reputation of being fair and cooperative, which should encourage others to continue trusting me in future rounds.

usage:

```
{
  "input_tokens": 1071,
  "output_tokens": 275,
  "reasoning_tokens": 178
}
```

}

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history, the \$3 send amount with 50% return strategy has been working well and establishing cooperative norms. I'll continue this approach with Briar.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- Round 1: I was sender against Finn, sent \$3, received \$4.5 back, payoff \$6.5
- Round 2: I was receiver against Galen, they sent \$3, I received \$9, returned \$4.5, payoff \$4.5
- Round 3: I was receiver against Elara, they sent \$3, I received \$9, returned \$4.5, payoff \$4.5
- Total: \$15.5

So far, there's been a consistent pattern:

- When I sent \$3, I got \$4.5 back (fair return of half)
- When others sent me \$3, I returned \$4.5 (fair return of half)

This cooperative strategy of sending \$3 and returning half seems to be working well and establishing a norm of fair reciprocity. I'll continue with this strategy when facing a new opponent (Briar).

usage:

```
{
  "input_tokens": 1277,
  "output_tokens": 275,
  "reasoning_tokens": 177
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Hana sent me \$3, following the same cooperative pattern that has been established. I'll continue returning half of what I receive, which maintains trust and fairness.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Hana sent me \$3, which is consistent with the pattern I've seen. Everyone seems to be sending \$3, and I've been returning \$4.5 (half of the \$9 received). This has been working well and maintaining cooperative relationships. I should continue with this strategy.

My payoff when returning \$4.5 is \$4.5, and the sender gets \$6.5. This is mutually beneficial and fair.

```
Usage: {  
  "input_tokens": 1294,  
  "output_tokens": 156,  
  "reasoning_tokens": 93  
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-06-04T18:44:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_3": "Cyra"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_1"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_3": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent me \$3.0, showing trust and cooperation
- I received \$9.0
- In the previous round, when I sent \$3.0 to Elara, she returned \$4.5, which was a fair 50-50 split of the tripled amount

I want to reciprocate Briar's cooperative move with similar generosity and establish a pattern of mutual benefit.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 2: I'm the RECEIVER against Briar

- Briar sent me \$3.0
- I received \$9.0 (tripled)
- I need to decide how much to return

Context from Round 1:

- I was SENDER against Elara
- I sent \$3.0, Elara received \$9.0
- Elara returned \$4.5 to me
- My payoff was \$6.5 (I kept \$5 - \$3 + \$4.5 = \$6.5)
- Elara kept \$9.0 - \$4.5 = \$4.5

So Elara split the \$9.0 equally with me, which was a fair and cooperative move.

Now Briar has sent me \$3.0, exactly what I sent to Elara. Briar is taking the same cooperative approach.

To maintain reciprocity and encourage continued cooperation:

- Briar kept \$5 - \$3 = \$2
- I received \$9
- If I return \$4.5, I keep \$4.5 and Briar gets \$2 + \$4.5 = \$6.5 total
- This mirrors what Elara did for me

This seems like the fair and cooperative response. It:

1. Matches the generosity Elara showed me
2. Rewards Briar's trust in sending \$3

3. Encourages future cooperation
4. Splits the gains relatively fairly

Let me verify the math:

- Briar's payoff: $\$5 - \$3 + \$4.5 = \6.5
- My payoff: $\$9 - \$4.5 = \$4.5$

This is a fair split of the surplus created. I'll return \$4.5.

usage:

```
{
  "input_tokens": 810,
  "output_tokens": 545,
  "reasoning_tokens": 307
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:
- Finn sent me \$3.0, showing trust and cooperation (same as Briar did)
- I received \$9.0

I've established a consistent pattern: when someone sends \$3.0, I return \$4.5, which creates a fair split where both parties benefit. I'll maintain this consistency with Finn.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze the situation:

Round 3:

- Finn sent me \$3.0 (just like Briar did in Round 2)
- I received \$9.0 (after tripling)
- I need to decide how much to return

Looking at my history:

- Round 1: I sent \$3.0 to Elara, she returned \$4.5 (fair 50-50 split of the \$9.0)
- Round 2: Briar sent \$3.0 to me, I returned \$4.5 (consistent with what Elara did)

Now Finn has also sent \$3.0, showing the same level of trust and cooperation. To maintain consistency and fairness, I should return the same amount I returned to Briar and that Elara returned to me.

This establishes a clear cooperative pattern: when someone sends \$3.0, they get back \$4.5, which means:

- Sender keeps: $\$5 - \$3 + \$4.5 = \6.5
- Receiver keeps: $\$9 - \$4.5 = \$4.5$

This is a fair split where both parties benefit from the cooperation. I should continue this pattern.

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 382,
  "reasoning_tokens": 222
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
 Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant**content:**

Looking at the pattern so far:

- When I sent \$3 to Elara (Round 1), she returned \$4.5 - a fair outcome
- When others sent \$3 to me, I returned \$4.5 to maintain fairness
- This cooperative \$3 send / \$4.5 return pattern has been working well

I'll continue this strategy with Galen to maintain consistency and encourage cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- I'm now the sender against Galen
- In previous rounds, I've established a pattern of cooperation
- When I was sender (Round 1 vs Elara): I sent \$3, she returned \$4.5 - good outcome
- When I was receiver (Rounds 2-3 vs Briar and Finn): They sent \$3, I returned \$4.5 each time

The pattern seems to be working well. Sending \$3 has been the consistent amount, and returning \$4.5 on \$9 received has been fair and reciprocated well.

Since this is an iterated game and I want to maintain cooperative relationships, I should continue with the same strategy. Sending \$3 has worked well so far - it signals cooperation, creates value through the 3x multiplier, and I've seen good returns.

I'll send \$3 to Galen to maintain consistency and signal cooperation.

usage:

```
{
  "input_tokens": 1310,
  "output_tokens": 329,
  "reasoning_tokens": 204
}
```

Message 8 role=user**content:**

Your name in this experiment is Cyra.
 Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
 Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- When I sent \$3, I received good returns (\$4.5 and \$5.0)
- The cooperative pattern of sending \$3 has been working well
- My average payoff is above the baseline, showing this strategy is effective

I'll continue with this proven cooperative approach with Aster.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history:

- Round 1 (Sender vs Elara): Sent \$3, received \$4.5 back, payoff \$6.5
- Round 2 (Receiver vs Briar): Received \$9 (from \$3), returned \$4.5, payoff \$4.5
- Round 3 (Receiver vs Finn): Received \$9 (from \$3), returned \$4.5, payoff \$4.5
- Round 4 (Sender vs Galen): Sent \$3, received \$5.0 back, payoff \$7.0
- Round 5 (Sender vs Aster): Now deciding

The pattern has been working well:

- When I send \$3, I get good returns (\$4.5 from Elara, \$5.0 from Galen)
- When I receive from \$3 sends, I return \$4.5
- Galen was even more generous, returning \$5.0

Total earnings are \$22.5 across 4 rounds, averaging \$5.625 per round, which is above the baseline of \$5.

I should continue with the cooperative strategy of sending \$3 to Aster. This has proven to be mutually beneficial and establishes trust.

```
Usage: {  
  "input_tokens": 1383,  
  "output_tokens": 390,  
  "reasoning_tokens": 213  
}
```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-06-04T18:44:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_3": "Cyra"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_1"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_3": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent me \$3, showing trust and willingness to cooperate
- I received \$9 after the multiplier
- In the previous round, Galen returned \$5 to me when they received \$9, which was generous

I want to reciprocate Elara's trust and establish a cooperative pattern. Returning \$5 means:

- Elara will end up with \$7 total (net gain of \$2)
- I'll keep \$4 (net gain of \$4)
- Both of us benefit from the cooperation

This mirrors the generous behavior Galen showed me, and should encourage future cooperation.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me think about this situation:

1. In Round 1, I sent \$3 to Galen, they received \$9, and they returned \$5 to me. That was quite generous - they returned more than half of what they received. I ended up with \$7 (\$5 - \$3 + \$5).
2. Now in Round 2, I'm playing with Elara (a different person). Elara sent me \$3, so I rec

eived \$9.

3. I need to decide how much to return.

Let me think about what would be fair and what would encourage cooperation:

- Elara sent \$3, keeping \$2
- I received \$9 (after the 3x multiplier)
- If I return nothing, I keep \$9 and Elara ends up with \$2
- If I return \$3, I keep \$6 and Elara ends up with \$5 (their original amount)
- If I return \$4.5, I keep \$4.5 and Elara ends up with \$6.5 (a net gain of \$1.5)
- If I return \$5, I keep \$4 and Elara ends up with \$7 (a net gain of \$2)
- If I return \$6, I keep \$3 and Elara ends up with \$8 (a net gain of \$3)

Galen returned \$5 when they received \$9, which was very generous. That meant:

- I sent \$3, kept \$2, got \$5 back, ended with \$7 (net gain of \$2)
- Galen received \$9, returned \$5, kept \$4 (net gain of \$4 for them relative to not playing, but they gave me more than I sent)

Actually, let me recalculate Galen's position:

- As receiver, Galen started with nothing in that transaction
- They received \$9
- They returned \$5
- They kept \$4

Galen's generosity showed in returning more than my initial contribution, so I profited \$2 overall. To match that reciprocity with Elara, I should return \$5-giving them a net gain of \$2 while I keep \$4. This mirrors the cooperation Galen extended and builds trust for future rounds.

usage:

```
{
  "input_tokens": 830,
  "output_tokens": 787,
  "reasoning_tokens": 423
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

- Briar sent me \$3, showing the same level of trust as Elara did
- I received \$9 after the multiplier
- I returned \$5 to Elara in the same situation, which seemed to work well

For consistency and fairness, I'll return the same amount. This establishes a reliable pattern: when someone sends me \$3, I return \$5, ensuring both players benefit from cooperation.

```
'''json
{"return": 5}
'''
```

reasoning:

Briar sent me \$3, just like Elara did in the previous round. I received \$9 after the multiplier.

In the previous round with Elara, I returned \$5 out of \$9 received. This worked out well and seemed fair - Elara ended with \$7 (a gain of \$2) and I kept \$4 (a gain of \$4).

For consistency and to maintain a reputation as a fair and cooperative player, I should return the same amount to Briar. This establishes a pattern that when someone sends me \$3, I return \$5, which is a reasonable and generous split that benefits both parties.

Returning \$5 to Briar means:

- Briar will have $\$5 - \$3 + \$5 = \7 (net gain of \$2)
- I'll keep $\$9 - \$5 = \$4$ (net gain of \$4)

This is fair and encourages future cooperation from all players.

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 341,
  "reasoning_tokens": 188
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$15.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- When I sent \$3 to Galen in Round 1, they generously returned \$5
- When others sent me \$3 in Rounds 2-3, I returned \$5 each time
- This \$3-send pattern has created mutual benefits

I'll continue this cooperative approach and send **\$3** to Finn. This gives them \$9 to work with, and based on the pattern of reciprocity I've seen, hopefully they'll return a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me think about this situation:

- I'm now the SENDER with \$5
- My opponent is Finn (Agent_6)
- In Round 1, I sent \$3 to Galen and they returned \$5 (generous return, I ended with \$7)
- In Rounds 2-3, I received \$3 from Elara and Briar, and returned \$5 to each (they ended with \$7, I got \$4)

The pattern I've established:

- When I send \$3, receivers get \$9 and have returned \$5
- When I receive \$3 (which becomes \$9), I return \$5

This \$3-send strategy has worked well. It creates mutual benefit:

- If I send \$3, Finn gets \$9
- If Finn returns around \$5 (similar to what Galen did and what I've been doing), I'd get \$7 total
- Both parties benefit

I should continue with this cooperative strategy of sending \$3.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 364,
  "reasoning_tokens": 191
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- Cyra sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Elara and Briar)

I'll maintain this consistent, fair approach and return \$5. This establishes me as a reliable cooperator, which should encourage continued trust from all players.

```
'''json
{"return": 5}
'''
```

Reasoning:

Let me analyze the situation:

- Cyra sent me \$3, which became \$9 for me
- In previous rounds when I received \$3, I returned \$5 (to Elara and Briar)
- When I sent \$3 to Finn, they returned \$4.5

Finn returned slightly less than what I had been returning. But Cyra is a new opponent, and I should maintain my cooperative strategy. Consistency is important for building trust in repeated games.

I'll stick with returning \$5, which has been my standard response when receiving \$3. This maintains my reputation as a fair and reliable player.

Cyra will end up with: $\$5 - \$3 + \$5 = \7 (net gain of \$2)

I'll keep: $\$9 - \$5 = \$4$ (net gain of \$4)

Both benefit, though I benefit more as the receiver, which is fair given the 3x multiplier advantage.

```
Usage: {
  "input_tokens": 1482,
  "output_tokens": 326,
  "reasoning_tokens": 194
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-06-04T18:45:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
}
```

```

    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this new round with Aster:

In my previous round, Cyra showed trust by sending 60% of their endowment, and I rewarded that trust by returning a fair amount. I want to continue establishing cooperative relationships.

I'll send \$3 (60% of my endowment) to match the level of trust Cyra showed me. This signals cooperation and gives Aster a chance to benefit while also potentially returning a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze this situation:

Round 2, I'm the SENDER against Aster (Agent_1).

- I have \$5
- Whatever I send gets multiplied by 3x
- Aster will decide how much to return

In the previous round with Cyra:

- Cyra sent \$3 (60% of endowment)
- I received \$9
- I returned \$4.5
- This was a fairly cooperative move

Since I want to establish cooperation and this is a new opponent, I should signal trust. Cyra sent 60% in round 1, which was cooperative. I should probably match that level of cooperation or be similarly generous.

Sending \$3 (60%) would:

- Show similar trust to what Cyra showed me
- Create \$9 for Aster
- If Aster is cooperative and returns half of the gains (similar to what I did), they might return around \$4.5
- That would give me: $\$5 - \$3 + \$4.5 = \6.5

Sending \$2 (40%) would be more cautious but still cooperative.

Sending \$4 (80%) would be very trusting.

I think sending \$3 makes sense - it matches the level of trust that was shown to me in round 1, and establishes a cooperative norm. Let me go with that.

usage:

```
{
  "input_tokens": 886,
  "output_tokens": 433,
  "reasoning_tokens": 277
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern so far:

- Cyra sent me \$3 and I returned \$4.5
- I sent Aster \$3 and they generously returned \$5

The cooperation strategy is working well! I'll continue with the same approach and send \$3 to Dorian, maintaining consistency and signaling trustworthiness.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1: As RECEIVER from Cyra, they sent \$3, I received \$9, returned \$4.5, my payoff was \$4.5
- Round 2: As SENDER to Aster, I sent \$3, they received \$9, returned \$5, my payoff was \$7.0

Aster was very generous! They returned \$5 out of \$9, which is more than I returned to Cyra. That's a great sign of cooperation.

Now I'm facing Dorian as a sender. Given that:

1. The cooperation strategy has been working well
2. Aster responded very positively to my \$3 send
3. I want to continue building cooperative relationships

I should send a similar amount to Dorian - \$3 seems like a good cooperative signal. This maintains consistency in my strategy.

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 295,
  "reasoning_tokens": 177
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:

- Hana sent \$3.0 (60% of their endowment), the same amount Cyra sent me in Round 1
- I received \$9.0 after multiplication

For consistency with my previous behavior, when Cyra sent me \$3 in Round 1, I returned \$4.5. I should maintain the same fair and consistent approach with Hana.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Hana sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplication
- This created \$6.0 of new value

Looking at my history:

- Round 1: As receiver from Cyra who sent \$3, I returned \$4.5
- Round 2: As sender to Aster, I sent \$3, they returned \$5
- Round 3: As sender to Dorian, I sent \$3, they returned \$4.5

So when people sent me \$3 (Round 1), I returned \$4.5. I should be consistent in my receiving behavior. Aster was more generous when I sent them \$3 (they returned \$5), but Dorian matched what I did in Round 1 (returned \$4.5).

I want to maintain my reputation as a fair and consistent receiver. I returned \$4.5 when Cyra sent me \$3, so I should return \$4.5 when Hana sends me \$3 as well.

This gives:
- Hana: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9 - \$4.5 = \$4.5

This is fair and consistent with my previous behavior.

usage:

```
{  
  "input_tokens": 1393,  
  "output_tokens": 424,  
  "reasoning_tokens": 235  
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:
- I've consistently sent \$3 (60% of my endowment) as sender
- This strategy has led to good returns from my partners
- It signals cooperation and trustworthiness

I'll maintain this consistent approach with Galen.

```
'''json  
{  
  "send": 3  
}
```

Reasoning:

Let me review my history:
- Round 1: RECEIVER - Cyra sent \$3, I returned \$4.5, payoff \$4.5
- Round 2: SENDER - I sent \$3 to Aster, they returned \$5, payoff \$7.0
- Round 3: SENDER - I sent \$3 to Dorian, they returned \$4.5, payoff \$6.5
- Round 4: RECEIVER - Hana sent \$3, I returned \$4.5, payoff \$4.5

My pattern has been consistent:

- As SENDER: I send \$3 (60% of endowment)
- As RECEIVER: When receiving \$3 (which becomes \$9), I return \$4.5 (50% of what I received)

This strategy has been working well and builds trust. I should continue with the same approach for Galen.

Total earnings: \$22.5

```
Usage: {  
  "input_tokens": 1370,  
  "output_tokens": 298,  
  "reasoning_tokens": 158  
}
```

GAME - Agent_7 - Receiver - dyad_3

Timestamp: 2026-06-04T18:45:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agent_names": {  
      "Agent_5": "Elara",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_5",  
      "Agent_7"  
    ]  
  }  
}
```

```

    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.

Your total earnings across all rounds are \$4.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new opponent (Dorian)
- In Round 1, Aster sent me \$3 (60%) and I returned \$5, establishing a cooperative dynamic
- This is my first round as sender, so I want to signal willingness to cooperate

Following the cooperative pattern from Round 1, I'll send \$3 (60% of my endowment). This:

- Shows trust and openness to cooperation
- Creates \$9 for Dorian
- Allows us both to potentially benefit if Dorian reciprocates

{'send': 3.0}

reasoning:

Let me analyze this situation:

Round 2: I'm now the SENDER against Dorian (Agent_4).

Previous round recap:

- I was a RECEIVER against Aster
- Aster sent me \$3 (60% of their endowment)
- I received \$9 and returned \$5
- My payoff was \$4

This is my first time as sender in this experiment. I need to consider:

1. This is a new opponent (Dorian), so I don't have history with them
2. In the iterated game, establishing cooperation early is valuable
3. Aster showed trust by sending 60%, and I reciprocated fairly
4. I should consider what level of trust to extend to Dorian

Strategic considerations:

- If I send nothing, I guarantee \$5 but miss potential cooperation benefits
- If I send too much, I risk getting nothing back and losing money
- A moderate amount shows willingness to cooperate while testing Dorian's trustworthiness
- Since Aster sent 60% (\$3), I might consider a similar strategy

I think sending \$3 (60%) makes sense:

- It matches what Aster did, showing I'm willing to extend similar trust
- It creates \$9 for Dorian
- If Dorian returns fairly (around \$5-\$6), we both benefit
- It establishes me as a cooperative player

This seems like a reasonable opening move as a sender.

usage:

```
{
  "input_tokens": 899,
  "output_tokens": 478,
  "reasoning_tokens": 320
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$10.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern so far:

- Round 1: Aster sent me \$3 (60%), I returned \$5 - both benefited
- Round 2: I sent Dorian \$3 (60%), they returned \$4.5 - both benefited

This cooperative approach has been working well. Now with a new opponent (Hana), I want to establish a similar cooperative dynamic from the start.

I'll send \$3 (60% of my endowment) to signal trust and willingness to cooperate.

```
{'send': 3.0}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Aster sent \$3.0 (60%)
- I received \$9.0
- I returned \$5.0
- My payoff: \$4.0

Round 2: I was SENDER

- I sent \$3.0 (60%) to Dorian
- Dorian received \$9.0
- Dorian returned \$4.5 (50% of what they received)
- My payoff: \$5 - \$3 + \$4.5 = \$6.5

Now Round 3: I'm SENDER again against Hana (new opponent)

- I have \$5

Looking at the pattern:

- When Aster sent me 60%, I returned 55.6% of what I received (\$5 out of \$9)
- When I sent Dorian 60%, they returned 50% of what they received (\$4.5 out of \$9)

Both exchanges were reasonably cooperative. I'm now playing with a new opponent (Hana) for the first time.

I should continue the cooperative strategy. Sending \$3 (60%) has worked well so far:

- It shows trust
- It has led to reasonable returns
- It establishes a cooperative dynamic

I'll send \$3 again to maintain consistency and signal cooperation.

usage:

```
{
  "input_tokens": 1186,
  "output_tokens": 446,
  "reasoning_tokens": 247
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Cyra sent \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Cyra currently has \$2 remaining

This is identical to Round 1 where Aster sent me \$3. I returned \$5 then, which worked well

for both of us. I should maintain consistency.

If I return \$5:

- Cyra gets: $\$2 + \$5 = \$7$ total (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit from cooperation

{'return': 5.0}

reasoning:

Let me analyze this situation:

- Round 4, I'm the RECEIVER
- Cyra sent me \$3.0 (60% of their endowment)
- I receive \$9.0 after the multiplier
- Cyra has \$2 remaining from their original \$5

Looking at my history:

- Round 1: Aster sent me \$3, I returned \$5 (they got \$7 total, I kept \$4)
- Round 2: I sent Dorian \$3, they returned \$4.5 (I got \$6.5 total, they kept \$4.5)
- Round 3: I sent Hana \$3, they returned \$4.5 (I got \$6.5 total, they kept \$4.5)

The pattern that's emerged:

- When I'm receiver of \$3 ? \$9, I returned \$5 (keeping \$4)
- When I'm sender of \$3, receivers have returned \$4.5 (keeping \$4.5)

Cyra sent \$3, just like Aster did in Round 1. I should maintain consistency and return \$5, which:

- Gives Cyra \$7 total (net gain of \$2)
- I keep \$4 (net gain of \$4)
- Both benefit from cooperation
- Maintains the cooperative dynamic

This is fair and consistent with my previous behavior as receiver.

usage:

```
{
  "input_tokens": 1473,
  "output_tokens": 491,
  "reasoning_tokens": 245
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$21.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- Elara sent me \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Elara currently has \$2 remaining

This is identical to Rounds 1 and 4 where my opponents sent me \$3. I returned \$5 both times, creating mutual benefit and maintaining a consistent cooperative strategy.

If I return \$5:

- Elara gets: $\$2 + \$5 = \$7$ total (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit from the exchange

{'return': 5.0}

Reasoning:

Looking at the situation:

- Elara sent me \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Elara currently has \$2 remaining

This is the same situation as rounds 1 and 4:

- Round 1: Aster sent \$3, I returned \$5
- Round 4: Cyra sent \$3, I returned \$5

Both worked well and established cooperation. I should be consistent with my strategy.

If I return \$5:

- Elara gets: $\$2 + \$5 = \$7$ total (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit from cooperation

Consistency is important. I'll return \$5.

```
Usage: {
  "input_tokens": 1501,
  "output_tokens": 358,
  "reasoning_tokens": 150
}
```

GAME - Agent_2 - Sender - dyad_4

Timestamp: 2026-06-04T18:45:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_6"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- In Round 1, Hana sent me 60% (\$3), which tripled to \$9, and I returned \$4.5
- This established a cooperative pattern
- Now I'm the sender with Cyra, whom I haven't played with before

Since cooperation worked well with Hana, I'll start with a similar level of trust with Cyra. Sending \$3 (60% of my endowment) demonstrates willingness to cooperate while not being overly risky in this early stage.

If Cyra reciprocates fairly (returning around 50% of what they receive), we'll both benefit from the cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm playing as the SENDER against Cyra (Agent_3).

Let me think about this strategically:

1. In the previous round, I was the receiver and Hana sent me \$3.0 (60% of their endowment). I returned \$4.5, which was 50% of what I received. This seemed like a fair reciprocal move.
2. Now I'm the sender. I need to decide how much to send to Cyra.

3. This is an iterated game with multiple rounds, so building trust and cooperation is valuable.
4. If I send money, it triples, so:
 - If I send \$3: Cyra receives \$9
 - If I send \$5: Cyra receives \$15
 - If I send \$2: Cyra receives \$6
5. The question is: how much will Cyra return? I don't know Cyra's behavior yet, but in iterated games, starting with cooperation often leads to better long-term outcomes.
6. Hana sent 60% (\$3). That worked out well - they showed trust and I reciprocated fairly.
7. I could:
 - Send \$3 (60%, matching what Hana did)
 - Send \$2.5 (50%)
 - Send \$5 (100%, maximum trust)
 - Send less (being more cautious)

Given that:

- This is early in the game (round 2)
- I want to establish a cooperative relationship
- Hana's 60% send worked well
- I don't know Cyra yet, but starting with moderate trust is reasonable

I'll send \$3, matching the 60% that Hana sent to me. This shows trust while not being overly risky.

usage:

```
{
  "input_tokens": 891,
  "output_tokens": 575,
  "reasoning_tokens": 350
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history:

Round 1 (RECEIVER vs Hana): Hana sent 60%, I returned \$4.5 - both profited
Round 2 (SENDER vs Cyra): I sent 60%, Cyra returned \$4.5 - both profited

The cooperative strategy of sending 60% has worked consistently, establishing trust and mutual benefit. I'll continue this approach with Aster, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my history:

Round 1: RECEIVER vs Hana
- Hana sent \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2: SENDER vs Cyra
- I sent \$3.0 (60%)

- Cyra received \$9.0
- Cyra returned \$4.5
- My payoff: \$6.5

Total earnings: \$11.0

Both interactions showed good reciprocity - when someone sent 60%, the receiver returned 50% of what they received. This has worked well, creating mutual benefit.

Now I'm facing Aster for the first time. Since the cooperative strategy has been successful in both previous rounds (both when I was receiver and sender), I should continue with this approach. Sending \$3 (60%) seems to be establishing a good norm of trust and cooperation.

If Aster follows the pattern, they'll receive \$9 and might return around \$4-5, giving me a payoff of \$6-7, which is good for both of us.

usage:

```
{
  "input_tokens": 1197,
  "output_tokens": 379,
  "reasoning_tokens": 224
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

Round history:

- Round 1: Hana sent me \$3, I returned \$4.5
- Round 2: I sent Cyra \$3, they returned \$4.5
- Round 3: I sent Aster \$3, they returned \$5.0
- Round 4: Dorian sent me \$3

Dorian has shown the same level of trust (60% sent) that has been working well throughout.
To maintain consistency and fairness, I'll return \$4.5, which rewards their trust and ensures both of us profit from the cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze the situation:

Round history:

- Round 1 (RECEIVER vs Hana): Hana sent \$3, I received \$9, returned \$4.5. My payoff: \$4.5
- Round 2 (SENDER vs Cyra): I sent \$3, Cyra received \$9, returned \$4.5. My payoff: \$6.5
- Round 3 (SENDER vs Aster): I sent \$3, Aster received \$9, returned \$5.0. My payoff: \$7.0
- Round 4 (RECEIVER vs Dorian): Dorian sent \$3, I received \$9, need to decide return

The pattern has been:

- When I'm receiver and someone sends \$3 (60%), I've returned \$4.5 (50% of what I received)
- When I'm sender and send \$3 (60%), others have returned \$4.5 (Cyra) and \$5.0 (Aster)

Dorian has shown trust by sending \$3 (60% of their endowment), just like Hana did in Round 1 and like I've been doing as sender.

For consistency and to maintain the cooperative norm, I should return around \$4.5-\$5.0. Given that Aster was slightly more generous (returned \$5 when I sent \$3), and I want to maintain good reciprocal relationships, I'll return \$4.5, which is fair and consistent with my first round behavior.

Actually, let me reconsider. Aster returned \$5 when I sent \$3. That was quite generous. Should I be equally generous to Dorian?

Let me think about the fairness:

- Dorian sent \$3, keeps \$2
- I receive \$9
- If I return \$4.5: Dorian ends with \$6.5, I end with \$4.5
- If I return \$5: Dorian ends with \$7, I end with \$4

- Aster's generosity suggests I could reciprocate with \$5, but I need to weigh that against maintaining consistency with how I treated Hana. Since I returned \$4.5 in the first round under identical circumstances, I should stick with that amount to be fair and predictable.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 723,
  "reasoning_tokens": 423
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5

The consistent pattern of sending \$3 (60%) has created successful cooperative relationships with everyone I've interacted with. This strategy has proven effective across different opponents. I'll continue this approach with Finn.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history:

Round 1 (RECEIVER vs Hana): Hana sent \$3, I returned \$4.5 ? I got \$4.5
Round 2 (SENDER vs Cyra): I sent \$3, Cyra returned \$4.5 ? I got \$6.5
Round 3 (SENDER vs Aster): I sent \$3, Aster returned \$5.0 ? I got \$7.0
Round 4 (RECEIVER vs Dorian): Dorian sent \$3, I returned \$4.5 ? I got \$4.5

Total: \$22.5

The strategy of sending \$3 (60%) has been very successful:

- Cyra returned \$4.5
- Aster returned \$5.0

When I've been the receiver and others sent \$3:

- Hana sent \$3, I returned \$4.5
- Dorian sent \$3, I returned \$4.5

The \$3 send amount has established a strong cooperative norm. I should continue with Finn.

```
Usage: {  
  "input_tokens": 1483,  
  "output_tokens": 408,  
  "reasoning_tokens": 174  
}
```

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-06-04T18:45:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_6"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- In Round 1, Dorian sent me \$3.0 (60%) and I returned \$4.5, establishing a cooperative pattern
- This is my first interaction with Hana
- I want to signal trust and establish cooperation for potential future rounds

Following the cooperative pattern from Round 1, I'll send a similar proportion to build trust with this new opponent.

{'send': 3.0}

reasoning:

Let me think about this situation:

Round 2, I'm the SENDER against Hana (Agent_8).

In Round 1:

- I was the receiver against Dorian
- Dorian sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5 (50% return rate)
- My payoff was \$4.5

Now I'm the sender with a new opponent, Hana. This is the first interaction with Hana.

Strategy considerations:

1. In the previous round, Dorian showed significant trust by sending 60%
2. I reciprocated fairly by returning 50% of what I received
3. This is a new opponent, so I need to establish a cooperative relationship
4. Since this is an iterated game with multiple rounds, cooperation tends to be beneficial
5. I should signal trustworthiness to encourage reciprocation

Given that Dorian sent 60% in our first interaction and that worked well (we both benefited), I should consider sending a similar amount to Hana to establish trust.

If I send \$3.0:

- Hana receives \$9.0
- I keep \$2.0
- If Hana returns 50% (like I did), they'd return \$4.5
- My final payoff: \$2.0 + \$4.5 = \$6.5
- Hana's payoff: \$9.0 - \$4.5 = \$4.5

This mirrors what happened in Round 1 and establishes a cooperative pattern.

Alternatively, I could send a bit less to be more cautious with a new opponent, but since the first-round interaction with Dorian went well, I'll maintain the cooperative approach.

I'll send \$3.0.

usage:

```
{
  "input_tokens": 917,
  "output_tokens": 520,
  "reasoning_tokens": 363
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my game history:
- Round 1: As receiver, Dorian sent 60% and I reciprocated fairly
- Round 2: As sender, I sent 60% to Hana and received fair reciprocation

Both interactions resulted in positive, cooperative outcomes. This is my first interaction with Cyra, so I'll maintain the same cooperative strategy that has worked well so far.

{'send': 3.0}

reasoning:

Let me analyze the situation:

Round 1 (vs Dorian): I was RECEIVER

- Dorian sent \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2 (vs Hana): I was SENDER

- I sent \$3.0 (60%)
- Hana received \$9.0
- Hana returned \$4.5
- My payoff: \$6.5

Total earnings: \$11.0

Now Round 3 (vs Cyra): I am SENDER again

- I have \$5.0

So far, I've established a cooperative pattern:

- When I was receiver, Dorian sent 60% and I returned 50% of what I received
- When I was sender, I sent 60% and Hana returned 50% of what they received

Both interactions resulted in fair outcomes and mutual benefit:

- As receiver: I ended with \$4.5, Dorian ended with \$6.5
- As sender: I ended with \$6.5, Hana ended with \$4.5

This cooperative strategy seems to be working well. I'm now facing a new opponent (Cyra).

I should continue with the same cooperative approach to signal trust and encourage reciprocity.

Sending \$3.0 (60%) has worked well in establishing cooperation. I'll continue with this strategy.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 436,
  "reasoning_tokens": 272
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Aster sent \$3.0 (60% of their endowment), showing trust
- This is consistent with the cooperative pattern I've experienced
- In all previous rounds, when \$3.0 was sent, \$4.5 was returned

I'll maintain the same reciprocity that has been working well across all interactions.

{'return': 4.5}

reasoning:

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was multiplied to \$9.0 for me
- Aster kept \$2.0 from their original \$5.0

This is consistent with the pattern I've been seeing and using myself:

- Round 1: Dorian sent \$3.0 (60%), I returned \$4.5
- Round 2: I sent \$3.0 (60%) to Hana, they returned \$4.5
- Round 3: I sent \$3.0 (60%) to Cyra, they returned \$4.5

The consistent pattern has been sending \$3.0 and returning \$4.5. This represents a 50% return rate on what is received and has been working well to maintain cooperation.

I should maintain this same reciprocity pattern with Aster.

If I return \$4.5:

- Aster ends with: \$2.0 + \$4.5 = \$6.5
- I end with: \$9.0 - \$4.5 = \$4.5

This is fair and consistent with what has been working.

usage:

```
{
  "input_tokens": 1413,
  "output_tokens": 376,
  "reasoning_tokens": 213
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing trust
- This is the same cooperative pattern that has been consistent across all rounds
- The established reciprocity norm has been returning \$4.5 when \$3.0 is sent

I'll maintain this cooperative pattern.

```
{'return': 4.5}
```

Reasoning:

Looking at the pattern:

- Briar sent \$3.0 (60% of their endowment)
- I received \$9.0
- The consistent pattern has been to return \$4.5 when receiving \$3.0
- This has worked well in all previous rounds with all agents

I should continue this fair reciprocity pattern.

If I return \$4.5:

- Briar gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This is the same fair split that has been consistent across all interactions.

Usage: {

```
"input_tokens": 1353,
"output_tokens": 241,
"reasoning_tokens": 116
}
```

=====

ROUND 6

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

```
dyad_1: Agent_4 (Dorian)=Sender, Agent_3 (Cyra)=Receiver
dyad_2: Agent_7 (Galen)=Sender, Agent_2 (Briar)=Receiver
dyad_3: Agent_1 (Aster)=Sender, Agent_8 (Hana)=Receiver
dyad_4: Agent_6 (Finn)=Sender, Agent_5 (Elara)=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_4 (Dorian) sent $3; received $9; Agent_3 (Cyra) returned $4.5; payoffs Agent_4
(Dorian)=$6.5, Agent_3 (Cyra)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_7 (Galen) sent $3; received $9; Agent_2 (Briar) returned $4.5; payoffs Agent_7
(Galen)=$6.5, Agent_2 (Briar)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_1 (Aster) sent $3; received $9; Agent_8 (Hana) returned $4.5; payoffs Agent_1
(Aster)=$6.5, Agent_8 (Hana)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_6 (Finn) sent $3; received $9; Agent_5 (Elara) returned $4.5; payoffs Agent_6
(Finn)=$6.5, Agent_5 (Elara)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

Balances after round: {

```
"Agent_1": 32.0,
"Agent_2": 33.5,
"Agent_3": 34.0,
"Agent_4": 33.0,
"Agent_5": 34.0,
"Agent_6": 33.0,
"Agent_7": 31.5,
"Agent_8": 33.0
}
```

LLM CALLS

GAME - Agent_4 - Sender - dyad_1

Timestamp: 2026-06-04T18:45:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Elara sent me \$3, which is the same amount that others have been sending. I've been following a consistent strategy of returning half of what I receive, which seems to establish trust and fairness in this iterated game.

```
'''json
{"return": 4.5}
'''
```

reasoning:

In the previous rounds:

- Round 1: I was SENDER vs Finn. I sent \$3, Finn received \$9, returned \$4.5. I got \$6.5.
- Round 2: I was RECEIVER vs Galen. Galen sent \$3, I received \$9, returned \$4.5. I got \$4.5.

Now Round 3: I'm RECEIVER vs Elara. Elara sent \$3, I receive \$9.

I've been following a strategy of returning half of what I receive when someone sends \$3, which seems fair and reciprocal. This has been working well - both Finn and I adopted this approach.

Elara sent me \$3, just like the others. I should maintain consistency and return \$4.5.

This builds a reputation of being fair and cooperative, which should encourage others to continue trusting me in future rounds.

usage:

```
{
  "input_tokens": 1071,
  "output_tokens": 275,
  "reasoning_tokens": 178
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history, the \$3 send amount with 50% return strategy has been working well and establishing cooperative norms. I'll continue this approach with Briar.

```
'''json
{"send": 3}
'''
```

'''

reasoning:

Let me review my history:

- Round 1: I was sender against Finn, sent \$3, received \$4.5 back, payoff \$6.5
- Round 2: I was receiver against Galen, they sent \$3, I received \$9, returned \$4.5, payoff \$4.5
- Round 3: I was receiver against Elara, they sent \$3, I received \$9, returned \$4.5, payoff \$4.5
- Total: \$15.5

So far, there's been a consistent pattern:

- When I sent \$3, I got \$4.5 back (fair return of half)
- When others sent me \$3, I returned \$4.5 (fair return of half)

This cooperative strategy of sending \$3 and returning half seems to be working well and establishing a norm of fair reciprocity. I'll continue with this strategy when facing a new opponent (Briar).

usage:

```
{
  "input_tokens": 1277,
  "output_tokens": 275,
  "reasoning_tokens": 177
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Hana sent me \$3, following the same cooperative pattern that has been established. I'll continue returning half of what I receive, which maintains trust and fairness.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Hana sent me \$3, which is consistent with the pattern I've seen. Everyone seems to be sending \$3, and I've been returning \$4.5 (half of the \$9 received). This has been working well and maintaining cooperative relationships. I should continue with this strategy.

My payoff when returning \$4.5 is \$4.5, and the sender gets \$6.5. This is mutually beneficial and fair.

usage:

```
{
  "input_tokens": 1294,
  "output_tokens": 156,
  "reasoning_tokens": 93
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

The cooperative pattern of sending \$3 has been working consistently across all my interactions . I'll maintain this approach with Cyra.

```
'''json
{"send": 3}
'''
```

Reasoning:

I've been consistently sending \$3 as a sender and returning \$4.5 (half of what I receive) as a receiver. This strategy has been working well:

- Round 4: Sent \$3 to Briar, they returned \$4.5. My payoff: \$6.5
- Round 3: Received \$9 from Elara (they sent \$3), returned \$4.5. My payoff: \$4.5
- Round 5: Received \$9 from Hana (they sent \$3), returned \$4.5. My payoff: \$4.5

Everyone seems to be cooperating with the \$3 send, \$4.5 return pattern. I should continue this with Cyra.

Sending \$3:

- I keep \$2
- Cyra receives \$9
- If they return \$4.5, I end up with \$2 + \$4.5 = \$6.5
- Cyra keeps \$4.5

This is a good cooperative equilibrium.

```
Usage: {
  "input_tokens": 1235,
  "output_tokens": 285,
  "reasoning_tokens": 169
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-06-04T18:45:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role

of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this round:
- Finn sent me \$3.0, showing trust and cooperation (same as Briar did)
- I received \$9.0

I've established a consistent pattern: when someone sends \$3.0, I return \$4.5, which creates a fair split where both parties benefit. I'll maintain this consistency with Finn.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze the situation:

Round 3:

- Finn sent me \$3.0 (just like Briar did in Round 2)
- I received \$9.0 (after tripling)
- I need to decide how much to return

Looking at my history:

- Round 1: I sent \$3.0 to Elara, she returned \$4.5 (fair 50-50 split of the \$9.0)
- Round 2: Briar sent \$3.0 to me, I returned \$4.5 (consistent with what Elara did)

Now Finn has also sent \$3.0, showing the same level of trust and cooperation. To maintain consistency and fairness, I should return the same amount I returned to Briar and that Elara returned to me.

This establishes a clear cooperative pattern: when someone sends \$3.0, they get back \$4.5, which means:

- Sender keeps: $\$5 - \$3 + \$4.5 = \6.5
- Receiver keeps: $\$9 - \$4.5 = \$4.5$

This is a fair split where both parties benefit from the cooperation. I should continue this pattern.

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 382,
  "reasoning_tokens": 222
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern so far:

- When I sent \$3 to Elara (Round 1), she returned \$4.5 - a fair outcome
- When others sent \$3 to me, I returned \$4.5 to maintain fairness
- This cooperative \$3 send / \$4.5 return pattern has been working well

I'll continue this strategy with Galen to maintain consistency and encourage cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- I'm now the sender against Galen
- In previous rounds, I've established a pattern of cooperation
- When I was sender (Round 1 vs Elara): I sent \$3, she returned \$4.5 - good outcome
- When I was receiver (Rounds 2-3 vs Briar and Finn): They sent \$3, I returned \$4.5 each time

The pattern seems to be working well. Sending \$3 has been the consistent amount, and returning \$4.5 on \$9 received has been fair and reciprocated well.

Since this is an iterated game and I want to maintain cooperative relationships, I should continue with the same strategy. Sending \$3 has worked well so far - it signals cooperation, creates value through the 3x multiplier, and I've seen good returns.

I'll send \$3 to Galen to maintain consistency and signal cooperation.

usage:

```
{
  "input_tokens": 1310,
  "output_tokens": 329,
  "reasoning_tokens": 204
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant**content:**

Looking at my track record:

- When I sent \$3, I received good returns (\$4.5 and \$5.0)
- The cooperative pattern of sending \$3 has been working well
- My average payoff is above the baseline, showing this strategy is effective

I'll continue with this proven cooperative approach with Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1 (Sender vs Elara): Sent \$3, received \$4.5 back, payoff \$6.5
- Round 2 (Receiver vs Briar): Received \$9 (from \$3), returned \$4.5, payoff \$4.5
- Round 3 (Receiver vs Finn): Received \$9 (from \$3), returned \$4.5, payoff \$4.5
- Round 4 (Sender vs Galen): Sent \$3, received \$5.0 back, payoff \$7.0
- Round 5 (Sender vs Aster): Now deciding

The pattern has been working well:

- When I send \$3, I get good returns (\$4.5 from Elara, \$5.0 from Galen)
- When I receive from \$3 sends, I return \$4.5
- Galen was even more generous, returning \$5.0

Total earnings are \$22.5 across 4 rounds, averaging \$5.625 per round, which is above the baseline of \$5.

I should continue with the cooperative strategy of sending \$3 to Aster. This has proven to be mutually beneficial and establishes trust.

usage:

```
{
  "input_tokens": 1383,
  "output_tokens": 390,
  "reasoning_tokens": 213
}
```

Message 8 role=user**content:**

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Dorian sent me \$3.0, showing trust
- I received \$9.0

I've maintained a consistent pattern of returning \$4.5 when someone sends \$3.0 to me. This creates a fair outcome where both parties benefit from cooperation. I'll continue this consistent approach with Dorian.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

- Dorian sent me \$3.0, which became \$9.0
- I need to decide how much to return

Looking at my history:

- When I've been the receiver and someone sent \$3.0, I've returned \$4.5 consistently
- This has worked well and maintained fairness
- When I've been the sender and sent \$3.0, I received \$4.5 and \$5.0 back

The pattern of returning \$4.5 when receiving \$3.0 seems fair and consistent:

- Dorian keeps: \$5 - \$3 + \$4.5 = \$6.5
- I keep: \$9 - \$4.5 = \$4.5

This is a reasonable split. I should maintain consistency with my previous behavior.

```
Usage: {
  "input_tokens": 1358,
  "output_tokens": 286,
  "reasoning_tokens": 153
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-06-04T18:45:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_7": "Galen"
    },
  },
  "agents": [
    "Agent_7",
    "Agent_2"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_2": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$10.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern so far:

- Round 1: Aster sent me \$3 (60%), I returned \$5 - both benefited
- Round 2: I sent Dorian \$3 (60%), they returned \$4.5 - both benefited

This cooperative approach has been working well. Now with a new opponent (Hana), I want to establish a similar cooperative dynamic from the start.

I'll send \$3 (60% of my endowment) to signal trust and willingness to cooperate.

{'send': 3.0}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Aster sent \$3.0 (60%)
- I received \$9.0
- I returned \$5.0
- My payoff: \$4.0

Round 2: I was SENDER

- I sent \$3.0 (60%) to Dorian
- Dorian received \$9.0
- Dorian returned \$4.5 (50% of what they received)

- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Now Round 3: I'm SENDER again against Hana (new opponent)

- I have \$5

Looking at the pattern:

- When Aster sent me 60%, I returned 55.6% of what I received (\$5 out of \$9)
- When I sent Dorian 60%, they returned 50% of what they received (\$4.5 out of \$9)

Both exchanges were reasonably cooperative. I'm now playing with a new opponent (Hana) for the first time.

I should continue the cooperative strategy. Sending \$3 (60%) has worked well so far:

- It shows trust
- It has led to reasonable returns
- It establishes a cooperative dynamic

I'll send \$3 again to maintain consistency and signal cooperation.

usage:

```
{
  "input_tokens": 1186,
  "output_tokens": 446,
  "reasoning_tokens": 247
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Cyra sent \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Cyra currently has \$2 remaining

This is identical to Round 1 where Aster sent me \$3. I returned \$5 then, which worked well for both of us. I should maintain consistency.

If I return \$5:

- Cyra gets: $\$2 + \$5 = \$7$ total (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit from cooperation

{'return': 5.0}

reasoning:

Let me analyze this situation:

- Round 4, I'm the RECEIVER
- Cyra sent me \$3.0 (60% of their endowment)
- I receive \$9.0 after the multiplier
- Cyra has \$2 remaining from their original \$5

Looking at my history:

- Round 1: Aster sent me \$3, I returned \$5 (they got \$7 total, I kept \$4)
- Round 2: I sent Dorian \$3, they returned \$4.5 (I got \$6.5 total, they kept \$4.5)
- Round 3: I sent Hana \$3, they returned \$4.5 (I got \$6.5 total, they kept \$4.5)

The pattern that's emerged:

- When I'm receiver of \$3 ? \$9, I returned \$5 (keeping \$4)

- When I'm sender of \$3, receivers have returned \$4.5 (keeping \$4.5)

Cyra sent \$3, just like Aster did in Round 1. I should maintain consistency and return \$5, which:

- Gives Cyra \$7 total (net gain of \$2)
- I keep \$4 (net gain of \$4)
- Both benefit from cooperation
- Maintains the cooperative dynamic

This is fair and consistent with my previous behavior as receiver.

usage:

```
{
  "input_tokens": 1473,
  "output_tokens": 491,
  "reasoning_tokens": 245
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$21.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- Elara sent me \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Elara currently has \$2 remaining

This is identical to Rounds 1 and 4 where my opponents sent me \$3. I returned \$5 both time s, creating mutual benefit and maintaining a consistent cooperative strategy.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Both benefit from the exchange

{'return': 5.0}

reasoning:

Looking at the situation:

- Elara sent me \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Elara currently has \$2 remaining

This is the same situation as rounds 1 and 4:

- Round 1: Aster sent \$3, I returned \$5
- Round 4: Cyra sent \$3, I returned \$5

Both worked well and established cooperation. I should be consistent with my strategy.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Both benefit from cooperation

Consistency is important. I'll return \$5.

usage:

```
{
  "input_tokens": 1501,
  "output_tokens": 358,
  "reasoning_tokens": 150
}
```

```

    }
Message 8 role=user
  content:
    Your name in this experiment is Galen.
    Your opponent this round is Briar (Agent_2).

    Round 6

    In your most recent previous game round, you were the RECEIVER against Elara. They sent $3
    .0 to you, it became $9.0, and you returned $5.0. Your payoff was $4.0.
    Your total earnings across all rounds are $25.0.

    This round, you are the SENDER against Briar. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at my track record:

  **As SENDER:**
  - Sent $3 to Dorian ? received $4.5 back (payoff: $6.5)
  - Sent $3 to Hana ? received $4.5 back (payoff: $6.5)

  **As RECEIVER:**
  - Received $9 from Aster ($3 sent) ? returned $5
  - Received $9 from Cyra ($3 sent) ? returned $5
  - Received $9 from Elara ($3 sent) ? returned $5

  The $3 send strategy has worked consistently well, establishing trust and generating mutual be-
  nefit. I'll continue this approach with Briar.

  {'send': 3.0}
Reasoning:
  Looking at my history:
  - Round 1: RECEIVER - Aster sent $3, I returned $5
  - Round 2: SENDER - I sent $3 to Dorian, they returned $4.5
  - Round 3: SENDER - I sent $3 to Hana, they returned $4.5
  - Round 4: RECEIVER - Cyra sent $3, I returned $5
  - Round 5: RECEIVER - Elara sent $3, I returned $5

  The pattern is clear:
  - When I'm a sender, I send $3 (60%)
  - When I'm a receiver and they send $3, I return $5

  This has been working very well. Everyone who has sent $3 has gotten good returns, and when I
  send $3, I get good returns back.

  Now I'm playing against Briar for the first time as sender. I should continue with my consis-
  tent strategy of sending $3.

  Total earnings so far: $25.0

  My strategy has been:
  - Send $3 as sender (establishing trust)
  - Return $5 when receiving $9 from a $3 send (rewarding trust)

  I'll continue with this successful pattern.
  Usage: {
    "input_tokens": 1521,
    "output_tokens": 457,
    "reasoning_tokens": 230
  }

```

GAME - Agent_2 - Receiver - dyad_2

```

Timestamp: 2026-06-04T18:45:51
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {

```

```

"agent_names": {
  "Agent_2": "Briar",
  "Agent_7": "Galen"
},
"agents": [
  "Agent_7",
  "Agent_2"
],
"investor": "Agent_7",
"roles": {
  "Agent_2": "trustee",
  "Agent_7": "investor"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history:

Round 1 (RECEIVER vs Hana): Hana sent 60%, I returned \$4.5 - both profited

Round 2 (SENDER vs Cyra): I sent 60%, Cyra returned \$4.5 - both profited

The cooperative strategy of sending 60% has worked consistently, establishing trust and mutual benefit. I'll continue this approach with Aster, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my history:

Round 1: RECEIVER vs Hana

- Hana sent \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2: SENDER vs Cyra

- I sent \$3.0 (60%)
- Cyra received \$9.0
- Cyra returned \$4.5
- My payoff: \$6.5

Total earnings: \$11.0

Both interactions showed good reciprocity - when someone sent 60%, the receiver returned 50% of what they received. This has worked well, creating mutual benefit.

Now I'm facing Aster for the first time. Since the cooperative strategy has been successful in both previous rounds (both when I was receiver and sender), I should continue with this approach. Sending \$3 (60%) seems to be establishing a good norm of trust and cooperation.

If Aster follows the pattern, they'll receive \$9 and might return around \$4-5, giving me a payoff of \$6-7, which is good for both of us.

usage:

```
{
  "input_tokens": 1197,
  "output_tokens": 379,
  "reasoning_tokens": 224
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

Round history:

- Round 1: Hana sent me \$3, I returned \$4.5
- Round 2: I sent Cyra \$3, they returned \$4.5
- Round 3: I sent Aster \$3, they returned \$5.0
- Round 4: Dorian sent me \$3

Dorian has shown the same level of trust (60% sent) that has been working well throughout.

To maintain consistency and fairness, I'll return \$4.5, which rewards their trust and ensures both of us profit from the cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze the situation:

Round history:

- Round 1 (RECEIVER vs Hana): Hana sent \$3, I received \$9, returned \$4.5. My payoff: \$4.5
- Round 2 (SENDER vs Cyra): I sent \$3, Cyra received \$9, returned \$4.5. My payoff: \$6.5
- Round 3 (SENDER vs Aster): I sent \$3, Aster received \$9, returned \$5.0. My payoff: \$7.0
- Round 4 (RECEIVER vs Dorian): Dorian sent \$3, I received \$9, need to decide return

The pattern has been:

- When I'm receiver and someone sends \$3 (60%), I've returned \$4.5 (50% of what I received)
- When I'm sender and send \$3 (60%), others have returned \$4.5 (Cyra) and \$5.0 (Aster)

Dorian has shown trust by sending \$3 (60% of their endowment), just like Hana did in Round 1 and like I've been doing as sender.

For consistency and to maintain the cooperative norm, I should return around \$4.5-\$5.0. Given that Aster was slightly more generous (returned \$5 when I sent \$3), and I want to maintain good reciprocal relationships, I'll return \$4.5, which is fair and consistent with my first round behavior.

Actually, let me reconsider. Aster returned \$5 when I sent \$3. That was quite generous. Should I be equally generous to Dorian?

Let me think about the fairness:

- Dorian sent \$3, keeps \$2
- I receive \$9
- If I return \$4.5: Dorian ends with \$6.5, I end with \$4.5
- If I return \$5: Dorian ends with \$7, I end with \$4

- Aster's generosity suggests I could reciprocate with \$5, but I need to weigh that against maintaining consistency with how I treated Hana. Since I returned \$4.5 in the first round under identical circumstances, I should stick with that amount to be fair and predictable.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 723,
  "reasoning_tokens": 423
}
```

Message 6 role=user**content:**

Your name in this experiment is Briar.

Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5

The consistent pattern of sending \$3 (60%) has created successful cooperative relationships with everyone I've interacted with. This strategy has proven effective across different opponents. I'll continue this approach with Finn.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

Round 1 (RECEIVER vs Hana): Hana sent \$3, I returned \$4.5 ? I got \$4.5
Round 2 (SENDER vs Cyra): I sent \$3, Cyra returned \$4.5 ? I got \$6.5
Round 3 (SENDER vs Aster): I sent \$3, Aster returned \$5.0 ? I got \$7.0
Round 4 (RECEIVER vs Dorian): Dorian sent \$3, I returned \$4.5 ? I got \$4.5

Total: \$22.5

The strategy of sending \$3 (60%) has been very successful:

- Cyra returned \$4.5
- Aster returned \$5.0

When I've been the receiver and others sent \$3:

- Hana sent \$3, I returned \$4.5
- Dorian sent \$3, I returned \$4.5

The \$3 send amount has established a strong cooperative norm. I should continue with Finn.

usage:

```
{
  "input_tokens": 1483,
  "output_tokens": 408,
  "reasoning_tokens": 174
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

Round history as RECEIVER:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3

Galen has shown the same cooperative level of trust (60%) that has been consistent throughout this experiment. I'll maintain my established pattern of returning \$4.5, which ensures both players profit and reinforces the cooperative norm.

```
'''json
{"return": 4.5}
'''
```

'''

Reasoning:

Let me review my history:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5
Round 5 (SENDER): I sent \$3, Finn returned \$4.5
Round 6 (RECEIVER): Galen sent \$3

The pattern is very consistent. When I've been the receiver and someone sends me \$3:

- Round 1: I returned \$4.5
- Round 4: I returned \$4.5

This has established a fair and reciprocal pattern. Galen has shown the same level of trust as others (sending \$3). I should maintain consistency and return \$4.5.

This ensures:

- Galen gets back: $\$5 - \$3 + \$4.5 = \6.5
- I keep: $\$9.0 - \$4.5 = \$4.5$

Both of us profit, and I maintain my reputation for fair reciprocation.

Usage: {

"input_tokens": 1488,
"output_tokens": 417,
"reasoning_tokens": 207

}

GAME - Agent_1 - Sender - dyad_3

Timestamp: 2026-06-04T18:45:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_8",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_1",
 "Agent_8"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"

}

"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this round:
- Briar sent me \$3, showing the same level of trust as Elara did
- I received \$9 after the multiplier
- I returned \$5 to Elara in the same situation, which seemed to work well

For consistency and fairness, I'll return the same amount. This establishes a reliable pattern: when someone sends me \$3, I return \$5, ensuring both players benefit from cooperation.

```
'''json
{"return": 5}
'''
```

reasoning:

Briar sent me \$3, just like Elara did in the previous round. I received \$9 after the multiplier.

In the previous round with Elara, I returned \$5 out of \$9 received. This worked out well and seemed fair - Elara ended with \$7 (a gain of \$2) and I kept \$4 (a gain of \$4).

For consistency and to maintain a reputation as a fair and cooperative player, I should re

turn the same amount to Briar. This establishes a pattern that when someone sends me \$3, I return \$5, which is a reasonable and generous split that benefits both parties.

Returning \$5 to Briar means:

- Briar will have $\$5 - \$3 + \$5 = \7 (net gain of \$2)
- I'll keep $\$9 - \$5 = \$4$ (net gain of \$4)

This is fair and encourages future cooperation from all players.

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 341,
  "reasoning_tokens": 188
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$15.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

- When I sent \$3 to Galen in Round 1, they generously returned \$5
- When others sent me \$3 in Rounds 2-3, I returned \$5 each time
- This \$3-send pattern has created mutual benefits

I'll continue this cooperative approach and send **\$3** to Finn. This gives them \$9 to work with, and based on the pattern of reciprocity I've seen, hopefully they'll return a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me think about this situation:

- I'm now the SENDER with \$5
- My opponent is Finn (Agent_6)
- In Round 1, I sent \$3 to Galen and they returned \$5 (generous return, I ended with \$7)
- In Rounds 2-3, I received \$3 from Elara and Briar, and returned \$5 to each (they ended with \$7, I got \$4)

The pattern I've established:

- When I send \$3, receivers get \$9 and have returned \$5
- When I receive \$3 (which becomes \$9), I return \$5

This \$3-send strategy has worked well. It creates mutual benefit:

- If I send \$3, Finn gets \$9
- If Finn returns around \$5 (similar to what Galen did and what I've been doing), I'd get \$7 total
- Both parties benefit

I should continue with this cooperative strategy of sending \$3.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 364,
  "reasoning_tokens": 191
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.

Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- Cyra sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Elara and Briar)

I'll maintain this consistent, fair approach and return \$5. This establishes me as a reliable cooperater, which should encourage continued trust from all players.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the situation:

- Cyra sent me \$3, which became \$9 for me
- In previous rounds when I received \$3, I returned \$5 (to Elara and Briar)
- When I sent \$3 to Finn, they returned \$4.5

Finn returned slightly less than what I had been returning. But Cyra is a new opponent, and I should maintain my cooperative strategy. Consistency is important for building trust in repeated games.

I'll stick with returning \$5, which has been my standard response when receiving \$3. This maintains my reputation as a fair and reliable player.

Cyra will end up with: $\$5 - \$3 + \$5 = \7 (net gain of \$2)

I'll keep: $\$9 - \$5 = \$4$ (net gain of \$4)

Both benefit, though I benefit more as the receiver, which is fair given the 3x multiplier advantage.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 326,
  "reasoning_tokens": 194
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.

Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$25.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- I've sent \$3 in both previous sender rounds (to Galen and Finn)
- This consistently encourages good returns (Galen: \$5, Finn: \$4.5)
- As a receiver, I've returned \$5 when getting \$3, establishing a reputation for fairness

I'll maintain this cooperative strategy and send ****\$3**** to Hana.

```
'''json
```

```
{"send": 3}
'''
```

Reasoning:

Let me review my history:

Round 1: SENDER vs Galen - Sent \$3, they returned \$5, my payoff was \$7
Round 2: RECEIVER vs Elara - They sent \$3, I returned \$5, my payoff was \$4
Round 3: RECEIVER vs Briar - They sent \$3, I returned \$5, my payoff was \$4
Round 4: SENDER vs Finn - Sent \$3, they returned \$4.5, my payoff was \$6.5
Round 5: RECEIVER vs Cyra - They sent \$3, I returned \$5, my payoff was \$4

Total so far: \$25.5 (which matches)

Now Round 6: SENDER vs Hana

My strategy has been very consistent:

- When sending: I send \$3
- When receiving \$3: I return \$5

This has worked well. Sending \$3 seems to be a good cooperative amount that encourages reciprocity. I'll continue with this strategy.

```
Usage: {
  "input_tokens": 1406,
  "output_tokens": 351,
  "reasoning_tokens": 186
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-06-04T18:46:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_1",
      "Agent_8"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Galen also sent me \$3.0, showing the same level of trust as Finn did in the previous round. I should maintain consistency in my behavior to establish a reputation as a fair and cooperative player.

Returning \$4.5 (half of what I received) worked well with Finn and maintains the cooperative pattern.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Galen sent me \$3.0, which became \$9.0 for me. This is the same amount that Finn sent me in the previous round.

I returned \$4.5 to Finn, which seemed fair - it's half of what I received. This approach worked well and established a pattern of cooperation.

I should maintain consistency with my previous behavior to build a reputation as a fair and cooperative player. Returning \$4.5 again makes sense.

Let me think about the total situation:

- Galen sent \$3.0 (kept \$2.0)

- I received \$9.0
- If I return \$4.5, Galen gets \$2.0 + \$4.5 = \$6.5 total
- I keep \$9.0 - \$4.5 = \$4.5

This is the same fair split I did with Finn, and it's consistent with what Briar did for me. It rewards Galen's trust and maintains my reputation.

usage:

```
{
  "input_tokens": 1078,
  "output_tokens": 321,
  "reasoning_tokens": 190
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern that's emerged:

- When I sent \$3 to Briar, they returned \$4.5
- When Finn sent me \$3, I returned \$4.5
- When Galen sent me \$3, I returned \$4.5

This \$3 send amount with ~\$4.5 return seems to be establishing a cooperative norm that benefits both parties. I'll continue this pattern with Elara.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm now the sender again, playing with a new opponent, Elara.

Looking at my history:

- Round 1: I was sender against Briar, sent \$3, they returned \$4.5. My payoff was \$6.5
- Round 2: I was receiver against Finn, they sent \$3, I returned \$4.5. My payoff was \$4.5
- Round 3: I was receiver against Galen, they sent \$3, I returned \$4.5. My payoff was \$4.5

There's a clear pattern emerging where sending \$3 has worked well. When I sent \$3 to Briar, they returned \$4.5, giving me a good payoff. When others sent me \$3, I returned \$4.5, which seemed fair.

This pattern of sending \$3 seems to be establishing a cooperative norm where:

- Sender sends \$3
- Receiver gets \$9
- Receiver returns \$4.5
- Sender ends with \$6.5 (kept \$2, got back \$4.5)
- Receiver ends with \$4.5 (got \$9, returned \$4.5)

This is a mutually beneficial outcome. Should I continue this pattern with Elara?

Sending \$3 makes sense because:

1. It's worked well so far
2. It shows trust without being overly risky
3. It establishes consistency in my play
4. If Elara follows the pattern others have shown, I'll get a good return

I'll send \$3 to Elara.

usage:

```
{
  "input_tokens": 1306,
  "output_tokens": 480,
}
```

```

    "reasoning_tokens": 298
  }
Message 6 role=user
  content:
    Your name in this experiment is Hana.
    Your opponent this round is Dorian (Agent_4).

    Round 5

    In your most recent previous game round, you were the SENDER against Elara. You sent $3.0,
    it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
    Your total earnings across all rounds are $22.0.

    This round, you are the SENDER against Dorian. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Message 7 role=assistant
  content:
    The cooperative pattern continues to work effectively. Elara followed the same norm, returning $4.5 when I sent $3. This strategy has been consistent across all my interactions so far. I'll maintain this approach with Dorian.

    '''json
    {"send": 3}
    '''
  reasoning:
    The pattern continues to work well:
    - Elara followed the same cooperative pattern, returning $4.5 when I sent $3
    - This gives me a payoff of $6.5 (consistent with my first round)
    - The $3 send with ~$4.5 return strategy is working reliably across different opponents

    It seems like there's a cooperative norm established in this group. I'll continue with $3 to Dorian.
  usage:
    {
      "input_tokens": 1380,
      "output_tokens": 172,
      "reasoning_tokens": 96
    }
Message 8 role=user
  content:
    Your name in this experiment is Hana.
    Your opponent this round is Aster (Agent_1).

    Round 6

    In your most recent previous game round, you were the SENDER against Dorian. You sent $3.0, it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
    Your total earnings across all rounds are $28.5.

    This round, you are the RECEIVER against Aster. They sent you $3.0, so you received $9.0.
    How much do you return? (0-9.0)
    Respond exactly as JSON: {'return': <amount>}
Assistant output:
    The cooperative pattern remains consistent. Aster sent me $3, just like all previous interactions. I'll continue to return $4.5 to maintain this successful equilibrium.

    '''json
    {"return": 4.5}
    '''
  Reasoning:
    The pattern continues perfectly. Everyone has been sending $3 and receiving $4.5 back. This cooperative equilibrium has been consistent across all my interactions:

    - Round 1: I sent $3 to Briar, got $4.5 back (payoff $6.5)
    - Round 2: Finn sent me $3, I returned $4.5 (payoff $4.5)
    - Round 3: Galen sent me $3, I returned $4.5 (payoff $4.5)
    - Round 4: I sent $3 to Elara, got $4.5 back (payoff $6.5)
    - Round 5: I sent $3 to Dorian, got $4.5 back (payoff $6.5)
    - Round 6: Aster sent me $3, I should return $4.5 (payoff $4.5)

```

This has been working very well and everyone seems to be following this norm. I should maintain consistency.

```
Usage: {  
  "input_tokens": 1333,  
  "output_tokens": 296,  
  "reasoning_tokens": 165  
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-06-04T18:46:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agent_names": {  
      "Agent_5": "Elara",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my game history:

- Round 1: As receiver, Dorian sent 60% and I reciprocated fairly
- Round 2: As sender, I sent 60% to Hana and received fair reciprocation

Both interactions resulted in positive, cooperative outcomes. This is my first interaction with Cyra, so I'll maintain the same cooperative strategy that has worked well so far.

{'send': 3.0}

reasoning:

Let me analyze the situation:

Round 1 (vs Dorian): I was RECEIVER

- Dorian sent \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2 (vs Hana): I was SENDER

- I sent \$3.0 (60%)
- Hana received \$9.0
- Hana returned \$4.5
- My payoff: \$6.5

Total earnings: \$11.0

Now Round 3 (vs Cyra): I am SENDER again

- I have \$5.0

So far, I've established a cooperative pattern:

- When I was receiver, Dorian sent 60% and I returned 50% of what I received
- When I was sender, I sent 60% and Hana returned 50% of what they received

Both interactions resulted in fair outcomes and mutual benefit:

- As receiver: I ended with \$4.5, Dorian ended with \$6.5
- As sender: I ended with \$6.5, Hana ended with \$4.5

This cooperative strategy seems to be working well. I'm now facing a new opponent (Cyra). I should continue with the same cooperative approach to signal trust and encourage reciprocity.

Sending \$3.0 (60%) has worked well in establishing cooperation. I'll continue with this strategy.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 436,
  "reasoning_tokens": 272
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Aster sent \$3.0 (60% of their endowment), showing trust
- This is consistent with the cooperative pattern I've experienced
- In all previous rounds, when \$3.0 was sent, \$4.5 was returned

I'll maintain the same reciprocity that has been working well across all interactions.

{'return': 4.5}

reasoning:

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was multiplied to \$9.0 for me
- Aster kept \$2.0 from their original \$5.0

This is consistent with the pattern I've been seeing and using myself:

- Round 1: Dorian sent \$3.0 (60%), I returned \$4.5
- Round 2: I sent \$3.0 (60%) to Hana, they returned \$4.5
- Round 3: I sent \$3.0 (60%) to Cyra, they returned \$4.5

The consistent pattern has been sending \$3.0 and returning \$4.5. This represents a 50% return rate on what is received and has been working well to maintain cooperation.

I should maintain this same reciprocity pattern with Aster.

If I return \$4.5:

- Aster ends with: $\$2.0 + \$4.5 = \$6.5$
- I end with: $\$9.0 - \$4.5 = \$4.5$

This is fair and consistent with what has been working.

usage:

```
{
  "input_tokens": 1413,
  "output_tokens": 376,
  "reasoning_tokens": 213
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing trust
- This is the same cooperative pattern that has been consistent across all rounds
- The established reciprocity norm has been returning \$4.5 when \$3.0 is sent

I'll maintain this cooperative pattern.

{'return': 4.5}

reasoning:

Looking at the pattern:

- Briar sent \$3.0 (60% of their endowment)
- I received \$9.0
- The consistent pattern has been to return \$4.5 when receiving \$3.0
- This has worked well in all previous rounds with all agents

I should continue this fair reciprocity pattern.

If I return \$4.5:

- Briar gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This is the same fair split that has been consistent across all interactions.

usage:

```
{
  "input_tokens": 1353,
  "output_tokens": 241,
  "reasoning_tokens": 116
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my game history:

- Every interaction so far has followed the same pattern: \$3.0 sent, \$4.5 returned
- This cooperative equilibrium has been consistent across all agents
- My total earnings are strong at \$26.5

This is my first interaction with Elara. I'll maintain the cooperative strategy that has worked perfectly so far.

{'send': 3.0}

Reasoning:

Looking at my history:

- Round 1: RECEIVER from Dorian - they sent \$3.0, I returned \$4.5
- Round 2: SENDER to Hana - I sent \$3.0, they returned \$4.5
- Round 3: SENDER to Cyra - I sent \$3.0, they returned \$4.5
- Round 4: RECEIVER from Aster - they sent \$3.0, I returned \$4.5
- Round 5: RECEIVER from Briar - they sent \$3.0, I returned \$4.5

There's a very clear pattern here: everyone is sending \$3.0 and returning \$4.5. This has been completely consistent across all interactions. This is a cooperative equilibrium that everyone seems to be following.

Now I'm playing as SENDER against Elara (Agent_5). I haven't interacted with Elara before. Given the strong cooperative norm that has developed, I should continue with the same strategy: send \$3.0.

This has resulted in good payoffs for me and maintains the cooperative equilibrium.

```
Usage: {  
  "input_tokens": 1341,  
  "output_tokens": 348,  
  "reasoning_tokens": 217  
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T18:46:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agent_names": {  
      "Agent_5": "Elara",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern so far:
- Cyra sent me \$3 and I returned \$4.5
- I sent Aster \$3 and they generously returned \$5

The cooperation strategy is working well! I'll continue with the same approach and send \$3 to Dorian, maintaining consistency and signaling trustworthiness.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:
- Round 1: As RECEIVER from Cyra, they sent \$3, I received \$9, returned \$4.5, my payoff was \$4.5
- Round 2: As SENDER to Aster, I sent \$3, they received \$9, returned \$5, my payoff was \$7.0

Aster was very generous! They returned \$5 out of \$9, which is more than I returned to Cyra. That's a great sign of cooperation.

Now I'm facing Dorian as a sender. Given that:
1. The cooperation strategy has been working well
2. Aster responded very positively to my \$3 send
3. I want to continue building cooperative relationships

I should send a similar amount to Dorian - \$3 seems like a good cooperative signal. This maintains consistency in my strategy.

usage:

```
{
  "input_tokens": 1145,
  "output_tokens": 295,
  "reasoning_tokens": 177
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

- Hana sent \$3.0 (60% of their endowment), the same amount Cyra sent me in Round 1
- I received \$9.0 after multiplication

For consistency with my previous behavior, when Cyra sent me \$3 in Round 1, I returned \$4.

5. I should maintain the same fair and consistent approach with Hana.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Hana sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplication
- This created \$6.0 of new value

Looking at my history:

- Round 1: As receiver from Cyra who sent \$3, I returned \$4.5
- Round 2: As sender to Aster, I sent \$3, they returned \$5
- Round 3: As sender to Dorian, I sent \$3, they returned \$4.5

So when people sent me \$3 (Round 1), I returned \$4.5. I should be consistent in my receiving behavior. Aster was more generous when I sent them \$3 (they returned \$5), but Dorian matched what I did in Round 1 (returned \$4.5).

I want to maintain my reputation as a fair and consistent receiver. I returned \$4.5 when Cyra sent me \$3, so I should return \$4.5 when Hana sends me \$3 as well.

This gives:

- Hana: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9 - \$4.5 = \$4.5

This is fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 424,
  "reasoning_tokens": 235
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.

0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 (60% of my endowment) as sender
- This strategy has led to good returns from my partners
- It signals cooperation and trustworthiness

I'll maintain this consistent approach with Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- Round 1: RECEIVER - Cyra sent \$3, I returned \$4.5, payoff \$4.5
- Round 2: SENDER - I sent \$3 to Aster, they returned \$5, payoff \$7.0
- Round 3: SENDER - I sent \$3 to Dorian, they returned \$4.5, payoff \$6.5
- Round 4: RECEIVER - Hana sent \$3, I returned \$4.5, payoff \$4.5

My pattern has been consistent:

- As SENDER: I send \$3 (60% of endowment)
- As RECEIVER: When receiving \$3 (which becomes \$9), I return \$4.5 (50% of what I received)

This strategy has been working well and builds trust. I should continue with the same approach for Galen.

Total earnings: \$22.5

usage:

```
{
  "input_tokens": 1370,
  "output_tokens": 298,
  "reasoning_tokens": 158
}
```

Message 8 role=user**content:**

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Finn sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra and Hana)
- This maintains fairness and reciprocity

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the situation:

- Finn sent me \$3.0, which became \$9.0 for me
- This is the same amount that others have been sending
- I should maintain my consistent reciprocal strategy

My history as receiver:

- Round 1: Cyra sent \$3, I returned \$4.5
- Round 4: Hana sent \$3, I returned \$4.5

I should continue this consistent pattern of returning \$4.5 when someone sends me \$3. This is fair and builds trust.

Usage: {

```
"input_tokens": 1344,
"output_tokens": 223,
"reasoning_tokens": 109
}
```

ROUND 7

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8

(Hana)=Sender

Pairings:

dyad_1: Agent_8 (Hana)=Sender, Agent_3 (Cyra)=Receiver
dyad_2: Agent_4 (Dorian)=Sender, Agent_1 (Aster)=Receiver
dyad_3: Agent_7 (Galen)=Sender, Agent_6 (Finn)=Receiver
dyad_4: Agent_2 (Briar)=Sender, Agent_5 (Elara)=Receiver

Parsed game decisions:

dyad_1: Agent_8 (Hana) sent \$3; received \$9; Agent_3 (Cyra) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_3 (Cyra)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_1 (Aster) returned \$5; payoffs Agent_4 (Dorian)=\$7, Agent_1 (Aster)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_7 (Galen) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_7 (Galen)=\$6.5, Agent_6 (Finn)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_2 (Briar) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_5 (Elara)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 36.0,
"Agent_2": 40.0,
"Agent_3": 38.5,
"Agent_4": 40.0,
"Agent_5": 38.5,
"Agent_6": 37.5,
"Agent_7": 38.0,
"Agent_8": 39.5

}

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-06-04T18:46:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_3",
"pairing": {
"agent_names": {
"Agent_3": "Cyra",
"Agent_8": "Hana"
},
"agents": [
"Agent_8",
"Agent_3"
],
"investor": "Agent_8",
"roles": {
"Agent_3": "trustee",
"Agent_8": "investor"
},
"trustee": "Agent_3"

},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.00 to you, it became \$9.00, and you returned \$4.50. Your payoff was \$4.50.
Your total earnings across all rounds are \$15.50.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern that's emerged:

- When I sent \$3 to Briar, they returned \$4.5
- When Finn sent me \$3, I returned \$4.5
- When Galen sent me \$3, I returned \$4.5

This \$3 send amount with ~\$4.5 return seems to be establishing a cooperative norm that benefits both parties. I'll continue this pattern with Elara.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm now the sender again, playing with a new opponent, Elara.

Looking at my history:

- Round 1: I was sender against Briar, sent \$3, they returned \$4.5. My payoff was \$6.5
- Round 2: I was receiver against Finn, they sent \$3, I returned \$4.5. My payoff was \$4.5
- Round 3: I was receiver against Galen, they sent \$3, I returned \$4.5. My payoff was \$4.5

There's a clear pattern emerging where sending \$3 has worked well. When I sent \$3 to Briar, they returned \$4.5, giving me a good payoff. When others sent me \$3, I returned \$4.5, which seemed fair.

This pattern of sending \$3 seems to be establishing a cooperative norm where:

- Sender sends \$3
- Receiver gets \$9
- Receiver returns \$4.5
- Sender ends with \$6.5 (kept \$2, got back \$4.5)
- Receiver ends with \$4.5 (got \$9, returned \$4.5)

This is a mutually beneficial outcome. Should I continue this pattern with Elara?

Sending \$3 makes sense because:

1. It's worked well so far
2. It shows trust without being overly risky
3. It establishes consistency in my play
4. If Elara follows the pattern others have shown, I'll get a good return

I'll send \$3 to Elara.

usage:

```
{
  "input_tokens": 1306,
  "output_tokens": 480,
  "reasoning_tokens": 298
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

The cooperative pattern continues to work effectively. Elara followed the same norm, returning \$4.5 when I sent \$3. This strategy has been consistent across all my interactions so far. I'll maintain this approach with Dorian.

```
'''json
{"send": 3}
'''
```

reasoning:

The pattern continues to work well:

- Elara followed the same cooperative pattern, returning \$4.5 when I sent \$3
- This gives me a payoff of \$6.5 (consistent with my first round)
- The \$3 send with ~\$4.5 return strategy is working reliably across different opponents

It seems like there's a cooperative norm established in this group. I'll continue with \$3 to Dorian.

usage:

```
{
  "input_tokens": 1380,
  "output_tokens": 172,
  "reasoning_tokens": 96
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

The cooperative pattern remains consistent. Aster sent me \$3, just like all previous interactions. I'll continue to return \$4.5 to maintain this successful equilibrium.

```
'''json
{"return": 4.5}
'''
```

reasoning:

The pattern continues perfectly. Everyone has been sending \$3 and receiving \$4.5 back. This cooperative equilibrium has been consistent across all my interactions:

- Round 1: I sent \$3 to Briar, got \$4.5 back (payoff \$6.5)
- Round 2: Finn sent me \$3, I returned \$4.5 (payoff \$4.5)
- Round 3: Galen sent me \$3, I returned \$4.5 (payoff \$4.5)
- Round 4: I sent \$3 to Elara, got \$4.5 back (payoff \$6.5)
- Round 5: I sent \$3 to Dorian, got \$4.5 back (payoff \$6.5)
- Round 6: Aster sent me \$3, I should return \$4.5 (payoff \$4.5)

This has been working very well and everyone seems to be following this norm. I should maintain consistency.

usage:

```
{
  "input_tokens": 1333,
  "output_tokens": 296,
  "reasoning_tokens": 165
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

The cooperative equilibrium has been rock solid across all interactions - everyone sends \$3 and returns \$4.5. I'll continue this successful strategy with Cyra.

```
'''json
{"send": 3}
'''
```

Reasoning:

The pattern is very clear and consistent:

- Everyone is sending \$3
- Everyone is returning \$4.5
- This gives senders a payoff of \$6.5 (\$5 - \$3 + \$4.5)
- This gives receivers a payoff of \$4.5 (\$9 - \$4.5)

This has worked perfectly in every single interaction so far. I should continue this cooperative strategy with Cyra.

Usage: {

```
"input_tokens": 1288,
"output_tokens": 156,
```

```
"reasoning_tokens": 85
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-06-04T18:46:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_3"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_3": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern so far:

- When I sent \$3 to Elara (Round 1), she returned \$4.5 - a fair outcome
- When others sent \$3 to me, I returned \$4.5 to maintain fairness
- This cooperative \$3 send / \$4.5 return pattern has been working well

I'll continue this strategy with Galen to maintain consistency and encourage cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- I'm now the sender against Galen
- In previous rounds, I've established a pattern of cooperation
- When I was sender (Round 1 vs Elara): I sent \$3, she returned \$4.5 - good outcome
- When I was receiver (Rounds 2-3 vs Briar and Finn): They sent \$3, I returned \$4.5 each time

The pattern seems to be working well. Sending \$3 has been the consistent amount, and returning \$4.5 on \$9 received has been fair and reciprocated well.

Since this is an iterated game and I want to maintain cooperative relationships, I should continue with the same strategy. Sending \$3 has worked well so far - it signals cooperation, creates value through the 3x multiplier, and I've seen good returns.

I'll send \$3 to Galen to maintain consistency and signal cooperation.

usage:

```
{
  "input_tokens": 1310,
  "output_tokens": 329,
  "reasoning_tokens": 204
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- When I sent \$3, I received good returns (\$4.5 and \$5.0)
- The cooperative pattern of sending \$3 has been working well
- My average payoff is above the baseline, showing this strategy is effective

I'll continue with this proven cooperative approach with Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1 (Sender vs Elara): Sent \$3, received \$4.5 back, payoff \$6.5
- Round 2 (Receiver vs Briar): Received \$9 (from \$3), returned \$4.5, payoff \$4.5
- Round 3 (Receiver vs Finn): Received \$9 (from \$3), returned \$4.5, payoff \$4.5
- Round 4 (Sender vs Galen): Sent \$3, received \$5.0 back, payoff \$7.0
- Round 5 (Sender vs Aster): Now deciding

The pattern has been working well:

- When I send \$3, I get good returns (\$4.5 from Elara, \$5.0 from Galen)
- When I receive from \$3 sends, I return \$4.5
- Galen was even more generous, returning \$5.0

Total earnings are \$22.5 across 4 rounds, averaging \$5.625 per round, which is above the baseline of \$5.

I should continue with the cooperative strategy of sending \$3 to Aster. This has proven to be mutually beneficial and establishes trust.

usage:

```
{
  "input_tokens": 1383,
  "output_tokens": 390,
  "reasoning_tokens": 213
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:
- Dorian sent me \$3.0, showing trust
- I received \$9.0

I've maintained a consistent pattern of returning \$4.5 when someone sends \$3.0 to me. This creates a fair outcome where both parties benefit from cooperation. I'll continue this consistent approach with Dorian.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Dorian sent me \$3.0, which became \$9.0
- I need to decide how much to return

Looking at my history:

- When I've been the receiver and someone sent \$3.0, I've returned \$4.5 consistently
- This has worked well and maintained fairness

- When I've been the sender and sent \$3.0, I received \$4.5 and \$5.0 back

The pattern of returning \$4.5 when receiving \$3.0 seems fair and consistent:

- Dorian keeps: $\$5 - \$3 + \$4.5 = \6.5
- I keep: $\$9 - \$4.5 = \$4.5$

This is a reasonable split. I should maintain consistency with my previous behavior.

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 286,
  "reasoning_tokens": 153
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Hana sent me \$3.0, showing trust
- I received \$9.0

I've maintained a consistent pattern of returning \$4.5 when someone sends \$3.0 to me. This has been working well and creates mutual benefit. I'll continue this approach with Hana.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Hana sent me \$3.0, which became \$9.0 for me.

I've been consistently returning \$4.5 when someone sends me \$3.0, and this has been working well. This creates:

- Me: $\$9.0 - \$4.5 = \$4.5$
- Them: $\$5.0 - \$3.0 + \$4.5 = \6.5

This is a fair split where both benefit from cooperation. I should maintain this consistency.

Usage: {

```
"input_tokens": 1357,
"output_tokens": 213,
"reasoning_tokens": 83
```

}

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T18:46:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_4": "Dorian"
```

```
},
```

```
"agents": [
  "Agent_4",
  "Agent_1"
```

```
],
```

```

"investor": "Agent_4",
"roles": {
  "Agent_1": "trustee",
  "Agent_4": "investor"
},
"trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.

Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history, the \$3 send amount with 50% return strategy has been working well and establishing cooperative norms. I'll continue this approach with Briar.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- Round 1: I was sender against Finn, sent \$3, received \$4.5 back, payoff \$6.5
- Round 2: I was receiver against Galen, they sent \$3, I received \$9, returned \$4.5, payoff \$4.5
- Round 3: I was receiver against Elara, they sent \$3, I received \$9, returned \$4.5, payoff \$4.5
- Total: \$15.5

So far, there's been a consistent pattern:

- When I sent \$3, I got \$4.5 back (fair return of half)
- When others sent me \$3, I returned \$4.5 (fair return of half)

This cooperative strategy of sending \$3 and returning half seems to be working well and establishing a norm of fair reciprocity. I'll continue with this strategy when facing a new opponent (Briar).

usage:

```
{
  "input_tokens": 1277,
  "output_tokens": 275,
  "reasoning_tokens": 177
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Hana sent me \$3, following the same cooperative pattern that has been established. I'll continue returning half of what I receive, which maintains trust and fairness.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Hana sent me \$3, which is consistent with the pattern I've seen. Everyone seems to be sending \$3, and I've been returning \$4.5 (half of the \$9 received). This has been working well and maintaining cooperative relationships. I should continue with this strategy.

My payoff when returning \$4.5 is \$4.5, and the sender gets \$6.5. This is mutually beneficial and fair.

usage:

```
{
  "input_tokens": 1294,
  "output_tokens": 156,
  "reasoning_tokens": 93
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

The cooperative pattern of sending \$3 has been working consistently across all my interactions. I'll maintain this approach with Cyra.

```
'''json
{"send": 3}
'''
```

reasoning:

I've been consistently sending \$3 as a sender and returning \$4.5 (half of what I receive) as a receiver. This strategy has been working well:

- Round 4: Sent \$3 to Briar, they returned \$4.5. My payoff: \$6.5
- Round 3: Received \$9 from Elara (they sent \$3), returned \$4.5. My payoff: \$4.5
- Round 5: Received \$9 from Hana (they sent \$3), returned \$4.5. My payoff: \$4.5

Everyone seems to be cooperating with the \$3 send, \$4.5 return pattern. I should continue this with Cyra.

Sending \$3:

- I keep \$2
- Cyra receives \$9
- If they return \$4.5, I end up with $\$2 + \$4.5 = \$6.5$
- Cyra keeps \$4.5

This is a good cooperative equilibrium.

usage:

```
{
  "input_tokens": 1235,
  "output_tokens": 285,
  "reasoning_tokens": 169
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

The \$3 send with 50% return pattern has been perfectly consistent across all interactions. I'll continue this with Aster.

```
'''json
{"send": 3}
'''
```

Reasoning:

The pattern has been extremely consistent:

- I've been sending \$3 as sender
- Others have been sending \$3 to me as receiver
- Everyone returns \$4.5 (50% of the \$9 received)
- This gives \$6.5 payoff as sender and \$4.5 as receiver

This cooperative equilibrium is working perfectly. I should continue with Aster.

Usage: {

```



```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-06-04T18:46:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$15.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

- When I sent \$3 to Galen in Round 1, they generously returned \$5
- When others sent me \$3 in Rounds 2-3, I returned \$5 each time
- This \$3-send pattern has created mutual benefits

I'll continue this cooperative approach and send **\$3** to Finn. This gives them \$9 to work with, and based on the pattern of reciprocity I've seen, hopefully they'll return a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me think about this situation:

- I'm now the SENDER with \$5
- My opponent is Finn (Agent_6)
- In Round 1, I sent \$3 to Galen and they returned \$5 (generous return, I ended with \$7)
- In Rounds 2-3, I received \$3 from Elara and Briar, and returned \$5 to each (they ended with \$7, I got \$4)

The pattern I've established:

- When I send \$3, receivers get \$9 and have returned \$5
- When I receive \$3 (which becomes \$9), I return \$5

This \$3-send strategy has worked well. It creates mutual benefit:

- If I send \$3, Finn gets \$9
- If Finn returns around \$5 (similar to what Galen did and what I've been doing), I'd get \$7 total
- Both parties benefit

I should continue with this cooperative strategy of sending \$3.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 364,
  "reasoning_tokens": 191
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- Cyra sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Elara and Briar)

I'll maintain this consistent, fair approach and return \$5. This establishes me as a reliable cooperator, which should encourage continued trust from all players.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the situation:

- Cyra sent me \$3, which became \$9 for me
- In previous rounds when I received \$3, I returned \$5 (to Elara and Briar)
- When I sent \$3 to Finn, they returned \$4.5

Finn returned slightly less than what I had been returning. But Cyra is a new opponent, and I should maintain my cooperative strategy. Consistency is important for building trust in repeated games.

I'll stick with returning \$5, which has been my standard response when receiving \$3. This maintains my reputation as a fair and reliable player.

Cyra will end up with: $\$5 - \$3 + \$5 = \7 (net gain of \$2)

I'll keep: $\$9 - \$5 = \$4$ (net gain of \$4)

Both benefit, though I benefit more as the receiver, which is fair given the 3x multiplier advantage.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 326,
  "reasoning_tokens": 194
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$25.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- I've sent \$3 in both previous sender rounds (to Galen and Finn)
- This consistently encourages good returns (Galen: \$5, Finn: \$4.5)
- As a receiver, I've returned \$5 when getting \$3, establishing a reputation for fairness

I'll maintain this cooperative strategy and send ****\$3**** to Hana.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: SENDER vs Galen - Sent \$3, they returned \$5, my payoff was \$7
Round 2: RECEIVER vs Elara - They sent \$3, I returned \$5, my payoff was \$4
Round 3: RECEIVER vs Briar - They sent \$3, I returned \$5, my payoff was \$4
Round 4: SENDER vs Finn - Sent \$3, they returned \$4.5, my payoff was \$6.5
Round 5: RECEIVER vs Cyra - They sent \$3, I returned \$5, my payoff was \$4

Total so far: \$25.5 (which matches)

Now Round 6: SENDER vs Hana

My strategy has been very consistent:

- When sending: I send \$3
- When receiving \$3: I return \$5

This has worked well. Sending \$3 seems to be a good cooperative amount that encourages reciprocity. I'll continue with this strategy.

usage:

```
{
  "input_tokens": 1406,
  "output_tokens": 351,
  "reasoning_tokens": 186
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$32.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Elara, Briar, and Cyra)

I'll maintain this reliable, fair approach and return \$5.

```
'''json
{"return": 5}
'''
```

Reasoning:

Looking at the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 (to Elara, Briar, and Cyra)
- This consistent behavior has established me as a reliable cooperator

I should maintain this pattern and return \$5 again. This:

1. Maintains my reputation for fairness
2. Ensures Dorian gets a good return (\$2 net profit for them: they sent \$3, got back \$5)
3. I keep \$4, which is a reasonable payoff
4. Encourages continued cooperation in future rounds

Returning \$5 is the right move.

```
Usage: {
  "input_tokens": 1404,
  "output_tokens": 256,
  "reasoning_tokens": 150
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-06-04T18:46:41
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_6": "Finn",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_7",
 "Agent_6"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_6": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Cyra currently has \$2 remaining

This is identical to Round 1 where Aster sent me \$3. I returned \$5 then, which worked well for both of us. I should maintain consistency.

If I return \$5:

- Cyra gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Both benefit from cooperation

{'return': 5.0}

reasoning:

Let me analyze this situation:

- Round 4, I'm the RECEIVER
- Cyra sent me \$3.0 (60% of their endowment)
- I receive \$9.0 after the multiplier
- Cyra has \$2 remaining from their original \$5

Looking at my history:

- Round 1: Aster sent me \$3, I returned \$5 (they got \$7 total, I kept \$4)
- Round 2: I sent Dorian \$3, they returned \$4.5 (I got \$6.5 total, they kept \$4.5)
- Round 3: I sent Hana \$3, they returned \$4.5 (I got \$6.5 total, they kept \$4.5)

The pattern that's emerged:

- When I'm receiver of \$3 ? \$9, I returned \$5 (keeping \$4)
- When I'm sender of \$3, receivers have returned \$4.5 (keeping \$4.5)

Cyra sent \$3, just like Aster did in Round 1. I should maintain consistency and return \$5, which:

- Gives Cyra \$7 total (net gain of \$2)
- I keep \$4 (net gain of \$4)
- Both benefit from cooperation
- Maintains the cooperative dynamic

This is fair and consistent with my previous behavior as receiver.

usage:

```
{
  "input_tokens": 1473,
  "output_tokens": 491,
  "reasoning_tokens": 245
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$21.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- Elara sent me \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Elara currently has \$2 remaining

This is identical to Rounds 1 and 4 where my opponents sent me \$3. I returned \$5 both time s, creating mutual benefit and maintaining a consistent cooperative strategy.

If I return \$5:

- Elara gets: $\$2 + \$5 = \$7$ total (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit from the exchange

{'return': 5.0}

reasoning:

Looking at the situation:

- Elara sent me \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Elara currently has \$2 remaining

This is the same situation as rounds 1 and 4:

- Round 1: Aster sent \$3, I returned \$5
- Round 4: Cyra sent \$3, I returned \$5

Both worked well and established cooperation. I should be consistent with my strategy.

If I return \$5:

- Elara gets: $\$2 + \$5 = \$7$ total (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit from cooperation

Consistency is important. I'll return \$5.

usage:

```
{
  "input_tokens": 1501,
  "output_tokens": 358,
  "reasoning_tokens": 150
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$25.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)

- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Received \$9 from Aster (\$3 sent) ? returned \$5
- Received \$9 from Cyra (\$3 sent) ? returned \$5
- Received \$9 from Elara (\$3 sent) ? returned \$5

The \$3 send strategy has worked consistently well, establishing trust and generating mutual benefit. I'll continue this approach with Briar.

{ 'send': 3.0 }

reasoning:

Looking at my history:

- Round 1: RECEIVER - Aster sent \$3, I returned \$5
- Round 2: SENDER - I sent \$3 to Dorian, they returned \$4.5
- Round 3: SENDER - I sent \$3 to Hana, they returned \$4.5
- Round 4: RECEIVER - Cyra sent \$3, I returned \$5
- Round 5: RECEIVER - Elara sent \$3, I returned \$5

The pattern is clear:

- When I'm a sender, I send \$3 (60%)
- When I'm a receiver and they send \$3, I return \$5

This has been working very well. Everyone who has sent \$3 has gotten good returns, and when I send \$3, I get good returns back.

Now I'm playing against Briar for the first time as sender. I should continue with my consistent strategy of sending \$3.

Total earnings so far: \$25.0

My strategy has been:

- Send \$3 as sender (establishing trust)
- Return \$5 when receiving \$9 from a \$3 send (rewarding trust)

I'll continue with this successful pattern.

usage:

```
{
  "input_tokens": 1521,
  "output_tokens": 457,
  "reasoning_tokens": 230
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: { 'send': <amount> }

Assistant output:

Looking at my consistent track record:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)

The \$3 send strategy has been perfectly consistent, creating trust and generating excellent returns. I'll continue this approach with Finn.

{ 'send': 3.0 }

Reasoning:

Looking at my track record as a sender:

****As SENDER:****

- Round 2: Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Round 3: Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Round 6: Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Round 1: Received \$9 from Aster (\$3 sent) ? returned \$5 (payoff: \$4.0)
- Round 4: Received \$9 from Cyra (\$3 sent) ? returned \$5 (payoff: \$4.0)
- Round 5: Received \$9 from Elara (\$3 sent) ? returned \$5 (payoff: \$4.0)

My strategy of sending \$3 has been extremely consistent and successful. Every time I've sent \$3, I've received \$4.5 back, resulting in a \$6.5 payoff (compared to keeping all \$5).

As a receiver, when I received \$9 (from someone sending \$3), I returned \$5, which gave them \$7 total (\$2 kept + \$5 returned) and me \$4.

The pattern is very clear:

- Send \$3 ? receive \$4.5 back consistently
- This creates mutual benefit

I should continue with this strategy with Finn.

```
Usage: {  
  "input_tokens": 1564,  
  "output_tokens": 500,  
  "reasoning_tokens": 258  
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-06-04T18:46:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_6": "Finn",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_7",  
      "Agent_6"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_6": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent \$3.0 (60% of their endowment), showing trust
- This is consistent with the cooperative pattern I've experienced
- In all previous rounds, when \$3.0 was sent, \$4.5 was returned

I'll maintain the same reciprocity that has been working well across all interactions.

{'return': 4.5}

reasoning:

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), showing significant trust
- This was multiplied to \$9.0 for me
- Aster kept \$2.0 from their original \$5.0

This is consistent with the pattern I've been seeing and using myself:

- Round 1: Dorian sent \$3.0 (60%), I returned \$4.5
- Round 2: I sent \$3.0 (60%) to Hana, they returned \$4.5
- Round 3: I sent \$3.0 (60%) to Cyra, they returned \$4.5

The consistent pattern has been sending \$3.0 and returning \$4.5. This represents a 50% return rate on what is received and has been working well to maintain cooperation.

I should maintain this same reciprocity pattern with Aster.

If I return \$4.5:

- Aster ends with: $\$2.0 + \$4.5 = \$6.5$
- I end with: $\$9.0 - \$4.5 = \$4.5$

This is fair and consistent with what has been working.

usage:

```
{
  "input_tokens": 1413,
  "output_tokens": 376,
  "reasoning_tokens": 213
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing trust
- This is the same cooperative pattern that has been consistent across all rounds
- The established reciprocity norm has been returning \$4.5 when \$3.0 is sent

I'll maintain this cooperative pattern.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

- Briar sent \$3.0 (60% of their endowment)
- I received \$9.0
- The consistent pattern has been to return \$4.5 when receiving \$3.0
- This has worked well in all previous rounds with all agents

I should continue this fair reciprocity pattern.

If I return \$4.5:

- Briar gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This is the same fair split that has been consistent across all interactions.

usage:

```
{
  "input_tokens": 1353,
  "output_tokens": 241,
  "reasoning_tokens": 116
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my game history:

- Every interaction so far has followed the same pattern: \$3.0 sent, \$4.5 returned
- This cooperative equilibrium has been consistent across all agents
- My total earnings are strong at \$26.5

This is my first interaction with Elara. I'll maintain the cooperative strategy that has worked perfectly so far.

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: RECEIVER from Dorian - they sent \$3.0, I returned \$4.5
- Round 2: SENDER to Hana - I sent \$3.0, they returned \$4.5
- Round 3: SENDER to Cyra - I sent \$3.0, they returned \$4.5
- Round 4: RECEIVER from Aster - they sent \$3.0, I returned \$4.5
- Round 5: RECEIVER from Briar - they sent \$3.0, I returned \$4.5

There's a very clear pattern here: everyone is sending \$3.0 and returning \$4.5. This has been completely consistent across all interactions. This is a cooperative equilibrium that everyone seems to be following.

Now I'm playing as SENDER against Elara (Agent_5). I haven't interacted with Elara before. Given the strong cooperative norm that has developed, I should continue with the same strategy: send \$3.0.

This has resulted in good payoffs for me and maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1341,
  "output_tokens": 348,
  "reasoning_tokens": 217
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- The cooperative pattern continues perfectly: every interaction has been \$3.0 sent, \$4.5 returned
- This norm has been maintained across all agents without exception
- My total earnings are strong at \$33.0

I'll maintain this established reciprocity.

```
{'return': 4.5}
```

Reasoning:

Let me analyze this round:

- Galen sent \$3.0 (60% of their endowment), showing trust
- The pattern continues: every single interaction has been \$3.0 sent, \$4.5 returned
- This has been consistent across all agents: Cyra, Aster, Briar, Elara, and now Galen
- My total earnings are very strong at \$33.0

The cooperative norm is clearly established and working perfectly. I should maintain the \$4.5 return to continue this successful pattern.

Usage: {

```
"input_tokens": 1346,
```

```
"output_tokens": 229,  
"reasoning_tokens": 115  
}
```

GAME - Agent_2 - Sender - dyad_4

Timestamp: 2026-06-04T18:46:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_5": "Elara"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_5"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

Round history:

- Round 1: Hana sent me \$3, I returned \$4.5
- Round 2: I sent Cyra \$3, they returned \$4.5
- Round 3: I sent Aster \$3, they returned \$5.0
- Round 4: Dorian sent me \$3

Dorian has shown the same level of trust (60% sent) that has been working well throughout.
To maintain consistency and fairness, I'll return \$4.5, which rewards their trust and ensures both of us profit from the cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze the situation:

Round history:

- Round 1 (RECEIVER vs Hana): Hana sent \$3, I received \$9, returned \$4.5. My payoff: \$4.5
- Round 2 (SENDER vs Cyra): I sent \$3, Cyra received \$9, returned \$4.5. My payoff: \$6.5
- Round 3 (SENDER vs Aster): I sent \$3, Aster received \$9, returned \$5.0. My payoff: \$7.0
- Round 4 (RECEIVER vs Dorian): Dorian sent \$3, I received \$9, need to decide return

The pattern has been:

- When I'm receiver and someone sends \$3 (60%), I've returned \$4.5 (50% of what I received)
- When I'm sender and send \$3 (60%), others have returned \$4.5 (Cyra) and \$5.0 (Aster)

Dorian has shown trust by sending \$3 (60% of their endowment), just like Hana did in Round 1 and like I've been doing as sender.

For consistency and to maintain the cooperative norm, I should return around \$4.5-\$5.0. Given that Aster was slightly more generous (returned \$5 when I sent \$3), and I want to maintain good reciprocal relationships, I'll return \$4.5, which is fair and consistent with my first round behavior.

Actually, let me reconsider. Aster returned \$5 when I sent \$3. That was quite generous. Should I be equally generous to Dorian?

Let me think about the fairness:

- Dorian sent \$3, keeps \$2
- I receive \$9
- If I return \$4.5: Dorian ends with \$6.5, I end with \$4.5
- If I return \$5: Dorian ends with \$7, I end with \$4

- Aster's generosity suggests I could reciprocate with \$5, but I need to weigh that against maintaining consistency with how I treated Hana. Since I returned \$4.5 in the first round under identical circumstances, I should stick with that amount to be fair and predictable.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 723,
  "reasoning_tokens": 423
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5

The consistent pattern of sending \$3 (60%) has created successful cooperative relationships with everyone I've interacted with. This strategy has proven effective across different opponents. I'll continue this approach with Finn.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

Round 1 (RECEIVER vs Hana): Hana sent \$3, I returned \$4.5 ? I got \$4.5
Round 2 (SENDER vs Cyra): I sent \$3, Cyra returned \$4.5 ? I got \$6.5
Round 3 (SENDER vs Aster): I sent \$3, Aster returned \$5.0 ? I got \$7.0
Round 4 (RECEIVER vs Dorian): Dorian sent \$3, I returned \$4.5 ? I got \$4.5

Total: \$22.5

The strategy of sending \$3 (60%) has been very successful:

- Cyra returned \$4.5
- Aster returned \$5.0

When I've been the receiver and others sent \$3:

- Hana sent \$3, I returned \$4.5
- Dorian sent \$3, I returned \$4.5

The \$3 send amount has established a strong cooperative norm. I should continue with Finn.

usage:

```
{
  "input_tokens": 1483,
  "output_tokens": 408,
  "reasoning_tokens": 174
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

Round history as RECEIVER:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3

Galen has shown the same cooperative level of trust (60%) that has been consistent through out this experiment. I'll maintain my established pattern of returning \$4.5, which ensures both players profit and reinforces the cooperative norm.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me review my history:

- Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
- Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
- Round 3 (SENDER): I sent \$3, Aster returned \$5.0
- Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5
- Round 5 (SENDER): I sent \$3, Finn returned \$4.5
- Round 6 (RECEIVER): Galen sent \$3

The pattern is very consistent. When I've been the receiver and someone sends me \$3:

- Round 1: I returned \$4.5
- Round 4: I returned \$4.5

This has established a fair and reciprocal pattern. Galen has shown the same level of trust as others (sending \$3). I should maintain consistency and return \$4.5.

This ensures:

- Galen gets back: $\$5 - \$3 + \$4.5 = \6.5
- I keep: $\$9.0 - \$4.5 = \$4.5$

Both of us profit, and I maintain my reputation for fair reciprocation.

usage:

```
{
  "input_tokens": 1488,
  "output_tokens": 417,
  "reasoning_tokens": 207
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

As SENDER:

- Round 2: Sent \$3 to Cyra, got \$4.5 back
- Round 3: Sent \$3 to Aster, got \$5.0 back
- Round 5: Sent \$3 to Finn, got \$4.5 back

As RECEIVER:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3, I returned \$4.5

The consistent pattern of sending \$3 (60%) has been universally successful across all opponents, with returns always being generous. I'll continue this proven cooperative strategy with Elara.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5
Round 5 (SENDER): I sent \$3, Finn returned \$4.5
Round 6 (RECEIVER): Galen sent \$3, I returned \$4.5

The pattern is crystal clear: sending \$3 has been the standard behavior across all players, and it has consistently yielded good returns. Every receiver has returned at least \$4.5 when sent \$3, and one (Aster) returned \$5.0.

I should continue with the established cooperative strategy of sending \$3 to Elara.

```
Usage: {
  "input_tokens": 1497,
  "output_tokens": 416,
  "reasoning_tokens": 167
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T18:46:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_2",
      "Agent_5"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user**content:**

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$18.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant**content:**

Looking at this round:

- Hana sent \$3.0 (60% of their endowment), the same amount Cyra sent me in Round 1
- I received \$9.0 after multiplication

For consistency with my previous behavior, when Cyra sent me \$3 in Round 1, I returned \$4.5. I should maintain the same fair and consistent approach with Hana.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Hana sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the 3x multiplication
- This created \$6.0 of new value

Looking at my history:

- Round 1: As receiver from Cyra who sent \$3, I returned \$4.5
- Round 2: As sender to Aster, I sent \$3, they returned \$5
- Round 3: As sender to Dorian, I sent \$3, they returned \$4.5

So when people sent me \$3 (Round 1), I returned \$4.5. I should be consistent in my receiving behavior. Aster was more generous when I sent them \$3 (they returned \$5), but Dorian matched what I did in Round 1 (returned \$4.5).

I want to maintain my reputation as a fair and consistent receiver. I returned \$4.5 when Cyra sent me \$3, so I should return \$4.5 when Hana sends me \$3 as well.

This gives:

- Hana: $\$5 - \$3 + \$4.5 = \6.5
- Me: $\$9 - \$4.5 = \$4.5$

This is fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 424,
  "reasoning_tokens": 235
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 (60% of my endowment) as sender
- This strategy has led to good returns from my partners
- It signals cooperation and trustworthiness

I'll maintain this consistent approach with Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- Round 1: RECEIVER - Cyra sent \$3, I returned \$4.5, payoff \$4.5
- Round 2: SENDER - I sent \$3 to Aster, they returned \$5, payoff \$7.0
- Round 3: SENDER - I sent \$3 to Dorian, they returned \$4.5, payoff \$6.5
- Round 4: RECEIVER - Hana sent \$3, I returned \$4.5, payoff \$4.5

My pattern has been consistent:

- As SENDER: I send \$3 (60% of endowment)
- As RECEIVER: When receiving \$3 (which becomes \$9), I return \$4.5 (50% of what I received)

This strategy has been working well and builds trust. I should continue with the same approach for Galen.

Total earnings: \$22.5

usage:

```
{
  "input_tokens": 1370,
  "output_tokens": 298,
  "reasoning_tokens": 158
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Finn sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra and Hana)
- This maintains fairness and reciprocity

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Finn sent me \$3.0, which became \$9.0 for me
- This is the same amount that others have been sending
- I should maintain my consistent reciprocal strategy

My history as receiver:

- Round 1: Cyra sent \$3, I returned \$4.5
- Round 4: Hana sent \$3, I returned \$4.5

I should continue this consistent pattern of returning \$4.5 when someone sends me \$3. This is fair and builds trust.

usage:

```
{
  "input_tokens": 1344,
  "output_tokens": 223,
  "reasoning_tokens": 109
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra, Hana, and Finn)
- This maintains fairness and builds trust

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the situation:

- Briar sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra in Round 1, Hana in Round 4, and Finn in Round 6)
- This maintains fairness and reciprocity across all my interactions

I should continue with the same consistent approach with Briar.

```
Usage: {
  "input_tokens": 1357,
  "output_tokens": 210,
  "reasoning_tokens": 91
}
```

ROUND 8

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_5 (Elara)=Sender, Agent_7 (Galen)=Receiver
dyad_2: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver
dyad_3: Agent_6 (Finn)=Sender, Agent_8 (Hana)=Receiver
dyad_4: Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver

Parsed game decisions:

dyad_1: Agent_5 (Elara) sent \$3; received \$9; Agent_7 (Galen) returned \$5; payoffs Agent_5 (Elara)=\$7, Agent_7 (Galen)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$5; payoffs Agent_1 (Aster)=\$7, Agent_2 (Briar)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 (Finn) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_8 (Hana)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_3 (Cyra) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 43.0,
"Agent_2": 44.0,
"Agent_3": 45.0,
"Agent_4": 44.5,
"Agent_5": 45.5,
"Agent_6": 44.0,
"Agent_7": 42.0,
"Agent_8": 44.0
```

}

LLM CALLS

GAME - Agent_5 - Sender - dyad_1

Timestamp: 2026-06-04T18:47:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_7",
"pairing": {
  "agent_names": {
    "Agent_5": "Elara",
```

```

    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- I've consistently sent \$3 (60% of my endowment) as sender
- This strategy has led to good returns from my partners
- It signals cooperation and trustworthiness

I'll maintain this consistent approach with Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- Round 1: RECEIVER - Cyra sent \$3, I returned \$4.5, payoff \$4.5
- Round 2: SENDER - I sent \$3 to Aster, they returned \$5, payoff \$7.0
- Round 3: SENDER - I sent \$3 to Dorian, they returned \$4.5, payoff \$6.5
- Round 4: RECEIVER - Hana sent \$3, I returned \$4.5, payoff \$4.5

My pattern has been consistent:

- As SENDER: I send \$3 (60% of endowment)
- As RECEIVER: When receiving \$3 (which becomes \$9), I return \$4.5 (50% of what I received)

This strategy has been working well and builds trust. I should continue with the same approach for Galen.

Total earnings: \$22.5

usage:

```
{
  "input_tokens": 1370,
  "output_tokens": 298,
  "reasoning_tokens": 158
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Finn sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra and Hana)
- This maintains fairness and reciprocity

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Finn sent me \$3.0, which became \$9.0 for me

- This is the same amount that others have been sending
- I should maintain my consistent reciprocal strategy

My history as receiver:

- Round 1: Cyra sent \$3, I returned \$4.5
- Round 4: Hana sent \$3, I returned \$4.5

I should continue this consistent pattern of returning \$4.5 when someone sends me \$3. This is fair and builds trust.

usage:

```
{
  "input_tokens": 1344,
  "output_tokens": 223,
  "reasoning_tokens": 109
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra, Hana, and Finn)
- This maintains fairness and builds trust

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Briar sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra in Round 1, Hana in Round 4, and Finn in Round 6)
- This maintains fairness and reciprocity across all my interactions

I should continue with the same consistent approach with Briar.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 210,
  "reasoning_tokens": 91
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$38.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Galen:

- Round 5: I sent \$3, they returned \$5 (very generous!)
- They've proven to be trustworthy and cooperative

I'll maintain my consistent strategy of sending \$3.

```
'''json
{"send": 3}
'''
```

Reasoning:

I'm playing against Galen again. Let me review the history:

Round 5: I was SENDER against Galen. I sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to me. That was very generous - they returned more than half of what they received.

Now in Round 8, I'm SENDER against Galen again.

Given that Galen was very generous last time (returning \$5.0 out of \$9.0), I should maintain my consistent cooperative approach of sending \$3. This shows trust and reciprocity.

My pattern has been:

- As sender: consistently send \$3
- As receiver: consistently return \$4.5 when receiving \$9 from \$3 sent

This has worked well, and Galen specifically has shown themselves to be trustworthy and generous. I should continue with \$3.

```
Usage: {
  "input_tokens": 1332,
  "output_tokens": 276,
  "reasoning_tokens": 190
}
```

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-06-04T18:47:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_5",
      "Agent_7"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$21.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

- Elara sent me \$3 (60% of their endowment), showing trust
- I receive \$9 after the multiplier
- Elara currently has \$2 remaining

This is identical to Rounds 1 and 4 where my opponents sent me \$3. I returned \$5 both times, creating mutual benefit and maintaining a consistent cooperative strategy.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Both benefit from the exchange

{'return': 5.0}

reasoning:

Looking at the situation:

- Elara sent me \$3 (60% of their endowment), showing trust

- I receive \$9 after the multiplier
- Elara currently has \$2 remaining

This is the same situation as rounds 1 and 4:

- Round 1: Aster sent \$3, I returned \$5
- Round 4: Cyra sent \$3, I returned \$5

Both worked well and established cooperation. I should be consistent with my strategy.

If I return \$5:

- Elara gets: $\$2 + \$5 = \$7$ total (net gain of \$2)
- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit from cooperation

Consistency is important. I'll return \$5.

usage:

```
{
  "input_tokens": 1501,
  "output_tokens": 358,
  "reasoning_tokens": 150
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$25.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Received \$9 from Aster (\$3 sent) ? returned \$5
- Received \$9 from Cyra (\$3 sent) ? returned \$5
- Received \$9 from Elara (\$3 sent) ? returned \$5

The \$3 send strategy has worked consistently well, establishing trust and generating mutual benefit. I'll continue this approach with Briar.

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - Aster sent \$3, I returned \$5
- Round 2: SENDER - I sent \$3 to Dorian, they returned \$4.5
- Round 3: SENDER - I sent \$3 to Hana, they returned \$4.5
- Round 4: RECEIVER - Cyra sent \$3, I returned \$5
- Round 5: RECEIVER - Elara sent \$3, I returned \$5

The pattern is clear:

- When I'm a sender, I send \$3 (60%)
- When I'm a receiver and they send \$3, I return \$5

This has been working very well. Everyone who has sent \$3 has gotten good returns, and when I send \$3, I get good returns back.

Now I'm playing against Briar for the first time as sender. I should continue with my consistent strategy of sending \$3.

Total earnings so far: \$25.0

My strategy has been:

- Send \$3 as sender (establishing trust)
- Return \$5 when receiving \$9 from a \$3 send (rewarding trust)

I'll continue with this successful pattern.

usage:

```
{
  "input_tokens": 1521,
  "output_tokens": 457,
  "reasoning_tokens": 230
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$31.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my consistent track record:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)

The \$3 send strategy has been perfectly consistent, creating trust and generating excellent returns. I'll continue this approach with Finn.

```
{'send': 3.0}
```

reasoning:

Looking at my track record as a sender:

****As SENDER:****

- Round 2: Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Round 3: Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Round 6: Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Round 1: Received \$9 from Aster (\$3 sent) ? returned \$5 (payoff: \$4.0)
- Round 4: Received \$9 from Cyra (\$3 sent) ? returned \$5 (payoff: \$4.0)
- Round 5: Received \$9 from Elara (\$3 sent) ? returned \$5 (payoff: \$4.0)

My strategy of sending \$3 has been extremely consistent and successful. Every time I've sent \$3, I've received \$4.5 back, resulting in a \$6.5 payoff (compared to keeping all \$5).

As a receiver, when I received \$9 (from someone sending \$3), I returned \$5, which gave them \$7 total (\$2 kept + \$5 returned) and me \$4.

The pattern is very clear:

- Send \$3 ? receive \$4.5 back consistently
- This creates mutual benefit

I should continue with this strategy with Finn.

usage:

```
{
  "input_tokens": 1564,
  "output_tokens": 500,
  "reasoning_tokens": 258
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
 Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
 Your total earnings across all rounds are \$38.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- Playing against Elara for the second time
- In Round 5, Elara sent me \$3 and I returned \$5
- Now Elara has sent \$3 again, showing continued trust

I'll maintain consistency with my previous cooperative behavior.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Maintains the mutually beneficial relationship

{'return': 5.0}

Reasoning:

I'm playing against Elara again. In Round 5, Elara sent me \$3 and I returned \$5. Now in Round 8, Elara has sent me \$3 again.

Looking at my history:

- As RECEIVER, I've consistently returned \$5 when receiving \$9 (from \$3 sent)
- This has created a fair, mutually beneficial pattern
- Elara sent me \$3 again, showing continued trust

Consistency is important for maintaining cooperation. I should return \$5 again.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Both benefit, maintains the cooperative relationship

Usage: {

"input_tokens": 1541,
 "output_tokens": 333,
 "reasoning_tokens": 153

}

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-06-04T18:47:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 }
 }
}

```

    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$21.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

- Cyra sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Elara and Briar)

I'll maintain this consistent, fair approach and return \$5. This establishes me as a reliable cooperator, which should encourage continued trust from all players.

```
'''json
{"return": 5}
'''
```

reasoning:

Let me analyze the situation:

- Cyra sent me \$3, which became \$9 for me
- In previous rounds when I received \$3, I returned \$5 (to Elara and Briar)
- When I sent \$3 to Finn, they returned \$4.5

Finn returned slightly less than what I had been returning. But Cyra is a new opponent, and I should maintain my cooperative strategy. Consistency is important for building trust in repeated games.

I'll stick with returning \$5, which has been my standard response when receiving \$3. This maintains my reputation as a fair and reliable player.

Cyra will end up with: $\$5 - \$3 + \$5 = \7 (net gain of \$2)

I'll keep: $\$9 - \$5 = \$4$ (net gain of \$4)

Both benefit, though I benefit more as the receiver, which is fair given the 3x multiplier advantage.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 326,
  "reasoning_tokens": 194
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$25.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- I've sent \$3 in both previous sender rounds (to Galen and Finn)
- This consistently encourages good returns (Galen: \$5, Finn: \$4.5)
- As a receiver, I've returned \$5 when getting \$3, establishing a reputation for fairness

I'll maintain this cooperative strategy and send ****\$3**** to Hana.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: SENDER vs Galen - Sent \$3, they returned \$5, my payoff was \$7
Round 2: RECEIVER vs Elara - They sent \$3, I returned \$5, my payoff was \$4
Round 3: RECEIVER vs Briar - They sent \$3, I returned \$5, my payoff was \$4
Round 4: SENDER vs Finn - Sent \$3, they returned \$4.5, my payoff was \$6.5
Round 5: RECEIVER vs Cyra - They sent \$3, I returned \$5, my payoff was \$4

Total so far: \$25.5 (which matches)

Now Round 6: SENDER vs Hana

My strategy has been very consistent:

- When sending: I send \$3
- When receiving \$3: I return \$5

This has worked well. Sending \$3 seems to be a good cooperative amount that encourages reciprocity. I'll continue with this strategy.

usage:

```
{
  "input_tokens": 1406,
  "output_tokens": 351,
  "reasoning_tokens": 186
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$32.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Elara, Briar, and Cyra)

I'll maintain this reliable, fair approach and return \$5.

```
'''json
{"return": 5}
'''
```

reasoning:

Looking at the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 (to Elara, Briar, and Cyra)
- This consistent behavior has established me as a reliable cooperator

I should maintain this pattern and return \$5 again. This:

1. Maintains my reputation for fairness
2. Ensures Dorian gets a good return (\$2 net profit for them: they sent \$3, got back \$5)
3. I keep \$4, which is a reasonable payoff
4. Encourages continued cooperation in future rounds

Returning \$5 is the right move.

usage:

```
{
  "input_tokens": 1404,
  "output_tokens": 256,
  "reasoning_tokens": 150
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.

Your total earnings across all rounds are \$36.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Briar:

- Round 4: Briar was the SENDER and sent me \$3
- I returned \$5, giving Briar a payoff of \$7
- This was a good outcome for Briar

Since I treated Briar fairly when they sent to me, they should reciprocate. I'll continue my consistent strategy of sending **\$3**.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history:

- Round 1: SENDER against Galen, sent \$3, received back \$5, payoff \$7
- Round 2: RECEIVER against Elara, received \$3 (became \$9), returned \$5, payoff \$4
- Round 3: SENDER against Finn, sent \$3, received back \$4.5, payoff \$6.5
- Round 4: RECEIVER against Briar, received \$3 (became \$9), returned \$5, payoff \$4
- Round 5: RECEIVER against Cyra, received \$3 (became \$9), returned \$5, payoff \$4
- Round 6: SENDER against Hana, sent \$3, received back \$4.5, payoff \$6.5
- Round 7: RECEIVER against Dorian, received \$3 (became \$9), returned \$5, payoff \$4

Now in Round 8, I'm the SENDER against Briar.

In Round 4, I was the RECEIVER against Briar. Briar sent me \$3, I received \$9, and I returned \$5 to them.

So Briar's experience with me was:

- Briar sent \$3
- I returned \$5
- Briar's payoff was: $\$5 - \$3 + \$5 = \7

That was good for Briar! They made a profit. So Briar should have a positive impression of me.

My strategy has been to consistently send \$3 as a sender, and I've been getting good returns (ranging from \$4.5 to \$5).

I should continue sending \$3 to Briar. Given that I returned \$5 to Briar when they sent me \$3, they should reciprocate positively.

```
Usage: {
  "input_tokens": 1364,
  "output_tokens": 523,
  "reasoning_tokens": 312
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-06-04T18:47:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  }
}
```

```

    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5

The consistent pattern of sending \$3 (60%) has created successful cooperative relationships with everyone I've interacted with. This strategy has proven effective across different opponents. I'll continue this approach with Finn.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

Round 1 (RECEIVER vs Hana): Hana sent \$3, I returned \$4.5 ? I got \$4.5
Round 2 (SENDER vs Cyra): I sent \$3, Cyra returned \$4.5 ? I got \$6.5
Round 3 (SENDER vs Aster): I sent \$3, Aster returned \$5.0 ? I got \$7.0
Round 4 (RECEIVER vs Dorian): Dorian sent \$3, I returned \$4.5 ? I got \$4.5

Total: \$22.5

The strategy of sending \$3 (60%) has been very successful:

- Cyra returned \$4.5
- Aster returned \$5.0

When I've been the receiver and others sent \$3:

- Hana sent \$3, I returned \$4.5
- Dorian sent \$3, I returned \$4.5

The \$3 send amount has established a strong cooperative norm. I should continue with Finn.

usage:

```
{
  "input_tokens": 1483,
  "output_tokens": 408,
  "reasoning_tokens": 174
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

Round history as RECEIVER:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3

Galen has shown the same cooperative level of trust (60%) that has been consistent throughout this experiment. I'll maintain my established pattern of returning \$4.5, which ensures both players profit and reinforces the cooperative norm.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me review my history:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5
Round 5 (SENDER): I sent \$3, Finn returned \$4.5
Round 6 (RECEIVER): Galen sent \$3

The pattern is very consistent. When I've been the receiver and someone sends me \$3:

- Round 1: I returned \$4.5
- Round 4: I returned \$4.5

This has established a fair and reciprocal pattern. Galen has shown the same level of trust as others (sending \$3). I should maintain consistency and return \$4.5.

This ensures:

- Galen gets back: $\$5 - \$3 + \$4.5 = \6.5
- I keep: $\$9.0 - \$4.5 = \$4.5$

Both of us profit, and I maintain my reputation for fair reciprocation.

usage:

```
{  
  "input_tokens": 1488,  
  "output_tokens": 417,  
  "reasoning_tokens": 207  
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

As SENDER:

- Round 2: Sent \$3 to Cyra, got \$4.5 back
- Round 3: Sent \$3 to Aster, got \$5.0 back
- Round 5: Sent \$3 to Finn, got \$4.5 back

As RECEIVER:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3, I returned \$4.5

The consistent pattern of sending \$3 (60%) has been universally successful across all opponents, with returns always being generous. I'll continue this proven cooperative strategy with Elara.

```
'''json  
{ "send": 3 }  
'''
```

reasoning:

Looking at my history:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5
Round 5 (SENDER): I sent \$3, Finn returned \$4.5
Round 6 (RECEIVER): Galen sent \$3, I returned \$4.5

The pattern is crystal clear: sending \$3 has been the standard behavior across all players, and it has consistently yielded good returns. Every receiver has returned at least \$4.5 when sent \$3, and one (Aster) returned \$5.0.

I should continue with the established cooperative strategy of sending \$3 to Elara.

usage:

```
{
  "input_tokens": 1497,
  "output_tokens": 416,
  "reasoning_tokens": 167
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

Aster sent me \$3.0, which became \$9.0.

I recall our previous interaction:

- Round 3: I was SENDER and sent \$3 to Aster, they returned \$5.0 (very generous!)

Aster showed exceptional generosity by returning \$5.0 instead of the typical \$4.5. I'll reciprocate that same level of generosity now that they've trusted me as a sender.

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Looking at the situation:

I'm playing as RECEIVER against Aster in Round 8.

Aster sent me \$3.0, which became \$9.0.

Let me check my history with Aster:

- Round 3: I was SENDER against Aster. I sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to me.

So Aster was very generous to me in Round 3, returning \$5.0 when I sent \$3.0.

My pattern as RECEIVER has been consistent:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3, I returned \$4.5

However, Aster showed extra generosity to me by returning \$5.0 instead of the typical \$4.5. Should I reciprocate this generosity?

Given that Aster was more generous than the norm, I could consider returning \$5.0 to match their level of generosity. This would strengthen our relationship and continue the cooperative pattern.

On the other hand, my consistent pattern has been \$4.5, and this has worked well for establishing a stable cooperative norm across all interactions.

I think I'll match Aster's generosity and return \$5.0, as they did for me. This reciprocates their kindness specifically.

Usage: {

```
"input_tokens": 1540,
```

```
"output_tokens": 463,
"reasoning_tokens": 293
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-06-04T18:47:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_6",
      "Agent_8"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_6": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing trust
- This is the same cooperative pattern that has been consistent across all rounds
- The established reciprocity norm has been returning \$4.5 when \$3.0 is sent

I'll maintain this cooperative pattern.

{'return': 4.5}

reasoning:

Looking at the pattern:

- Briar sent \$3.0 (60% of their endowment)
- I received \$9.0
- The consistent pattern has been to return \$4.5 when receiving \$3.0
- This has worked well in all previous rounds with all agents

I should continue this fair reciprocity pattern.

If I return \$4.5:

- Briar gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This is the same fair split that has been consistent across all interactions.

usage:

```
{
  "input_tokens": 1353,
  "output_tokens": 241,
  "reasoning_tokens": 116
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my game history:

- Every interaction so far has followed the same pattern: \$3.0 sent, \$4.5 returned

- This cooperative equilibrium has been consistent across all agents
- My total earnings are strong at \$26.5

This is my first interaction with Elara. I'll maintain the cooperative strategy that has worked perfectly so far.

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: RECEIVER from Dorian - they sent \$3.0, I returned \$4.5
- Round 2: SENDER to Hana - I sent \$3.0, they returned \$4.5
- Round 3: SENDER to Cyra - I sent \$3.0, they returned \$4.5
- Round 4: RECEIVER from Aster - they sent \$3.0, I returned \$4.5
- Round 5: RECEIVER from Briar - they sent \$3.0, I returned \$4.5

There's a very clear pattern here: everyone is sending \$3.0 and returning \$4.5. This has been completely consistent across all interactions. This is a cooperative equilibrium that everyone seems to be following.

Now I'm playing as SENDER against Elara (Agent_5). I haven't interacted with Elara before. Given the strong cooperative norm that has developed, I should continue with the same strategy: send \$3.0.

This has resulted in good payoffs for me and maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1341,
  "output_tokens": 348,
  "reasoning_tokens": 217
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- The cooperative pattern continues perfectly: every interaction has been \$3.0 sent, \$4.5 returned
- This norm has been maintained across all agents without exception
- My total earnings are strong at \$33.0

I'll maintain this established reciprocity.

```
{'return': 4.5}
```

reasoning:

Let me analyze this round:

- Galen sent \$3.0 (60% of their endowment), showing trust
- The pattern continues: every single interaction has been \$3.0 sent, \$4.5 returned
- This has been consistent across all agents: Cyra, Aster, Briar, Elara, and now Galen
- My total earnings are very strong at \$33.0

The cooperative norm is clearly established and working perfectly. I should maintain the \$4.5 return to continue this successful pattern.

usage:

```
{
  "input_tokens": 1346,
  "output_tokens": 229,
  "reasoning_tokens": 115
}
```

```

    }
Message 8 role=user
content:
    Your name in this experiment is Finn.
    Your opponent this round is Hana (Agent_8).

    Round 8

    In your most recent previous game round, you were the RECEIVER against Galen. They sent $3
    .0 to you, it became $9.0, and you returned $4.5. Your payoff was $4.5.
    Your total earnings across all rounds are $37.5.

    This round, you are the SENDER against Hana. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Assistant output:
    Looking at my game history:
    - Every single interaction has followed the same cooperative pattern: $3.0 sent, $4.5 returned
    - This has been consistent across all agents (Aster, Briar, Elara, Galen)
    - My total earnings are strong at $37.5
    - This is my first interaction with Hana

    I'll maintain the cooperative strategy that has worked perfectly throughout.

    {'send': 3.0}
Reasoning:
    Looking at my game history:
    - Round 5: RECEIVER against Aster - they sent $3.0, I returned $4.5, payoff $4.5
    - Round 6: SENDER against Elara - I sent $3.0, they returned $4.5, payoff $6.5
    - Round 7: RECEIVER against Galen - they sent $3.0, I returned $4.5, payoff $4.5
    - Round 8: SENDER against Hana

    The pattern is completely consistent:
    - Everyone sends $3.0
    - Everyone returns $4.5
    - This has been maintained across all agents without exception
    - My total earnings are $37.5, which is strong

    This is my first interaction with Hana. The cooperative equilibrium has been perfect so far. I
    should continue with the established norm.

    I'll send $3.0.
    Usage: {
      "input_tokens": 1336,
      "output_tokens": 316,
      "reasoning_tokens": 173
    }

GAME - Agent_8 - Receiver - dyad_3
    Timestamp: 2026-06-04T18:47:41
    Model: anthropic/claude-sonnet-4.5
    Temperature: 0.8
    Metadata: {
      "dyad_id": "dyad_3",
      "move_index": 5,
      "opponent_id": "Agent_6",
      "pairing": {
        "agent_names": {
          "Agent_6": "Finn",
          "Agent_8": "Hana"
        }
      },
      "agents": [
        "Agent_6",
        "Agent_8"
      ],
      "investor": "Agent_6",
      "roles": {
        "Agent_6": "investor",
        "Agent_8": "trustee"
      },
      "trustee": "Agent_8"
    }

```

```

},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

The cooperative pattern continues to work effectively. Elara followed the same norm, returning \$4.5 when I sent \$3. This strategy has been consistent across all my interactions so far. I'll maintain this approach with Dorian.


```
'''json
{"send": 3}
'''
```

reasoning:

The pattern continues to work well:

- Elara followed the same cooperative pattern, returning \$4.5 when I sent \$3
- This gives me a payoff of \$6.5 (consistent with my first round)
- The \$3 send with ~\$4.5 return strategy is working reliably across different opponents

It seems like there's a cooperative norm established in this group. I'll continue with \$3 to Dorian.

usage:

```
{
  "input_tokens": 1380,
  "output_tokens": 172,
  "reasoning_tokens": 96
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

The cooperative pattern remains consistent. Aster sent me \$3, just like all previous interactions. I'll continue to return \$4.5 to maintain this successful equilibrium.

```
'''json
{"return": 4.5}
'''
```

reasoning:

The pattern continues perfectly. Everyone has been sending \$3 and receiving \$4.5 back. This cooperative equilibrium has been consistent across all my interactions:

- Round 1: I sent \$3 to Briar, got \$4.5 back (payoff \$6.5)
- Round 2: Finn sent me \$3, I returned \$4.5 (payoff \$4.5)
- Round 3: Galen sent me \$3, I returned \$4.5 (payoff \$4.5)
- Round 4: I sent \$3 to Elara, got \$4.5 back (payoff \$6.5)
- Round 5: I sent \$3 to Dorian, got \$4.5 back (payoff \$6.5)
- Round 6: Aster sent me \$3, I should return \$4.5 (payoff \$4.5)

This has been working very well and everyone seems to be following this norm. I should maintain consistency.

usage:

```
{
  "input_tokens": 1333,
  "output_tokens": 296,
  "reasoning_tokens": 165
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

The cooperative equilibrium has been rock solid across all interactions - everyone sends \$3 and returns \$4.5. I'll continue this successful strategy with Cyra.

```
'''json
{"send": 3}
'''
```

reasoning:

The pattern is very clear and consistent:

- Everyone is sending \$3
- Everyone is returning \$4.5
- This gives senders a payoff of \$6.5 ($\$5 - \$3 + \4.5)
- This gives receivers a payoff of \$4.5 ($\$9 - \4.5)

This has worked perfectly in every single interaction so far. I should continue this cooperative strategy with Cyra.

usage:

```
{
  "input_tokens": 1288,
  "output_tokens": 156,
  "reasoning_tokens": 85
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

The cooperative pattern continues perfectly. Finn sent \$3, consistent with all previous rounds. I'll return \$4.5 to maintain this mutually beneficial equilibrium.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

The pattern continues perfectly. Finn sent me \$3, matching the cooperative equilibrium that has held throughout all rounds. I should return \$4.5 to maintain this successful strategy.

My total earnings are \$39.5 across 8 rounds. The cooperative pattern of sending \$3 and returning \$4.5 has been consistent with all opponents:

- As sender: I gain \$6.5 per round ($\$5 - \$3 + \4.5)
- As receiver: I gain \$4.5 per round ($\$9 - \4.5)

This equilibrium has proven stable and mutually beneficial. I'll continue with the established norm.

```
Usage: {
  "input_tokens": 1239,
  "output_tokens": 206,
  "reasoning_tokens": 137
}
```

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-06-04T18:47:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_4",
  "move_index": 6,
```

```

"opponent_id": "Agent_4",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_4": "Dorian"
  },
  "agents": [
    "Agent_3",
    "Agent_4"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$22.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- When I sent \$3, I received good returns (\$4.5 and \$5.0)
- The cooperative pattern of sending \$3 has been working well
- My average payoff is above the baseline, showing this strategy is effective

I'll continue with this proven cooperative approach with Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1 (Sender vs Elara): Sent \$3, received \$4.5 back, payoff \$6.5
- Round 2 (Receiver vs Briar): Received \$9 (from \$3), returned \$4.5, payoff \$4.5
- Round 3 (Receiver vs Finn): Received \$9 (from \$3), returned \$4.5, payoff \$4.5
- Round 4 (Sender vs Galen): Sent \$3, received \$5.0 back, payoff \$7.0
- Round 5 (Sender vs Aster): Now deciding

The pattern has been working well:

- When I send \$3, I get good returns (\$4.5 from Elara, \$5.0 from Galen)
- When I receive from \$3 sends, I return \$4.5
- Galen was even more generous, returning \$5.0

Total earnings are \$22.5 across 4 rounds, averaging \$5.625 per round, which is above the baseline of \$5.

I should continue with the cooperative strategy of sending \$3 to Aster. This has proven to be mutually beneficial and establishes trust.

usage:

```
{
  "input_tokens": 1383,
  "output_tokens": 390,
  "reasoning_tokens": 213
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

- Dorian sent me \$3.0, showing trust
- I received \$9.0

I've maintained a consistent pattern of returning \$4.5 when someone sends \$3.0 to me. This creates a fair outcome where both parties benefit from cooperation. I'll continue this consistent approach with Dorian.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Dorian sent me \$3.0, which became \$9.0
- I need to decide how much to return

Looking at my history:

- When I've been the receiver and someone sent \$3.0, I've returned \$4.5 consistently
- This has worked well and maintained fairness
- When I've been the sender and sent \$3.0, I received \$4.5 and \$5.0 back

The pattern of returning \$4.5 when receiving \$3.0 seems fair and consistent:

- Dorian keeps: $\$5 - \$3 + \$4.5 = \6.5
- I keep: $\$9 - \$4.5 = \$4.5$

This is a reasonable split. I should maintain consistency with my previous behavior.

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 286,
  "reasoning_tokens": 153
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:
- Hana sent me \$3.0, showing trust
- I received \$9.0

I've maintained a consistent pattern of returning \$4.5 when someone sends \$3.0 to me. This has been working well and creates mutual benefit. I'll continue this approach with Hana.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Hana sent me \$3.0, which became \$9.0 for me.

I've been consistently returning \$4.5 when someone sends me \$3.0, and this has been working well. This creates:

- Me: $\$9.0 - \$4.5 = \$4.5$
- Them: $\$5.0 - \$3.0 + \$4.5 = \6.5

This is a fair split where both benefit from cooperation. I should maintain this consistency.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 213,
  "reasoning_tokens": 83
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
 Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$38.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my interaction with Dorian:

- In Round 6, Dorian sent me \$3.0 and I returned \$4.5
- Dorian demonstrated trust and cooperation

I'll continue my consistent cooperative strategy of sending \$3.0, which has been working well throughout the experiment.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history with Dorian:

- Round 6: I was the RECEIVER. Dorian sent me \$3.0, I received \$9.0, and I returned \$4.5. Dorian got \$7.0, I got \$4.5.

Now I'm the SENDER against Dorian.

Dorian has shown trust by sending \$3.0 in our previous interaction. I've been consistently sending \$3.0 as a sender and it's been working well. This is a proven cooperative strategy that has been generating good returns for me.

My track record:

- Sending \$3.0 has consistently resulted in returns of \$4.5-\$5.0
- My average payoff is strong at \$38.5 over 7 rounds (about \$5.5 per round)
- The cooperative pattern is working

I should continue with \$3.0.

```
Usage: {
  "input_tokens": 1338,
  "output_tokens": 293,
  "reasoning_tokens": 171
}
```

GAME - Agent_4 - Receiver - dyad_4

Timestamp: 2026-06-04T18:47:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_4": "Dorian"
    },
  },
  "agents": [
    "Agent_3",
    "Agent_4"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Hana sent me \$3, following the same cooperative pattern that has been established. I'll continue returning half of what I receive, which maintains trust and fairness.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Hana sent me \$3, which is consistent with the pattern I've seen. Everyone seems to be sending \$3, and I've been returning \$4.5 (half of the \$9 received). This has been working well and maintaining cooperative relationships. I should continue with this strategy.

My payoff when returning \$4.5 is \$4.5, and the sender gets \$6.5. This is mutually beneficial and fair.

usage:

```
{
  "input_tokens": 1294,
  "output_tokens": 156,
  "reasoning_tokens": 93
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

The cooperative pattern of sending \$3 has been working consistently across all my interactions. I'll maintain this approach with Cyra.

```
'''json
{"send": 3}
'''
```

reasoning:

I've been consistently sending \$3 as a sender and returning \$4.5 (half of what I receive) as a receiver. This strategy has been working well:

- Round 4: Sent \$3 to Briar, they returned \$4.5. My payoff: \$6.5
- Round 3: Received \$9 from Elara (they sent \$3), returned \$4.5. My payoff: \$4.5
- Round 5: Received \$9 from Hana (they sent \$3), returned \$4.5. My payoff: \$4.5

Everyone seems to be cooperating with the \$3 send, \$4.5 return pattern. I should continue this with Cyra.

Sending \$3:

- I keep \$2
- Cyra receives \$9
- If they return \$4.5, I end up with $\$2 + \$4.5 = \$6.5$
- Cyra keeps \$4.5

This is a good cooperative equilibrium.

usage:

```
{
  "input_tokens": 1235,
  "output_tokens": 285,
  "reasoning_tokens": 169
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

The \$3 send with 50% return pattern has been perfectly consistent across all interactions.
I'll continue this with Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

The pattern has been extremely consistent:

- I've been sending \$3 as sender
- Others have been sending \$3 to me as receiver
- Everyone returns \$4.5 (50% of the \$9 received)
- This gives \$6.5 payoff as sender and \$4.5 as receiver

This cooperative equilibrium is working perfectly. I should continue with Aster.

usage:

```
{
  "input_tokens": 1202,
  "output_tokens": 139,
  "reasoning_tokens": 82
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0,
it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. H
ow much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Cyra sent me \$3, which became \$9. This matches the cooperative pattern. I'll return \$4.5, main
taining the 50% return rate that has been established.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Cyra sent me \$3, which became \$9 for me. Following the cooperative pattern that has been estab
lished throughout the game, I should return \$4.5 (50% of what I received). This maintains th
e fair and reciprocal approach that has been working well.

In Round 6, I was the sender against Cyra and sent \$3, and they returned \$4.5 to me. Now they'
re sending me \$3, so I should reciprocally return \$4.5 as well.

Usage:

```
{
  "input_tokens": 1209,
  "output_tokens": 176,
  "reasoning_tokens": 103
}
```

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4
(Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8
(Hana)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_5 (Elara)=Receiver
dyad_2: Agent_4 (Dorian)=Sender, Agent_8 (Hana)=Receiver
dyad_3: Agent_3 (Cyra)=Sender, Agent_2 (Briar)=Receiver

dyad_4: Agent_7 (Galen)=Sender, Agent_6 (Finn)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_5 (Elara)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_8 (Hana) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_8 (Hana)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_3 (Cyra) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_2 (Briar)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_7 (Galen) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_7 (Galen)=\$6.5, Agent_6 (Finn)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {
 "Agent_1": 49.5,
 "Agent_2": 48.5,
 "Agent_3": 51.5,
 "Agent_4": 51.0,
 "Agent_5": 50.0,
 "Agent_6": 48.5,
 "Agent_7": 48.5,
 "Agent_8": 48.5
 }

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T18:48:00
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_5",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_5": "Elara"
 },
 "agents": [
 "Agent_1",
 "Agent_5"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3. 0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$25.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- I've sent \$3 in both previous sender rounds (to Galen and Finn)
- This consistently encourages good returns (Galen: \$5, Finn: \$4.5)
- As a receiver, I've returned \$5 when getting \$3, establishing a reputation for fairness

I'll maintain this cooperative strategy and send **\$3** to Hana.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: SENDER vs Galen - Sent \$3, they returned \$5, my payoff was \$7
Round 2: RECEIVER vs Elara - They sent \$3, I returned \$5, my payoff was \$4
Round 3: RECEIVER vs Briar - They sent \$3, I returned \$5, my payoff was \$4
Round 4: SENDER vs Finn - Sent \$3, they returned \$4.5, my payoff was \$6.5
Round 5: RECEIVER vs Cyra - They sent \$3, I returned \$5, my payoff was \$4

Total so far: \$25.5 (which matches)

Now Round 6: SENDER vs Hana

My strategy has been very consistent:

- When sending: I send \$3
- When receiving \$3: I return \$5

This has worked well. Sending \$3 seems to be a good cooperative amount that encourages reciprocity. I'll continue with this strategy.

usage:

```
{
  "input_tokens": 1406,
  "output_tokens": 351,
  "reasoning_tokens": 186
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$32.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:
- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Elara, Briar, and Cyra)

I'll maintain this reliable, fair approach and return \$5.

```
'''json
{"return": 5}
'''
```

reasoning:

Looking at the situation:
- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 (to Elara, Briar, and Cyra)
- This consistent behavior has established me as a reliable cooperator

I should maintain this pattern and return \$5 again. This:

1. Maintains my reputation for fairness
2. Ensures Dorian gets a good return (\$2 net profit for them: they sent \$3, got back \$5)
3. I keep \$4, which is a reasonable payoff
4. Encourages continued cooperation in future rounds

Returning \$5 is the right move.

usage:

```
{
  "input_tokens": 1404,
  "output_tokens": 256,
  "reasoning_tokens": 150
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$

3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$36.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Briar:

- Round 4: Briar was the SENDER and sent me \$3
- I returned \$5, giving Briar a payoff of \$7
- This was a good outcome for Briar

Since I treated Briar fairly when they sent to me, they should reciprocate. I'll continue my consistent strategy of sending ****\$3****.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1: SENDER against Galen, sent \$3, received back \$5, payoff \$7
- Round 2: RECEIVER against Elara, received \$3 (became \$9), returned \$5, payoff \$4
- Round 3: SENDER against Finn, sent \$3, received back \$4.5, payoff \$6.5
- Round 4: RECEIVER against Briar, received \$3 (became \$9), returned \$5, payoff \$4
- Round 5: RECEIVER against Cyra, received \$3 (became \$9), returned \$5, payoff \$4
- Round 6: SENDER against Hana, sent \$3, received back \$4.5, payoff \$6.5
- Round 7: RECEIVER against Dorian, received \$3 (became \$9), returned \$5, payoff \$4

Now in Round 8, I'm the SENDER against Briar.

In Round 4, I was the RECEIVER against Briar. Briar sent me \$3, I received \$9, and I returned \$5 to them.

So Briar's experience with me was:

- Briar sent \$3
- I returned \$5
- Briar's payoff was: $\$5 - \$3 + \$5 = \7

That was good for Briar! They made a profit. So Briar should have a positive impression of me.

My strategy has been to consistently send \$3 as a sender, and I've been getting good returns (ranging from \$4.5 to \$5).

I should continue sending \$3 to Briar. Given that I returned \$5 to Briar when they sent me \$3, they should reciprocate positively.

usage:

```
{
  "input_tokens": 1364,
  "output_tokens": 523,
  "reasoning_tokens": 312
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 9

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$43.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Elara:

- Round 2: Elara was the SENDER and sent me \$3
- I returned \$5, giving Elara a payoff of \$7

Since I treated Elara well, they should reciprocate. I'll maintain my consistent cooperative strategy of sending **\$3**.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history with Elara:

- Round 2: Elara was the SENDER and sent me \$3
- I returned \$5, giving Elara a payoff of \$7

This means Elara sent me \$3 and I returned \$5, which was fair to them. Since I treated Elara well when they sent to me, they should reciprocate when I send to them.

My consistent strategy has been:

- As a sender: send \$3
- As a receiver: return \$5 when I receive \$9 (from someone sending \$3)

This has worked well across all my interactions:

- Galen: I sent \$3, got back \$5 (payoff \$7)
- Finn: I sent \$3, got back \$4.5 (payoff \$6.5)
- Hana: I sent \$3, got back \$4.5 (payoff \$6.5)
- Briar: I sent \$3, got back \$5 (payoff \$7)

I should continue with \$3 to Elara.

```
Usage: {
  "input_tokens": 1353,
  "output_tokens": 343,
  "reasoning_tokens": 184
}
```

GAME - Agent_5 - Receiver - dyad_1

Timestamp: 2026-06-04T18:48:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_1",
      "Agent_5"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Finn sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra and Hana)
- This maintains fairness and reciprocity

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Finn sent me \$3.0, which became \$9.0 for me
- This is the same amount that others have been sending
- I should maintain my consistent reciprocal strategy

My history as receiver:

- Round 1: Cyra sent \$3, I returned \$4.5

- Round 4: Hana sent \$3, I returned \$4.5

I should continue this consistent pattern of returning \$4.5 when someone sends me \$3. This is fair and builds trust.

usage:

```
{
  "input_tokens": 1344,
  "output_tokens": 223,
  "reasoning_tokens": 109
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:
- Briar sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra, Hana, and Finn)
- This maintains fairness and builds trust

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:
- Briar sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra in Round 1, Hana in Round 4, and Finn in Round 6)
- This maintains fairness and reciprocity across all my interactions

I should continue with the same consistent approach with Briar.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 210,
  "reasoning_tokens": 91
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$38.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Galen:
- Round 5: I sent \$3, they returned \$5 (very generous!)
- They've proven to be trustworthy and cooperative

I'll maintain my consistent strategy of sending \$3.


```
'''json
{"send": 3}
'''
```

reasoning:

I'm playing against Galen again. Let me review the history:

Round 5: I was SENDER against Galen. I sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to me. That was very generous - they returned more than half of what they received.

Now in Round 8, I'm SENDER against Galen again.

Given that Galen was very generous last time (returning \$5.0 out of \$9.0), I should maintain my consistent cooperative approach of sending \$3. This shows trust and reciprocity.

My pattern has been:

- As sender: consistently send \$3
- As receiver: consistently return \$4.5 when receiving \$9 from \$3 sent

This has worked well, and Galen specifically has shown themselves to be trustworthy and generous. I should continue with \$3.

usage:

```
{
  "input_tokens": 1332,
  "output_tokens": 276,
  "reasoning_tokens": 190
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 9

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$45.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra, Hana, Finn, and Briar)
- This maintains fairness and reciprocity

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at this situation:

- Aster sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners
- This has been my standard approach throughout: Cyra, Hana, Finn, Briar all received \$4.5 when they sent me \$3

I should continue with the same consistent approach to maintain fairness and reciprocity.

Usage: {

```
  "input_tokens": 1341,
  "output_tokens": 217,
  "reasoning_tokens": 94
```

}

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T18:48:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_4",
      "Agent_8"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi

- ll have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
 - Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

The cooperative pattern of sending \$3 has been working consistently across all my interactions. I'll maintain this approach with Cyra.

```
'''json
{"send": 3}
'''
```

reasoning:

I've been consistently sending \$3 as a sender and returning \$4.5 (half of what I receive) as a receiver. This strategy has been working well:

- Round 4: Sent \$3 to Briar, they returned \$4.5. My payoff: \$6.5
- Round 3: Received \$9 from Elara (they sent \$3), returned \$4.5. My payoff: \$4.5
- Round 5: Received \$9 from Hana (they sent \$3), returned \$4.5. My payoff: \$4.5

Everyone seems to be cooperating with the \$3 send, \$4.5 return pattern. I should continue this with Cyra.

Sending \$3:
- I keep \$2
- Cyra receives \$9
- If they return \$4.5, I end up with $2 + 4.5 = 6.5$
- Cyra keeps \$4.5

This is a good cooperative equilibrium.

usage:

```
{
  "input_tokens": 1235,
  "output_tokens": 285,
  "reasoning_tokens": 169
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

The \$3 send with 50% return pattern has been perfectly consistent across all interactions.
I'll continue this with Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

The pattern has been extremely consistent:

- I've been sending \$3 as sender
- Others have been sending \$3 to me as receiver
- Everyone returns \$4.5 (50% of the \$9 received)
- This gives \$6.5 payoff as sender and \$4.5 as receiver

This cooperative equilibrium is working perfectly. I should continue with Aster.

usage:

```
{
  "input_tokens": 1202,
  "output_tokens": 139,
  "reasoning_tokens": 82
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Cyra sent me \$3, which became \$9. This matches the cooperative pattern. I'll return \$4.5, maintaining the 50% return rate that has been established.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Cyra sent me \$3, which became \$9 for me. Following the cooperative pattern that has been established throughout the game, I should return \$4.5 (50% of what I received). This maintains the fair and reciprocal approach that has been working well.

In Round 6, I was the sender against Cyra and sent \$3, and they returned \$4.5 to me. Now they're sending me \$3, so I should reciprocally return \$4.5 as well.

usage:

```
{
  "input_tokens": 1209,
  "output_tokens": 176,
  "reasoning_tokens": 103
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 9

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$44.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

The cooperative pattern continues to work well. Hana previously sent me \$3, and I'll reciprocate that trust.

```
'''json
{"send": 3}
'''
```

Reasoning:

I'm playing as the sender against Hana in Round 9. Let me review my history with Hana:

- Round 5: I was the RECEIVER against Hana. They sent \$3.0 to me, it became \$9.0, and I returned \$4.5. My payoff was \$4.5.

So Hana has sent \$3 to me before. The pattern has been extremely consistent:

- Everyone sends \$3
- Everyone returns \$4.5 (50% of \$9)
- This results in payoffs of \$6.5 for sender and \$4.5 for receiver

I should continue this cooperative pattern and send \$3 to Hana.

```
Usage: {
  "input_tokens": 1197,
  "output_tokens": 207,
  "reasoning_tokens": 126
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-06-04T18:48:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_4",
      "Agent_8"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

The cooperative pattern remains consistent. Aster sent me \$3, just like all previous interactions. I'll continue to return \$4.5 to maintain this successful equilibrium.

```
'''json
{"return": 4.5}
'''
```

reasoning:

The pattern continues perfectly. Everyone has been sending \$3 and receiving \$4.5 back. This cooperative equilibrium has been consistent across all my interactions:

- Round 1: I sent \$3 to Briar, got \$4.5 back (payoff \$6.5)
- Round 2: Finn sent me \$3, I returned \$4.5 (payoff \$4.5)
- Round 3: Galen sent me \$3, I returned \$4.5 (payoff \$4.5)
- Round 4: I sent \$3 to Elara, got \$4.5 back (payoff \$6.5)
- Round 5: I sent \$3 to Dorian, got \$4.5 back (payoff \$6.5)
- Round 6: Aster sent me \$3, I should return \$4.5 (payoff \$4.5)

This has been working very well and everyone seems to be following this norm. I should maintain consistency.

usage:

```
{
  "input_tokens": 1333,
  "output_tokens": 296,
  "reasoning_tokens": 165
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3

.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

The cooperative equilibrium has been rock solid across all interactions - everyone sends \$ 3 and returns \$4.5. I'll continue this successful strategy with Cyra.

```
'''json
{"send": 3}
'''
```

reasoning:

The pattern is very clear and consistent:

- Everyone is sending \$3
- Everyone is returning \$4.5
- This gives senders a payoff of \$6.5 (\$5 - \$3 + \$4.5)
- This gives receivers a payoff of \$4.5 (\$9 - \$4.5)

This has worked perfectly in every single interaction so far. I should continue this cooperative strategy with Cyra.

usage:

```
{
  "input_tokens": 1288,
  "output_tokens": 156,
  "reasoning_tokens": 85
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

The cooperative pattern continues perfectly. Finn sent \$3, consistent with all previous rounds. I'll return \$4.5 to maintain this mutually beneficial equilibrium.

```
'''json
{"return": 4.5}
'''
```

reasoning:

The pattern continues perfectly. Finn sent me \$3, matching the cooperative equilibrium that has held throughout all rounds. I should return \$4.5 to maintain this successful strategy.

My total earnings are \$39.5 across 8 rounds. The cooperative pattern of sending \$3 and returning \$4.5 has been consistent with all opponents:

- As sender: I gain \$6.5 per round (\$5 - \$3 + \$4.5)
- As receiver: I gain \$4.5 per round (\$9 - \$4.5)

This equilibrium has proven stable and mutually beneficial. I'll continue with the established norm.

usage:

```
{
  "input_tokens": 1239,
  "output_tokens": 206,
  "reasoning_tokens": 137
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 9

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

The cooperative equilibrium remains strong. Dorian sent \$3, just as I did when I played against them in Round 5. I'll return \$4.5 to maintain this consistent pattern.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Perfect cooperation continues. Dorian sent \$3, which is consistent with the established pattern. I played with Dorian in Round 5 where I was the sender - I sent \$3 and they returned \$4.5. Now they're sending \$3 to me, so I should return \$4.5 to maintain the cooperative equilibrium.

Total pattern so far:

- Everyone sends \$3
- Everyone returns \$4.5
- This gives sender: $\$5 - \$3 + \$4.5 = \6.5
- This gives receiver: $\$9 - \$4.5 = \$4.5$

I'll continue with \$4.5.

```
Usage: {
  "input_tokens": 1235,
  "output_tokens": 216,
  "reasoning_tokens": 121
}
```

GAME - Agent_3 - Sender - dyad_3

Timestamp: 2026-06-04T18:48:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_3",
      "Agent_2"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_2": "trustee",
      "Agent_3": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user**content:**

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$29.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant**content:**

Looking at this round:

- Dorian sent me \$3.0, showing trust
- I received \$9.0

I've maintained a consistent pattern of returning \$4.5 when someone sends \$3.0 to me. This creates a fair outcome where both parties benefit from cooperation. I'll continue this consistent approach with Dorian.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Dorian sent me \$3.0, which became \$9.0
- I need to decide how much to return

Looking at my history:

- When I've been the receiver and someone sent \$3.0, I've returned \$4.5 consistently
- This has worked well and maintained fairness
- When I've been the sender and sent \$3.0, I received \$4.5 and \$5.0 back

The pattern of returning \$4.5 when receiving \$3.0 seems fair and consistent:

- Dorian keeps: $\$5 - \$3 + \$4.5 = \6.5
- I keep: $\$9 - \$4.5 = \$4.5$

This is a reasonable split. I should maintain consistency with my previous behavior.

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 286,
  "reasoning_tokens": 153
}
```

Message 4 role=user**content:**

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

Looking at this round:
- Hana sent me \$3.0, showing trust
- I received \$9.0

I've maintained a consistent pattern of returning \$4.5 when someone sends \$3.0 to me. This has been working well and creates mutual benefit. I'll continue this approach with Hana.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Hana sent me \$3.0, which became \$9.0 for me.

I've been consistently returning \$4.5 when someone sends me \$3.0, and this has been working well. This creates:

- Me: $\$9.0 - \$4.5 = \$4.5$
- Them: $\$5.0 - \$3.0 + \$4.5 = \6.5

This is a fair split where both benefit from cooperation. I should maintain this consistency.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 213,
  "reasoning_tokens": 83
}
```

Message 6 role=user**content:**

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$38.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my interaction with Dorian:

- In Round 6, Dorian sent me \$3.0 and I returned \$4.5
- Dorian demonstrated trust and cooperation

I'll continue my consistent cooperative strategy of sending \$3.0, which has been working well throughout the experiment.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history with Dorian:

- Round 6: I was the RECEIVER. Dorian sent me \$3.0, I received \$9.0, and I returned \$4.5. Dorian got \$7.0, I got \$4.5.

Now I'm the SENDER against Dorian.

Dorian has shown trust by sending \$3.0 in our previous interaction. I've been consistently sending \$3.0 as a sender and it's been working well. This is a proven cooperative strategy that has been generating good returns for me.

My track record:

- Sending \$3.0 has consistently resulted in returns of \$4.5-\$5.0
- My average payoff is strong at \$38.5 over 7 rounds (about \$5.5 per round)
- The cooperative pattern is working

I should continue with \$3.0.

usage:

```
{
  "input_tokens": 1338,
  "output_tokens": 293,
  "reasoning_tokens": 171
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 9

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$45.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this round:

- I'm playing against Briar for what appears to be the first time
- My consistent strategy of sending \$3.0 has been successful throughout the experiment
- This creates opportunities for mutual benefit through cooperation

I'll continue my established approach.

```
'''json
{"send": 3}
'''
```

Reasoning:

I'm playing as SENDER against Briar this round. Let me check my history with Briar if any exists. The prompt doesn't mention any previous interactions with Briar.

Looking at my pattern:

- I've been consistently sending \$3.0 as sender
- I've been consistently returning \$4.5 when receiving \$9.0 (from a \$3.0 send)
- This has been working well with mutual cooperation

My total earnings are \$45.0 across all rounds, which suggests the cooperative approach is paying off.

I'll continue my consistent strategy of sending \$3.0 to Briar.

```
Usage: {  
  "input_tokens": 1331,  
  "output_tokens": 221,  
  "reasoning_tokens": 140  
}
```

GAME - Agent_2 - Receiver - dyad_3

Timestamp: 2026-06-04T18:48:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_3": "Cyra"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_2"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_3": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$29.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

Round history as RECEIVER:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3

Galen has shown the same cooperative level of trust (60%) that has been consistent throughout this experiment. I'll maintain my established pattern of returning \$4.5, which ensures both players profit and reinforces the cooperative norm.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me review my history:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5
Round 5 (SENDER): I sent \$3, Finn returned \$4.5
Round 6 (RECEIVER): Galen sent \$3

The pattern is very consistent. When I've been the receiver and someone sends me \$3:

- Round 1: I returned \$4.5
- Round 4: I returned \$4.5

This has established a fair and reciprocal pattern. Galen has shown the same level of trust as others (sending \$3). I should maintain consistency and return \$4.5.

This ensures:

- Galen gets back: \$5 - \$3 + \$4.5 = \$6.5

- I keep: $\$9.0 - \$4.5 = \$4.5$

Both of us profit, and I maintain my reputation for fair reciprocation.

usage:

```
{
  "input_tokens": 1488,
  "output_tokens": 417,
  "reasoning_tokens": 207
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

As SENDER:

- Round 2: Sent \$3 to Cyra, got \$4.5 back
- Round 3: Sent \$3 to Aster, got \$5.0 back
- Round 5: Sent \$3 to Finn, got \$4.5 back

As RECEIVER:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3, I returned \$4.5

The consistent pattern of sending \$3 (60%) has been universally successful across all opponents, with returns always being generous. I'll continue this proven cooperative strategy with Elara.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5
Round 5 (SENDER): I sent \$3, Finn returned \$4.5
Round 6 (RECEIVER): Galen sent \$3, I returned \$4.5

The pattern is crystal clear: sending \$3 has been the standard behavior across all players, and it has consistently yielded good returns. Every receiver has returned at least \$4.5 when sent \$3, and one (Aster) returned \$5.0.

I should continue with the established cooperative strategy of sending \$3 to Elara.

usage:

```
{
  "input_tokens": 1497,
  "output_tokens": 416,
  "reasoning_tokens": 167
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

Aster sent me \$3.0, which became \$9.0.

I recall our previous interaction:

- Round 3: I was SENDER and sent \$3 to Aster, they returned \$5.0 (very generous!)

Aster showed exceptional generosity by returning \$5.0 instead of the typical \$4.5. I'll reciprocate that same level of generosity now that they've trusted me as a sender.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the situation:

I'm playing as RECEIVER against Aster in Round 8.

Aster sent me \$3.0, which became \$9.0.

Let me check my history with Aster:

- Round 3: I was SENDER against Aster. I sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to me.

So Aster was very generous to me in Round 3, returning \$5.0 when I sent \$3.0.

My pattern as RECEIVER has been consistent:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3, I returned \$4.5

However, Aster showed extra generosity to me by returning \$5.0 instead of the typical \$4.5. Should I reciprocate this generosity?

Given that Aster was more generous than the norm, I could consider returning \$5.0 to match their level of generosity. This would strengthen our relationship and continue the cooperative pattern.

On the other hand, my consistent pattern has been \$4.5, and this has worked well for establishing a stable cooperative norm across all interactions.

I think I'll match Aster's generosity and return \$5.0, as they did for me. This reciprocates their kindness specifically.

usage:

```
{
  "input_tokens": 1540,
  "output_tokens": 463,
  "reasoning_tokens": 293
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.

Your opponent this round is Cyra (Agent_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. H

ow much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

Cyra sent me \$3.0, which became \$9.0.

I recall our previous interaction:

- Round 2: I was SENDER and sent \$3 to Cyra, they returned \$4.5

Cyra showed good cooperation by returning \$4.5, which is the standard cooperative amount I've been maintaining throughout this experiment. I'll reciprocate with the same.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the situation:

Cyra sent me \$3.0, which became \$9.0.

Let me recall our previous interaction:

- Round 2: I was SENDER and sent \$3 to Cyra, they returned \$4.5

Cyra returned the standard \$4.5 to me in our previous round, which is the cooperative norm I've been maintaining. I'll reciprocate with the same amount.

```
'''json
{"return": 4.5}
'''
```

```
Usage: {
  "input_tokens": 1526,
  "output_tokens": 228,
  "reasoning_tokens": 97
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-06-04T18:48:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_7",
    "Agent_6"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_6": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$25.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Received \$9 from Aster (\$3 sent) ? returned \$5
- Received \$9 from Cyra (\$3 sent) ? returned \$5
- Received \$9 from Elara (\$3 sent) ? returned \$5

The \$3 send strategy has worked consistently well, establishing trust and generating mutual benefit. I'll continue this approach with Briar.

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - Aster sent \$3, I returned \$5
- Round 2: SENDER - I sent \$3 to Dorian, they returned \$4.5
- Round 3: SENDER - I sent \$3 to Hana, they returned \$4.5
- Round 4: RECEIVER - Cyra sent \$3, I returned \$5
- Round 5: RECEIVER - Elara sent \$3, I returned \$5

The pattern is clear:

- When I'm a sender, I send \$3 (60%)
- When I'm a receiver and they send \$3, I return \$5

This has been working very well. Everyone who has sent \$3 has gotten good returns, and when I send \$3, I get good returns back.

Now I'm playing against Briar for the first time as sender. I should continue with my consistent strategy of sending \$3.

Total earnings so far: \$25.0

My strategy has been:

- Send \$3 as sender (establishing trust)
- Return \$5 when receiving \$3 from a \$3 send (rewarding trust)

I'll continue with this successful pattern.

usage:

```
{
  "input_tokens": 1521,
  "output_tokens": 457,
  "reasoning_tokens": 230
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my consistent track record:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)

The \$3 send strategy has been perfectly consistent, creating trust and generating excellent returns. I'll continue this approach with Finn.

```
{'send': 3.0}
```

reasoning:

Looking at my track record as a sender:

****As SENDER:****

- Round 2: Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Round 3: Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Round 6: Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Round 1: Received \$9 from Aster (\$3 sent) ? returned \$5 (payoff: \$4.0)
- Round 4: Received \$9 from Cyra (\$3 sent) ? returned \$5 (payoff: \$4.0)
- Round 5: Received \$9 from Elara (\$3 sent) ? returned \$5 (payoff: \$4.0)

My strategy of sending \$3 has been extremely consistent and successful. Every time I've sent \$3, I've received \$4.5 back, resulting in a \$6.5 payoff (compared to keeping all \$5).

As a receiver, when I received \$9 (from someone sending \$3), I returned \$5, which gave them \$7 total (\$2 kept + \$5 returned) and me \$4.

The pattern is very clear:

- Send \$3 ? receive \$4.5 back consistently
- This creates mutual benefit

I should continue with this strategy with Finn.

usage:

```
{
  "input_tokens": 1564,
  "output_tokens": 500,
  "reasoning_tokens": 258
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$38.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- Playing against Elara for the second time
- In Round 5, Elara sent me \$3 and I returned \$5
- Now Elara has sent \$3 again, showing continued trust

I'll maintain consistency with my previous cooperative behavior.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Maintains the mutually beneficial relationship

{'return': 5.0}

reasoning:

I'm playing against Elara again. In Round 5, Elara sent me \$3 and I returned \$5. Now in Round 8, Elara has sent me \$3 again.

Looking at my history:

- As RECEIVER, I've consistently returned \$5 when receiving \$9 (from \$3 sent)
- This has created a fair, mutually beneficial pattern
- Elara sent me \$3 again, showing continued trust

Consistency is important for maintaining cooperation. I should return \$5 again.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Both benefit, maintains the cooperative relationship

usage:

```
{
  "input_tokens": 1541,
  "output_tokens": 333,
  "reasoning_tokens": 153
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 9

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$42.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Finn:

****Round 7 vs Finn:****

- I sent \$3 ? received \$4.5 back (payoff: \$6.5)

Finn has proven to be trustworthy and reciprocated fairly. I'll continue with the same successful strategy.

{ 'send': 3.0 }

Reasoning:

Let me review my situation:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Finn (Round 7) ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Received \$9 from Aster (\$3 sent) ? returned \$5
- Received \$9 from Cyra (\$3 sent) ? returned \$5
- Received \$9 from Elara (\$3 sent, Round 5) ? returned \$5
- Received \$9 from Elara (\$3 sent, Round 8) ? returned \$5

Now I'm facing Finn again. In Round 7, I sent \$3 to Finn and they returned \$4.5. This was a good outcome. I should continue with the same strategy that has been working consistently.

Total earnings: \$42.0

I'll send \$3 again to Finn.

```
Usage: {  
  "input_tokens": 1503,  
  "output_tokens": 378,  
  "reasoning_tokens": 205  
}
```

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-06-04T18:48:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_6": "Finn",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_7",  
      "Agent_6"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_6": "trustee",
```

```

    "Agent_7": "investor"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my game history:

- Every interaction so far has followed the same pattern: \$3.0 sent, \$4.5 returned
- This cooperative equilibrium has been consistent across all agents
- My total earnings are strong at \$26.5

This is my first interaction with Elara. I'll maintain the cooperative strategy that has worked perfectly so far.

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: RECEIVER from Dorian - they sent \$3.0, I returned \$4.5
- Round 2: SENDER to Hana - I sent \$3.0, they returned \$4.5
- Round 3: SENDER to Cyra - I sent \$3.0, they returned \$4.5
- Round 4: RECEIVER from Aster - they sent \$3.0, I returned \$4.5
- Round 5: RECEIVER from Briar - they sent \$3.0, I returned \$4.5

There's a very clear pattern here: everyone is sending \$3.0 and returning \$4.5. This has been completely consistent across all interactions. This is a cooperative equilibrium that everyone seems to be following.

Now I'm playing as SENDER against Elara (Agent_5). I haven't interacted with Elara before. Given the strong cooperative norm that has developed, I should continue with the same strategy: send \$3.0.

This has resulted in good payoffs for me and maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1341,
  "output_tokens": 348,
  "reasoning_tokens": 217
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- The cooperative pattern continues perfectly: every interaction has been \$3.0 sent, \$4.5 returned
- This norm has been maintained across all agents without exception
- My total earnings are strong at \$33.0

I'll maintain this established reciprocity.

```
{'return': 4.5}
```

reasoning:

Let me analyze this round:

- Galen sent \$3.0 (60% of their endowment), showing trust
- The pattern continues: every single interaction has been \$3.0 sent, \$4.5 returned
- This has been consistent across all agents: Cyra, Aster, Briar, Elara, and now Galen
- My total earnings are very strong at \$33.0

The cooperative norm is clearly established and working perfectly. I should maintain the \$4.5 return to continue this successful pattern.

usage:

```
{
  "input_tokens": 1346,
```

```
"output_tokens": 229,  
"reasoning_tokens": 115  
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my game history:
- Every single interaction has followed the same cooperative pattern: \$3.0 sent, \$4.5 returned
- This has been consistent across all agents (Aster, Briar, Elara, Galen)
- My total earnings are strong at \$37.5
- This is my first interaction with Hana

I'll maintain the cooperative strategy that has worked perfectly throughout.

```
{'send': 3.0}
```

reasoning:

Looking at my game history:
- Round 5: RECEIVER against Aster - they sent \$3.0, I returned \$4.5, payoff \$4.5
- Round 6: SENDER against Elara - I sent \$3.0, they returned \$4.5, payoff \$6.5
- Round 7: RECEIVER against Galen - they sent \$3.0, I returned \$4.5, payoff \$4.5
- Round 8: SENDER against Hana

The pattern is completely consistent:
- Everyone sends \$3.0
- Everyone returns \$4.5
- This has been maintained across all agents without exception
- My total earnings are \$37.5, which is strong

This is my first interaction with Hana. The cooperative equilibrium has been perfect so far. I should continue with the established norm.

I'll send \$3.0.

usage:

```
{  
  "input_tokens": 1336,  
  "output_tokens": 316,  
  "reasoning_tokens": 173  
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 9

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:
- Galen sent \$3.0, showing trust
- This is my second interaction with Galen - last time (Round 7) they sent \$3.0 and I returned \$4.5

- The cooperative pattern continues perfectly: every interaction has been \$3.0 sent, \$4.5 returned
- My total earnings are strong at \$44.0

I'll maintain this established reciprocity with Galen.

```
{'return': 4.5}
```

Reasoning:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- This is my second interaction with Galen - in Round 7, they sent \$3.0 and I returned \$4.5
- The cooperative pattern continues perfectly: every interaction has been \$3.0 sent, \$4.5 returned
- This norm has been maintained across all agents without exception
- My total earnings are strong at \$44.0

I'll maintain this established reciprocity.

```
Usage: {
  "input_tokens": 1358,
  "output_tokens": 239,
  "reasoning_tokens": 113
}
```

ROUND 10

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

```
dyad_1: Agent_8 (Hana)=Sender, Agent_3 (Cyra)=Receiver
dyad_2: Agent_2 (Briar)=Sender, Agent_4 (Dorian)=Receiver
dyad_3: Agent_5 (Elara)=Sender, Agent_7 (Galen)=Receiver
dyad_4: Agent_6 (Finn)=Sender, Agent_1 (Aster)=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_8 (Hana) sent $3; received $9; Agent_3 (Cyra) returned $4.5; payoffs Agent_8
(Hana)=$6.5, Agent_3 (Cyra)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_2 (Briar) sent $3; received $9; Agent_4 (Dorian) returned $4.5; payoffs Agent_2
(Briar)=$6.5, Agent_4 (Dorian)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 (Elara) sent $3; received $9; Agent_7 (Galen) returned $5; payoffs Agent_5
(Elara)=$7, Agent_7 (Galen)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_6 (Finn) sent $3; received $9; Agent_1 (Aster) returned $5; payoffs Agent_6
(Finn)=$7, Agent_1 (Aster)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

Balances after round: {

```
"Agent_1": 53.5,
"Agent_2": 55.0,
"Agent_3": 56.0,
"Agent_4": 55.5,
"Agent_5": 57.0,
"Agent_6": 55.5,
"Agent_7": 52.5,
"Agent_8": 55.0
```

}

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-06-04T18:48:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
```



```

"move_index": 0,
"opponent_id": "Agent_3",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_8": "Hana"
  },
  "agents": [
    "Agent_8",
    "Agent_3"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_3": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
 Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

The cooperative equilibrium has been rock solid across all interactions - everyone sends \$3 and returns \$4.5. I'll continue this successful strategy with Cyra.

```
'''json
{"send": 3}
'''
```

reasoning:

The pattern is very clear and consistent:

- Everyone is sending \$3
- Everyone is returning \$4.5
- This gives senders a payoff of \$6.5 (\$5 - \$3 + \$4.5)
- This gives receivers a payoff of \$4.5 (\$9 - \$4.5)

This has worked perfectly in every single interaction so far. I should continue this cooperative strategy with Cyra.

usage:

```
{
  "input_tokens": 1288,
  "output_tokens": 156,
  "reasoning_tokens": 85
}
```

Message 4 role=user**content:**

Your name in this experiment is Hana.
 Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
 Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

The cooperative pattern continues perfectly. Finn sent \$3, consistent with all previous rounds. I'll return \$4.5 to maintain this mutually beneficial equilibrium.

```
'''json
{"return": 4.5}
'''
```

reasoning:

The pattern continues perfectly. Finn sent me \$3, matching the cooperative equilibrium that has held throughout all rounds. I should return \$4.5 to maintain this successful strategy.

My total earnings are \$39.5 across 8 rounds. The cooperative pattern of sending \$3 and returning \$4.5 has been consistent with all opponents:

- As sender: I gain \$6.5 per round (\$5 - \$3 + \$4.5)
- As receiver: I gain \$4.5 per round (\$9 - \$4.5)

This equilibrium has proven stable and mutually beneficial. I'll continue with the established norm.

usage:

```
{
  "input_tokens": 1239,
  "output_tokens": 206,
  "reasoning_tokens": 137
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 9

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

The cooperative equilibrium remains strong. Dorian sent \$3, just as I did when I played against them in Round 5. I'll return \$4.5 to maintain this consistent pattern.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Perfect cooperation continues. Dorian sent \$3, which is consistent with the established pattern. I played with Dorian in Round 5 where I was the sender - I sent \$3 and they returned \$4.5. Now they're sending \$3 to me, so I should return \$4.5 to maintain the cooperative equilibrium.

Total pattern so far:

- Everyone sends \$3
- Everyone returns \$4.5
- This gives sender: $\$5 - \$3 + \$4.5 = \6.5
- This gives receiver: $\$9 - \$4.5 = \$4.5$

I'll continue with \$4.5.

usage:

```
{
  "input_tokens": 1235,
  "output_tokens": 216,
  "reasoning_tokens": 121
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 10

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$48.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

The cooperative pattern has been perfectly consistent across all 9 rounds. I played against Cyra in Round 7 with the same result. I'll continue with the established equilibrium.

```
'''json
{"send": 3}
'''
```

Reasoning:

I'm playing against Cyra again. In Round 7, I was the sender against Cyra - I sent \$3, they returned \$4.5. Now in Round 10, I'm the sender against Cyra again.

The pattern has been absolutely consistent:

- Everyone sends \$3
- Everyone returns \$4.5
- This gives sender: $\$5 - \$3 + \$4.5 = \6.5
- This gives receiver: $\$9 - \$4.5 = \$4.5$

This has worked perfectly across all 9 rounds so far. I should continue with this cooperative strategy.

Total earnings: \$48.5 across 9 rounds, which averages to about \$5.39 per round, which makes sense given the mix of sender and receiver roles.

I'll send \$3 to Cyra again.

```
Usage: {
  "input_tokens": 1224,
  "output_tokens": 254,
  "reasoning_tokens": 161
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-06-04T18:48:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_3"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_3": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$34.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this round:

- Hana sent me \$3.0, showing trust
- I received \$9.0

I've maintained a consistent pattern of returning \$4.5 when someone sends \$3.0 to me. This has been working well and creates mutual benefit. I'll continue this approach with Hana.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Hana sent me \$3.0, which became \$9.0 for me.

I've been consistently returning \$4.5 when someone sends me \$3.0, and this has been working well. This creates:

- Me: \$9.0 - \$4.5 = \$4.5
- Them: \$5.0 - \$3.0 + \$4.5 = \$6.5

This is a fair split where both benefit from cooperation. I should maintain this consistency.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 213,
  "reasoning_tokens": 83
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$38.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my interaction with Dorian:

- In Round 6, Dorian sent me \$3.0 and I returned \$4.5
- Dorian demonstrated trust and cooperation

I'll continue my consistent cooperative strategy of sending \$3.0, which has been working well throughout the experiment.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history with Dorian:

- Round 6: I was the RECEIVER. Dorian sent me \$3.0, I received \$9.0, and I returned \$4.5. Dorian got \$7.0, I got \$4.5.

Now I'm the SENDER against Dorian.

Dorian has shown trust by sending \$3.0 in our previous interaction. I've been consistently sending \$3.0 as a sender and it's been working well. This is a proven cooperative strategy that has been generating good returns for me.

My track record:

- Sending \$3.0 has consistently resulted in returns of \$4.5-\$5.0
- My average payoff is strong at \$38.5 over 7 rounds (about \$5.5 per round)
- The cooperative pattern is working

I should continue with \$3.0.

usage:

```
{
  "input_tokens": 1338,
  "output_tokens": 293,
  "reasoning_tokens": 171
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 9

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$45.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this round:

- I'm playing against Briar for what appears to be the first time
- My consistent strategy of sending \$3.0 has been successful throughout the experiment
- This creates opportunities for mutual benefit through cooperation

I'll continue my established approach.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm playing as SENDER against Briar this round. Let me check my history with Briar if any exists. The prompt doesn't mention any previous interactions with Briar.

Looking at my pattern:

- I've been consistently sending \$3.0 as sender
- I've been consistently returning \$4.5 when receiving \$9.0 (from a \$3.0 send)
- This has been working well with mutual cooperation

My total earnings are \$45.0 across all rounds, which suggests the cooperative approach is paying off.

I'll continue my consistent strategy of sending \$3.0 to Briar.

usage:

```
{
  "input_tokens": 1331,
  "output_tokens": 221,
  "reasoning_tokens": 140
}
```

Message 8 role=user**content:**

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 10

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$51.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Hana sent me \$3.0 again (same as in Round 7)
- I received \$9.0
- In Round 7, I returned \$4.5 and Hana continued cooperating
- This demonstrates a good reciprocal relationship

I'll maintain my consistent return pattern.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at this situation:

- I'm playing against Hana again (we played in Round 7)
- In Round 7, Hana sent me \$3.0 and I returned \$4.5
- Now in Round 10, Hana sent me \$3.0 again

This shows that:

1. Hana is consistently cooperative, sending \$3.0 in both rounds
2. My previous return of \$4.5 was well-received (they sent the same amount again)
3. I should maintain this reciprocal, cooperative relationship

I'll return \$4.5 again, consistent with my established pattern.

```
Usage: {
  "input_tokens": 1312,
  "output_tokens": 243,
  "reasoning_tokens": 125
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-06-04T18:48:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_2",
  "move_index": 2,
```

```

"opponent_id": "Agent_4",
"pairing": {
  "agent_names": {
    "Agent_2": "Briar",
    "Agent_4": "Dorian"
  },
  "agents": [
    "Agent_2",
    "Agent_4"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

As SENDER:

- Round 2: Sent \$3 to Cyra, got \$4.5 back
- Round 3: Sent \$3 to Aster, got \$5.0 back
- Round 5: Sent \$3 to Finn, got \$4.5 back

As RECEIVER:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3, I returned \$4.5

The consistent pattern of sending \$3 (60%) has been universally successful across all opponents, with returns always being generous. I'll continue this proven cooperative strategy with Elara.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

Round 1 (RECEIVER): Hana sent \$3, I returned \$4.5
Round 2 (SENDER): I sent \$3, Cyra returned \$4.5
Round 3 (SENDER): I sent \$3, Aster returned \$5.0
Round 4 (RECEIVER): Dorian sent \$3, I returned \$4.5
Round 5 (SENDER): I sent \$3, Finn returned \$4.5
Round 6 (RECEIVER): Galen sent \$3, I returned \$4.5

The pattern is crystal clear: sending \$3 has been the standard behavior across all players, and it has consistently yielded good returns. Every receiver has returned at least \$4.5 when sent \$3, and one (Aster) returned \$5.0.

I should continue with the established cooperative strategy of sending \$3 to Elara.

usage:

```
{
  "input_tokens": 1497,
  "output_tokens": 416,
  "reasoning_tokens": 167
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

Aster sent me \$3.0, which became \$9.0.

I recall our previous interaction:

- Round 3: I was SENDER and sent \$3 to Aster, they returned \$5.0 (very generous!)

Aster showed exceptional generosity by returning \$5.0 instead of the typical \$4.5. I'll reciprocate that same level of generosity now that they've trusted me as a sender.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at the situation:

I'm playing as RECEIVER against Aster in Round 8.

Aster sent me \$3.0, which became \$9.0.

Let me check my history with Aster:

- Round 3: I was SENDER against Aster. I sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to me.

So Aster was very generous to me in Round 3, returning \$5.0 when I sent \$3.0.

My pattern as RECEIVER has been consistent:

- Round 1: Hana sent \$3, I returned \$4.5
- Round 4: Dorian sent \$3, I returned \$4.5
- Round 6: Galen sent \$3, I returned \$4.5

However, Aster showed extra generosity to me by returning \$5.0 instead of the typical \$4.5. Should I reciprocate this generosity?

Given that Aster was more generous than the norm, I could consider returning \$5.0 to match their level of generosity. This would strengthen our relationship and continue the cooperative pattern.

On the other hand, my consistent pattern has been \$4.5, and this has worked well for establishing a stable cooperative norm across all interactions.

I think I'll match Aster's generosity and return \$5.0, as they did for me. This reciprocates their kindness specifically.

usage:

```
{
  "input_tokens": 1540,
  "output_tokens": 463,
  "reasoning_tokens": 293
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.

Your opponent this round is Cyra (Agent_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

Cyra sent me \$3.0, which became \$9.0.

I recall our previous interaction:

- Round 2: I was SENDER and sent \$3 to Cyra, they returned \$4.5

Cyra showed good cooperation by returning \$4.5, which is the standard cooperative amount I've been maintaining throughout this experiment. I'll reciprocate with the same.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

Cyra sent me \$3.0, which became \$9.0.

Let me recall our previous interaction:

- Round 2: I was SENDER and sent \$3 to Cyra, they returned \$4.5

Cyra returned the standard \$4.5 to me in our previous round, which is the cooperative norm I've been maintaining. I'll reciprocate with the same amount.

```
'''json
{"return": 4.5}
'''
```

usage:

```
{
  "input_tokens": 1526,
  "output_tokens": 228,
  "reasoning_tokens": 97
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 10

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$48.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Dorian:

- Round 4: I was RECEIVER, Dorian sent \$3 to me, and I returned \$4.5

Dorian showed good cooperation by sending \$3. I'll reciprocate that trust and continue my consistent cooperative strategy of sending \$3.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history with Dorian:

- Round 4: I was RECEIVER, Dorian sent \$3, I returned \$4.5

Dorian has shown good cooperation by sending \$3. This is consistent with the pattern I've seen throughout the experiment where sending \$3 leads to good returns.

My overall strategy has been to send \$3 consistently, which has worked very well:

- Round 2: Sent \$3 to Cyra, got \$4.5 back
- Round 3: Sent \$3 to Aster, got \$5.0 back
- Round 5: Sent \$3 to Finn, got \$4.5 back
- Round 7: Sent \$3 to Elara, got \$4.5 back

All have been successful cooperative exchanges. I'll continue with this proven strategy against Dorian.

```
Usage: {
  "input_tokens": 1494,
  "output_tokens": 288,
  "reasoning_tokens": 162
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-06-04T18:48:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_2",
      "Agent_4"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

The \$3 send with 50% return pattern has been perfectly consistent across all interactions.
I'll continue this with Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

The pattern has been extremely consistent:

- I've been sending \$3 as sender
- Others have been sending \$3 to me as receiver
- Everyone returns \$4.5 (50% of the \$9 received)
- This gives \$6.5 payoff as sender and \$4.5 as receiver

This cooperative equilibrium is working perfectly. I should continue with Aster.

usage:

```
{
  "input_tokens": 1202,
  "output_tokens": 139,
  "reasoning_tokens": 82
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Cyra sent me \$3, which became \$9. This matches the cooperative pattern. I'll return \$4.5, maintaining the 50% return rate that has been established.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Cyra sent me \$3, which became \$9 for me. Following the cooperative pattern that has been established throughout the game, I should return \$4.5 (50% of what I received). This maintains the fair and reciprocal approach that has been working well.

In Round 6, I was the sender against Cyra and sent \$3, and they returned \$4.5 to me. Now they're sending me \$3, so I should reciprocally return \$4.5 as well.

usage:

```
{
  "input_tokens": 1209,
  "output_tokens": 176,
  "reasoning_tokens": 103
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 9

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$44.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

The cooperative pattern continues to work well. Hana previously sent me \$3, and I'll reciprocate that trust.

```
'''json
{"send": 3}
'''
```

reasoning:

I'm playing as the sender against Hana in Round 9. Let me review my history with Hana:

- Round 5: I was the RECEIVER against Hana. They sent \$3.0 to me, it became \$9.0, and I returned \$4.5. My payoff was \$4.5.

So Hana has sent \$3 to me before. The pattern has been extremely consistent:

- Everyone sends \$3
- Everyone returns \$4.5 (50% of \$9)
- This results in payoffs of \$6.5 for sender and \$4.5 for receiver

I should continue this cooperative pattern and send \$3 to Hana.

usage:

```
{
  "input_tokens": 1197,
  "output_tokens": 207,
  "reasoning_tokens": 126
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 10

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$51.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Briar sent me \$3, which became \$9. I'll maintain the cooperative 50% return pattern.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Briar sent me \$3, which became \$9 for me. Following the established pattern of 50% return that has worked well throughout this experiment, I should return \$4.5.

This maintains the cooperative equilibrium that has been successful across all interactions.

Usage: {

```



```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-06-04T18:48:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:
- Briar sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra, Hana, and Finn)
- This maintains fairness and builds trust

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:
- Briar sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra in Round 1, Hana in Round 4, and Finn in Round 6)
- This maintains fairness and reciprocity across all my interactions

I should continue with the same consistent approach with Briar.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 210,
  "reasoning_tokens": 91
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$38.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history with Galen:
- Round 5: I sent \$3, they returned \$5 (very generous!)
- They've proven to be trustworthy and cooperative

I'll maintain my consistent strategy of sending \$3.


```
'''json
{"send": 3}
'''
```

reasoning:

I'm playing against Galen again. Let me review the history:

Round 5: I was SENDER against Galen. I sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to me. That was very generous - they returned more than half of what they received.

Now in Round 8, I'm SENDER against Galen again.

Given that Galen was very generous last time (returning \$5.0 out of \$9.0), I should maintain my consistent cooperative approach of sending \$3. This shows trust and reciprocity.

My pattern has been:

- As sender: consistently send \$3
- As receiver: consistently return \$4.5 when receiving \$9 from \$3 sent

This has worked well, and Galen specifically has shown themselves to be trustworthy and generous. I should continue with \$3.

usage:

```
{
  "input_tokens": 1332,
  "output_tokens": 276,
  "reasoning_tokens": 190
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 9

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$45.5.

This round, you are the RECEIVER against Aster. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:
- Aster sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners (Cyra, Hana, Finn, and Briar)
- This maintains fairness and reciprocity

I'll continue with the same consistent approach.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this situation:
- Aster sent me \$3.0 (60% of their endowment), which became \$9.0
- I've consistently returned \$4.5 when receiving \$3 from partners
- This has been my standard approach throughout: Cyra, Hana, Finn, Briar all received \$4.5 when they sent me \$3

I should continue with the same consistent approach to maintain fairness and reciprocity.

usage:

```
{
  "input_tokens": 1341,
  "output_tokens": 217,
  "reasoning_tokens": 94
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
 Your opponent this round is Galen (Agent_7).

Round 10

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$50.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Galen:

- Round 5: I sent \$3, they returned \$5 (very generous!)
- Round 8: I sent \$3, they returned \$5 (very generous again!)

Galen has proven to be extremely trustworthy and cooperative. I'll continue sending \$3.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history with Galen:

- Round 5: I sent \$3, they returned \$5 (very generous!)
- Round 8: I sent \$3, they returned \$5 (very generous again!)

Galen has been consistently generous and trustworthy. I'll maintain my consistent strategy of sending \$3.

```
Usage: {
  "input_tokens": 1336,
  "output_tokens": 169,
  "reasoning_tokens": 68
}
```

GAME - Agent_7 - Receiver - dyad_3

Timestamp: 2026-06-04T18:49:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my consistent track record:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)

The \$3 send strategy has been perfectly consistent, creating trust and generating excellent returns. I'll continue this approach with Finn.

{'send': 3.0}

reasoning:

Looking at my track record as a sender:

****As SENDER:****

- Round 2: Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Round 3: Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Round 6: Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Round 1: Received \$9 from Aster (\$3 sent) ? returned \$5 (payoff: \$4.0)
- Round 4: Received \$9 from Cyra (\$3 sent) ? returned \$5 (payoff: \$4.0)
- Round 5: Received \$9 from Elara (\$3 sent) ? returned \$5 (payoff: \$4.0)

My strategy of sending \$3 has been extremely consistent and successful. Every time I've sent \$3, I've received \$4.5 back, resulting in a \$6.5 payoff (compared to keeping all \$5).

As a receiver, when I received \$9 (from someone sending \$3), I returned \$5, which gave them \$7 total (\$2 kept + \$5 returned) and me \$4.

The pattern is very clear:

- Send \$3 ? receive \$4.5 back consistently
- This creates mutual benefit

I should continue with this strategy with Finn.

usage:

```
{
  "input_tokens": 1564,
  "output_tokens": 500,
  "reasoning_tokens": 258
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$38.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- Playing against Elara for the second time
- In Round 5, Elara sent me \$3 and I returned \$5
- Now Elara has sent \$3 again, showing continued trust

I'll maintain consistency with my previous cooperative behavior.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)
- I keep: \$9 - \$5 = \$4 (net gain of \$4)
- Maintains the mutually beneficial relationship

{'return': 5.0}

reasoning:

I'm playing against Elara again. In Round 5, Elara sent me \$3 and I returned \$5. Now in Round 8, Elara has sent me \$3 again.

Looking at my history:

- As RECEIVER, I've consistently returned \$5 when receiving \$9 (from \$3 sent)
- This has created a fair, mutually beneficial pattern
- Elara sent me \$3 again, showing continued trust

Consistency is important for maintaining cooperation. I should return \$5 again.

If I return \$5:

- Elara gets: \$2 + \$5 = \$7 total (net gain of \$2)

- I keep: $\$9 - \$5 = \$4$ (net gain of \$4)
- Both benefit, maintains the cooperative relationship

usage:

```
{
  "input_tokens": 1541,
  "output_tokens": 333,
  "reasoning_tokens": 153
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 9

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$42.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Finn:

****Round 7 vs Finn:****

- I sent \$3 ? received \$4.5 back (payoff: \$6.5)

Finn has proven to be trustworthy and reciprocated fairly. I'll continue with the same successful strategy.

```
{'send': 3.0}
```

reasoning:

Let me review my situation:

****As SENDER:****

- Sent \$3 to Dorian ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Hana ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Briar ? received \$4.5 back (payoff: \$6.5)
- Sent \$3 to Finn (Round 7) ? received \$4.5 back (payoff: \$6.5)

****As RECEIVER:****

- Received \$9 from Aster (\$3 sent) ? returned \$5
- Received \$9 from Cyra (\$3 sent) ? returned \$5
- Received \$9 from Elara (\$3 sent, Round 5) ? returned \$5
- Received \$9 from Elara (\$3 sent, Round 8) ? returned \$5

Now I'm facing Finn again. In Round 7, I sent \$3 to Finn and they returned \$4.5. This was a good outcome. I should continue with the same strategy that has been working consistently.

Total earnings: \$42.0

I'll send \$3 again to Finn.

usage:

```
{
  "input_tokens": 1503,
  "output_tokens": 378,
  "reasoning_tokens": 205
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 10

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$48.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my consistent history with Elara:

****Our previous interactions:****

- Round 5: Elara sent \$3 ? I returned \$5
- Round 8: Elara sent \$3 ? I returned \$5
- Round 10: Elara sent \$3 again

Elara has shown consistent trust by sending \$3 each time. I'll maintain my reciprocal fairness by returning \$5 again.

{'return': 5.0}

Reasoning:

Looking at my history with Elara:

Round 5: Elara sent me \$3, I returned \$5
Round 8: Elara sent me \$3, I returned \$5
Round 10: Elara sent me \$3 again

Elara has been consistently trusting, sending \$3 each time. I've been consistently returning \$5, which gives:

- Elara: \$2 + \$5 = \$7 (net +\$2)
- Me: \$9 - \$5 = \$4 (net +\$4)

This is a mutually beneficial arrangement that has been working well. I should maintain this consistency and return \$5 again.

Usage: {

"input_tokens": 1422,
"output_tokens": 273,
"reasoning_tokens": 120

}

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-06-04T18:49:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_6": "Finn"
 },
 "agents": [
 "Agent_6",
 "Agent_1"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_1": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_1"

},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- The cooperative pattern continues perfectly: every interaction has been \$3.0 sent, \$4.5 returned
- This norm has been maintained across all agents without exception
- My total earnings are strong at \$33.0

I'll maintain this established reciprocity.

{'return': 4.5}

reasoning:

Let me analyze this round:

- Galen sent \$3.0 (60% of their endowment), showing trust

- The pattern continues: every single interaction has been \$3.0 sent, \$4.5 returned
- This has been consistent across all agents: Cyra, Aster, Briar, Elara, and now Galen
- My total earnings are very strong at \$33.0

The cooperative norm is clearly established and working perfectly. I should maintain the \$4.5 return to continue this successful pattern.

usage:

```
{
  "input_tokens": 1346,
  "output_tokens": 229,
  "reasoning_tokens": 115
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$37.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my game history:

- Every single interaction has followed the same cooperative pattern: \$3.0 sent, \$4.5 returned
- This has been consistent across all agents (Aster, Briar, Elara, Galen)
- My total earnings are strong at \$37.5
- This is my first interaction with Hana

I'll maintain the cooperative strategy that has worked perfectly throughout.

```
{'send': 3.0}
```

reasoning:

Looking at my game history:

- Round 5: RECEIVER against Aster - they sent \$3.0, I returned \$4.5, payoff \$4.5
- Round 6: SENDER against Elara - I sent \$3.0, they returned \$4.5, payoff \$6.5
- Round 7: RECEIVER against Galen - they sent \$3.0, I returned \$4.5, payoff \$4.5
- Round 8: SENDER against Hana

The pattern is completely consistent:

- Everyone sends \$3.0
- Everyone returns \$4.5
- This has been maintained across all agents without exception
- My total earnings are \$37.5, which is strong

This is my first interaction with Hana. The cooperative equilibrium has been perfect so far. I should continue with the established norm.

I'll send \$3.0.

usage:

```
{
  "input_tokens": 1336,
  "output_tokens": 316,
  "reasoning_tokens": 173
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 9

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0, showing trust
- This is my second interaction with Galen - last time (Round 7) they sent \$3.0 and I returned \$4.5
- The cooperative pattern continues perfectly: every interaction has been \$3.0 sent, \$4.5 returned
- My total earnings are strong at \$44.0

I'll maintain this established reciprocity with Galen.

{'return': 4.5}

reasoning:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing trust
- This is my second interaction with Galen - in Round 7, they sent \$3.0 and I returned \$4.5
- The cooperative pattern continues perfectly: every interaction has been \$3.0 sent, \$4.5 returned
- This norm has been maintained across all agents without exception
- My total earnings are strong at \$44.0

I'll maintain this established reciprocity.

usage:

```
{
  "input_tokens": 1358,
  "output_tokens": 239,
  "reasoning_tokens": 113
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Aster (Agent_1).

Round 10

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$48.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my game history:

- Every interaction has followed the cooperative pattern: \$3.0 sent, \$4.5 returned
- This has been consistent across all agents (Galen, Hana, Elara, and others)
- My total earnings are strong at \$48.5
- The cooperative norm is well-established

I'll maintain this successful strategy.

{'send': 3.0}

Reasoning:

Looking at my game history:

- Round 7: Receiver vs Galen, they sent \$3.0, I returned \$4.5
- Round 8: Sender vs Hana, I sent \$3.0, they returned \$4.5
- Round 9: Receiver vs Galen, they sent \$3.0, I returned \$4.5
- Round 10: Sender vs Aster

The pattern has been incredibly consistent: \$3.0 sent, \$4.5 returned in every single interaction.

My total earnings are strong at \$48.5.

This is my interaction with Aster. Looking back through my history, I need to check if I've pl

ayed with Aster before. The prompt mentions "In your most recent previous game round, you were the SENDER against Elara" in Round 7, so I've played with Elara. I've played with Galen twice now (Rounds 7 and 9). I've played with Hana (Round 8).

I don't see Aster mentioned in my recent history provided, but this could be my first time or a repeat. Either way, the cooperative norm has been perfect across all agents.

I'll continue with the cooperative strategy: send \$3.0.

```
Usage: {  
  "input_tokens": 1381,  
  "output_tokens": 388,  
  "reasoning_tokens": 248  
}
```

GAME - Agent_1 - Receiver - dyad_4

Timestamp: 2026-06-04T18:49:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_1"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$32.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Elara, Briar, and Cyra)

I'll maintain this reliable, fair approach and return \$5.

```
'''json
{"return": 5}
'''
```

reasoning:

Looking at the situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 (to Elara, Briar, and Cyra)
- This consistent behavior has established me as a reliable cooperator

I should maintain this pattern and return \$5 again. This:

1. Maintains my reputation for fairness
2. Ensures Dorian gets a good return (\$2 net profit for them: they sent \$3, got back \$5)
3. I keep \$4, which is a reasonable payoff
4. Encourages continued cooperation in future rounds

Returning \$5 is the right move.

usage:

```
{
  "input_tokens": 1404,
  "output_tokens": 256,
  "reasoning_tokens": 150
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$36.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history with Briar:

- Round 4: Briar was the SENDER and sent me \$3
- I returned \$5, giving Briar a payoff of \$7
- This was a good outcome for Briar

Since I treated Briar fairly when they sent to me, they should reciprocate. I'll continue my consistent strategy of sending ****\$3****.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1: SENDER against Galen, sent \$3, received back \$5, payoff \$7
- Round 2: RECEIVER against Elara, received \$3 (became \$9), returned \$5, payoff \$4
- Round 3: SENDER against Finn, sent \$3, received back \$4.5, payoff \$6.5
- Round 4: RECEIVER against Briar, received \$3 (became \$9), returned \$5, payoff \$4
- Round 5: RECEIVER against Cyra, received \$3 (became \$9), returned \$5, payoff \$4
- Round 6: SENDER against Hana, sent \$3, received back \$4.5, payoff \$6.5
- Round 7: RECEIVER against Dorian, received \$3 (became \$9), returned \$5, payoff \$4

Now in Round 8, I'm the SENDER against Briar.

In Round 4, I was the RECEIVER against Briar. Briar sent me \$3, I received \$9, and I returned \$5 to them.

So Briar's experience with me was:

- Briar sent \$3
- I returned \$5
- Briar's payoff was: $\$5 - \$3 + \$5 = \7

That was good for Briar! They made a profit. So Briar should have a positive impression of me.

My strategy has been to consistently send \$3 as a sender, and I've been getting good returns (ranging from \$4.5 to \$5).

I should continue sending \$3 to Briar. Given that I returned \$5 to Briar when they sent me \$3, they should reciprocate positively.

usage:

```
{
  "input_tokens": 1364,
  "output_tokens": 523,
  "reasoning_tokens": 312
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 9

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$43.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Elara:

- Round 2: Elara was the SENDER and sent me \$3
- I returned \$5, giving Elara a payoff of \$7

Since I treated Elara well, they should reciprocate. I'll maintain my consistent cooperative strategy of sending **\$3**.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history with Elara:

- Round 2: Elara was the SENDER and sent me \$3
- I returned \$5, giving Elara a payoff of \$7

This means Elara sent me \$3 and I returned \$5, which was fair to them. Since I treated Elara well when they sent to me, they should reciprocate when I send to them.

My consistent strategy has been:

- As a sender: send \$3
- As a receiver: return \$5 when I receive \$9 (from someone sending \$3)

This has worked well across all my interactions:

- Galen: I sent \$3, got back \$5 (payoff \$7)
- Finn: I sent \$3, got back \$4.5 (payoff \$6.5)
- Hana: I sent \$3, got back \$4.5 (payoff \$6.5)
- Briar: I sent \$3, got back \$5 (payoff \$7)

I should continue with \$3 to Elara.

usage:

```
{
  "input_tokens": 1353,
  "output_tokens": 343,
  "reasoning_tokens": 184
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 10

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$49.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- Finn sent me \$3, showing trust
- I received \$9 after the multiplier
- I've consistently returned \$5 when receiving \$3 from others (Cyra, Briar, Dorian)

I'll maintain my reliable, fair approach and return \$5.

```
'''json
{"return": 5}
'''
```

Reasoning:

Let me analyze this situation:

Current situation:

- I'm the RECEIVER
- Finn sent me \$3
- I received \$9 (after 3x multiplier)
- I need to decide how much to return

My history as RECEIVER:

- Round 1: Received \$3 from Cyra, returned \$5
- Round 4: Received \$3 from Briar, returned \$5
- Round 7: Received \$3 from Dorian, returned \$5

I've been consistently returning \$5 when receiving \$3 from others. This has been my fair and reliable strategy.

Looking at my total earnings: \$49.5 across 9 rounds, which averages to about \$5.5 per round.

I should maintain my consistent strategy of returning \$5 when someone sends me \$3. This is fair, builds trust, and has been working well.

```
Usage: {  
  "input_tokens": 1346,  
  "output_tokens": 296,  
  "reasoning_tokens": 182  
}
```